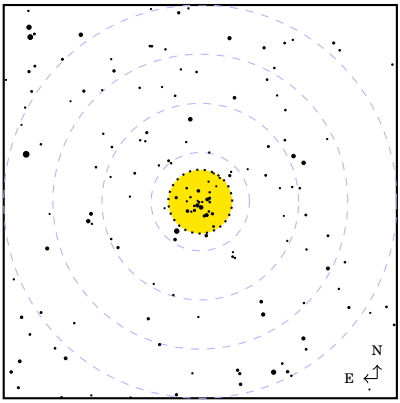
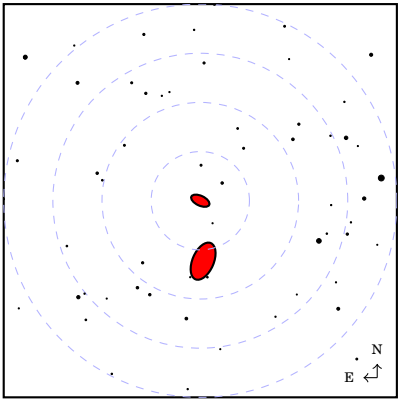
Pocket Finder-Chart Atlas





Correct-Image Version

Alan Watson Forster

Pocket Finder-Chart Atlas

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alanwatsonforster@gmail.com

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Correct-image version of 2022-03-26

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Preface

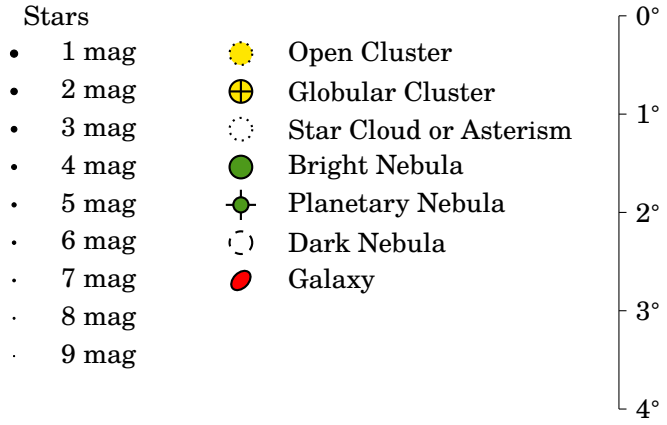
This is an atlas of finder charts for the Messier, Caldwell, and Astronomical League Urban Observing Program deep-sky objects.

The charts have a field of 4 degrees at a scale of 13 mm per degree. Circles show fields with diameters of 1, 2, 3, and 4 degrees. The charts show stars to magnitude 9.5, and represent deep-sky objects following the current convention. The correspondence between the magnitude of a star and its size on the charts is designed to better match the view in a telescope eyepiece.

There are versions of the atlas with inverted-image and correct-image charts. This version has correct-image charts, with north up and east to the left, corresponding to the view through a refractor or Cassegrain telescope either without a diagonal or with a correct-image diagonal.

The size and binding of this atlas are designed for convenience at the telescope.

Chart Legend



I created this atlas to assist me as I observed under the light-polluted skies of Mexico City with a 70 mm f/6 refractor. I don't use a conventional finder with this telescope, but instead star-hop using a 32 mm eyepiece with a true field of about 4 degrees. Under my usual observing conditions, very few objects are immediately obvious, and the charts tell me when I've successfully reached the desired field and where to focus my efforts.

The obvious question is, but why not just use use an all-sky atlas? I do indeed use the *Pocket Sky Atlas* for star-hopping, but I find that for confirming a field none of my all-sky atlases combine adequate depth with convenience at the telescope, and none really give me the sense of what I see through the eyepiece.

To some degree, the relation between these charts and the *Pocket Sky Atlas* is similar to the relation between the larger-scale and smaller-scale charts in *The Observer's Sky Atlas*. The smaller-scale charts are for star-hopping to the field and the larger-scale charts are to confirm the field and locate the object. I don't use *The Observer's Sky Atlas* at the telescope, because the smaller-scale charts are too shallow and too narrow for my needs, the large-scale charts aren't inverted, and the binding is inconvenient.

The positions and magnitudes of stars are taken from the Tycho-2 catalog, supplemented by the Hipparcos catalog for the very brightest stars. The positions, types, and sizes of deep sky objects are taken from the CDS SIMBAD database, supplemented by information from the NASA/IPAC Extragalactic Database (NED), O'Meara's *Deep-Sky Companions* volumes, and the Astronomical League's Urban Observing Program list.

One flaw of these charts is that all deep-sky objects are drawn as either ellipses (galaxies and star clouds) or circles (everything else). This means, for example, that they do not correctly show the shapes of irregular nebulae such as M42 and M43.

The Messier Objects

The Messier objects are probably the most famous deep-sky objects. I find O'Meara's *Deep-Sky Companion: The Messier Objects* to be excellent on the origin of the catalog, the appearance of the objects, and their nature.

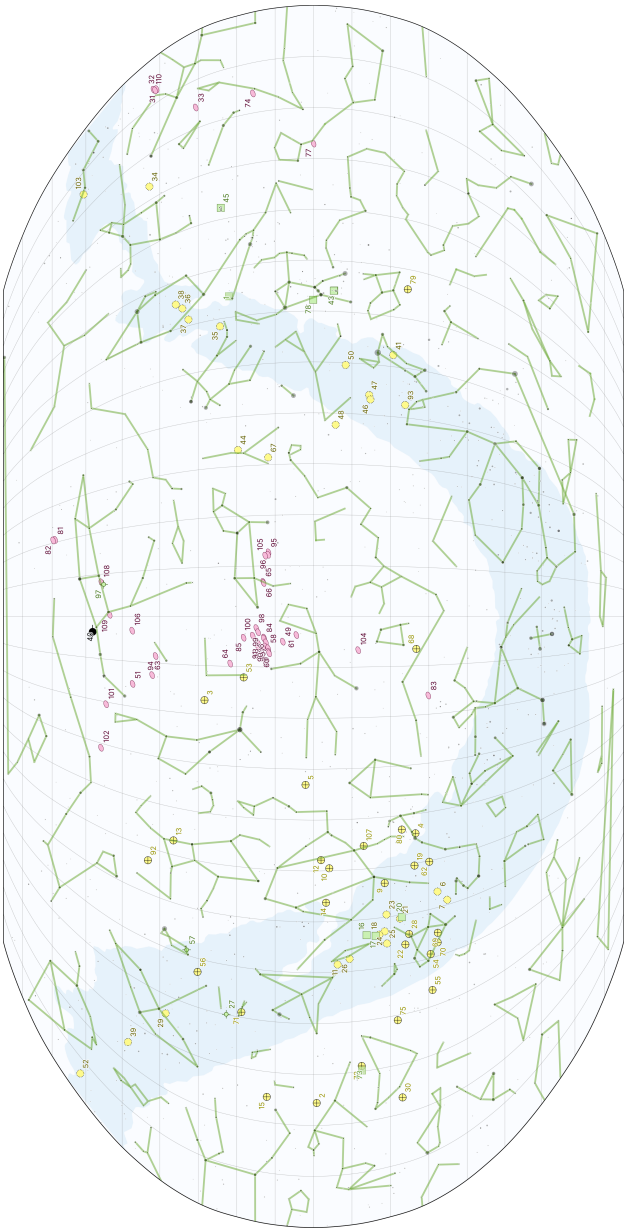
The following table lists the objects with their J2000 positions (decimal hours of right ascension and degrees of declination), the charts on which they appear in the *Pocket Sky Atlas*, their types, and other names. I follow O'Meara in identifying M102 as NGC 5866.

For completeness, I include finder charts for all Messier objects, even bright ones like M45.

Name	Position	Type	Other Names
M1	05.6 +22	BN	NGC 1952 = BSC 119 = Crab Nebula
M2	21.6 −01	GC	NGC 7089 = U81 = BSC 56
M3	13.7 +28	GC	NGC 5272 = U51 = BSC 53
M4	16.4 −27	GC	NGC 6121 = U53 = BSC 43
M5	15.3 +02	GC	NGC 5904 = U52 = BSC 66
M6	17.7 −32	OC	NGC 6405 = U60 = BSC 18 = Butterfly Nebula
M7	17.9 −35	OC	NGC 6475 = U62 = BSC 10
M8	18.1 −24	BN	NGC 6523/6530 = U64 = BSC 45 = Lagoon Nebula
M9	17.3 −19	GC	NGC 6333 = BSC 123
M10	17.0 −04	GC	NGC 6254 = U57 = BSC 61
M11	18.9 −06	OC	NGC 6705 = U69 = Mel 213 = Cr 391 = Wild Duck Cluster
M12	16.8 −02	GC	NGC 6218 = U56 = BSC 94
M13	16.7 +36	GC	NGC 6205 = U54 = BSC 40 = Hercules Cluster
M14	17.6 −03	GC	NGC 6402 = BSC 116
M15	21.5 +12	GC	NGC 7078 = U80 = BSC 54
M16	18.3 −14	BN	NGC 6611 = IC 4703 = BSC 46 = Eagle Nebula
M17	18.3 −16	BN	NGC 6618 = U65 = BSC 47 = Omega Nebula
M18	18.3 −17	OC	NGC 6613 = BSC 87
M19	17.0 −26	GC	NGC 6273 = BSC 86
M20	18.0 −23	BN	NGC 6514 = BSC 57 = Cr 360 = Trifid Nebula
M21	18.1 −23	OC	NGC 6531 = BSC 65
M22	18.6 −24	GC	NGC 6656 = U67 = BSC 28
M23	17.9 −19	OC	NGC 6494 = BSC 69
M24	18.3 −19	SC	IC 4715 = BSC 23
M25	18.5 −19	OC	IC 4725 = BSC 24
M26	18.8 −09	OC	NGC 6694 = BSC 104
M27	20.0 +23	PN	NGC 6853 = U75 = BSC 88 = Dumbell Nebula
M28	18.4 −25	GC	NGC 6626 = BSC 93
M29	20.4 +38	OC	NGC 6913 = BSC 75
M30	21.7 −23	GC	NGC 7099 = BSC 95
M31	00.7 +41	Gal	NGC 224 = U3 = BSC 11 = Andromeda Galaxy
M32	00.7 +41	Gal	NGC 221 = U2 = BSC 110
M33	01.6 +31	Gal	NGC 598 = BSC 39
M34	02.7 +43	OC	NGC 1039 = BSC 35
M35	06.2 +24	OC	NGC 2168 = U27 = BSC 31
M36	05.6 +34	OC	NGC 1960 = U23 = BSC 58
M37	05.9 +33	OC	NGC 2099 = U26 = BSC 55
M38	05.5 +36	OC	NGC 1912 = U22 = BSC 82
M39	21.5 +48	OC	NGC 7092 = U82 = BSC 36
M40	12.4 +58	DS	
M41	06.8 −21	OC	NGC 2287 = U33 = BSC 21
M42	05.6 −05	BN	NGC 1976 = U24 = BSC 15 = Orion Nebula
M43	05.6 −05	BN	NGC 1982 = BSC 136
M44	08.7 +20	OC	NGC 2632 = U39 = BSC 12 = Mel 88 = Beehive Cluster
M45	03.8 +24	OC	U17 = BSC 4 = Mel 22 = Pleiades
M46	07.7 −15	OC	NGC 2437 = BSC 50
M47	07.6 −14	OC	NGC 2422 = BSC 19
M48	08.2 −06	OC	NGC 2548 = U38 = BSC 37
M49	12.5 +08	Gal	NGC 4472 = BSC 150
M50	07.0 −08	OC	NGC 2323 = U35 = BSC 44

Name	Position	Type	Other Names
M51	13.5 +47	Gal	NGC 5194 = BSC 120 = Whirlpool Galaxy
M52	23.4 +62	OC	NGC 7654 = BSC 27
M53	13.2 +18	GC	NGC 5024 = BSC 117
M54	18.9 −30	GC	NGC 6715 = BSC 118
M55	19.7 −31	GC	NGC 6809 = BSC 85
M56	19.3 +30	GC	NGC 6779 = BSC 113
M57	18.9 +33	PN	NGC 6720 = U71 = BSC 127 = Ring Nebula
M58	12.6 +12	Gal	NGC 4579
M59	12.7 +12	Gal	NGC 4621
M60	12.7 +12	Gal	NGC 4649
M61	12.4 +04	Gal	NGC 4303
M62	17.0 −30	GC	NGC 6266 = U58 = BSC 81
M63	13.3 +42	Gal	NGC 5055 = BSC 145 = Sunflower Galaxy
M64	12.9 +22	Gal	NGC 4826 = U50 = BSC 149 = Black-Eye Galaxy
M65	11.3 +13	Gal	NGC 3623
M66	11.3 +13	Gal	NGC 3627 = BSC 130
M67	08.9 +12	OC	NGC 2682 = U40 = BSC 51
M68	12.7 −27	GC	NGC 4590
M69	18.5 −32	GC	NGC 6637
M70	18.7 −32	GC	NGC 6681 = BSC 105
M71	19.9 +19	GC	NGC 6838 = BSC 52
M72	20.9 −13	GC	NGC 6981 = BSC 148
M73	21.0 −13	AST	NGC 6994 = Cr 426
M74	01.6 +16	Gal	NGC 628
M75	20.1 −22	GC	NGC 6864 = BSC 144
M76	01.7 +52	PN	NGC 651/650 = Little Dumbell
M77	02.7 −00	Gal	NGC 1068 = U12
M78	05.8 +00	BN	NGC 2068 = BSC 114
M79	05.4 −25	GC	NGC 1904 = BSC 126
M80	16.3 −23	GC	NGC 6093 = BSC 101
M81	09.9 +69	Gal	NGC 3031 = U41 = BSC 72
M82	09.9 +70	Gal	NGC 3034 = U42 = BSC 122
M83	13.6 −30	Gal	NGC 5236 = BSC 89
M84	12.4 +13	Gal	NGC 4374 = U45
M85	12.4 +18	Gal	NGC 4382
M86	12.4 +13	Gal	NGC 4406 = U46
M87	12.5 +12	Gal	NGC 4486 = U47
M88	12.5 +14	Gal	NGC 4501
M89	12.6 +13	Gal	NGC 4552
M90	12.6 +13	Gal	NGC 4569
M91	12.6 +14	Gal	NGC 4548
M92	17.3 +43	GC	NGC 6341 = U59 = BSC 59
M93	07.7 −24	OC	NGC 2447 = BSC 48 = Cr 160
M94	12.8 +41	Gal	NGC 4736 = U49 = BSC 134
M95	10.7 +12	Gal	NGC 3351
M96	10.8 +12	Gal	NGC 3368
M97	11.2 +55	PN	NGC 3587
M98	12.2 +15	Gal	NGC 4192
M99	12.3 +14	Gal	NGC 4254
M100	12.4 +16	Gal	NGC 4321

Name	Position	Type	Other Names
M101	14.1 +54	Gal	NGC 5457 = BSC 100
M102	15.1 +56	Gal	NGC 5866
M103	01.6 +61	OC	NGC 581 = BSC 83
M104	12.7 −12	Gal	NGC 4594 = U48 = BSC 133 = Sombrero Galaxy
M105	10.8 +13	Gal	NGC 3379
M106	12.3 +47	Gal	NGC 4258 = BSC 141
M107	16.5 −13	GC	NGC 6171 = BSC 129
M108	11.2 +56	Gal	NGC 3556
M109	12.0 +53	Gal	NGC 3992
M110	00.7 +42	Gal	NGC 205 = BSC 132



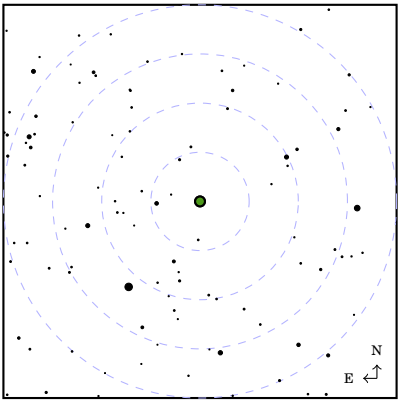
The distribution of Messier objects over the whole sky

M1

NGC 1952 = BSC 119 = Crab Nebula

RA : 05.6^h	PSA: 14	Type: BN
Dec: $+22°$	OSA: E3	Mag : 8.0
Con: Tau	CSA: 3/9	Size : $6.0' \times 4.0'$

See:

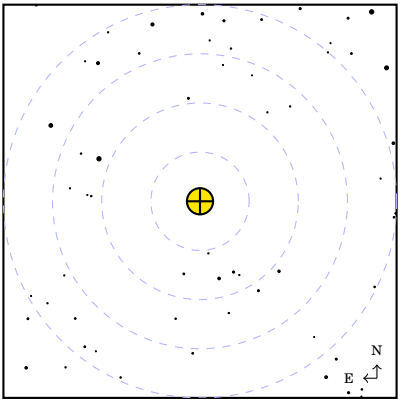


M2

NGC 7089 = U81 = BSC 56

RA : 21.6^h	PSA: 75/77	Type: GC
Dec: $-01°$	OSA: E24	Mag : 6.3
Con: Aqr	CSA: 13	Size : $16.0'$

See:

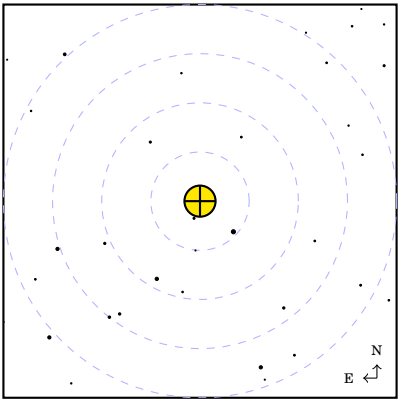


M3

NGC 5272 = U51 = BSC 53

RA : 13.7^h	PSA: 43/44	Type: GC
Dec: $+28°$	OSA: E15	Mag : 5.9
Con: CVn	CSA: 5	Size : $19.0'$

See:

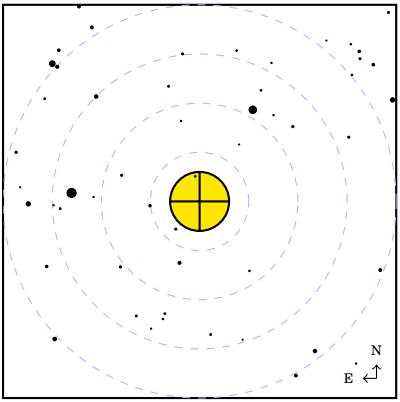


M4

NGC 6121 = U53 = BSC 43

RA : 16.4^h	PSA: 56/58	Type: GC
Dec: $-27°$	OSA: E18	Mag : 5.4
Con: Sco	CSA: 11/12/17/18	Size : 36.0′

See:

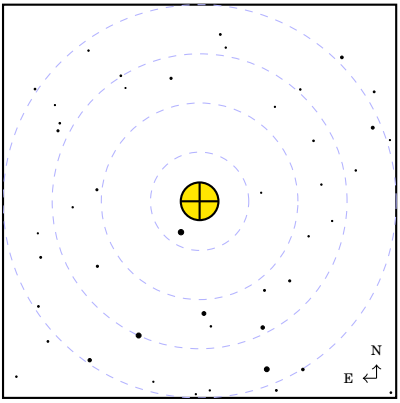


M5

NGC 5904 = U52 = BSC 66

RA : 15.3^h	PSA: 55/57	Type: GC
Dec: $+02°$	OSA: E15	Mag : 5.7
Con: Ser	CSA: 11/12	Size : 23.0′

See:



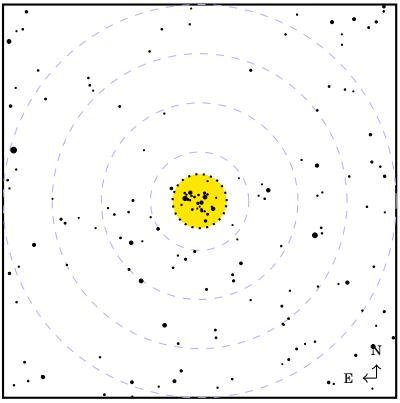
M6

NGC 6405 = U60 = BSC 18 = Butterfly Nebula

RA : 17.7^h	PSA: 58/69	Type: OC
Dec: $-32°$	OSA: E18	Mag : 4.2
Con: Sco	CSA: 18	Size : 33.0′

See:

λ Sco Shaula (mag 1.6)
→ 5° N
→ 1° E to M6/U60



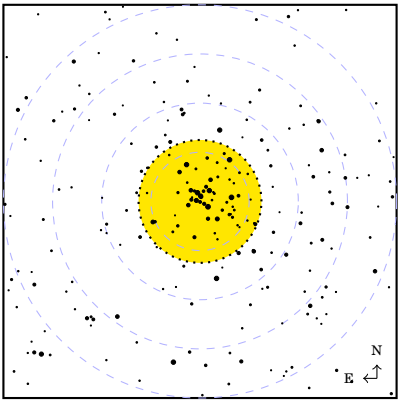
M7

NGC 6475 = U62 = BSC 10

RA : 17.9^h	PSA: 58/69	Type: OC
Dec: $-35°$	OSA: E18	Mag : 2.8
Con: Sco	CSA: 18	Size : 75.0′

See:

ε Sgr Kaus Australis (mag 1.8)
→ 6° W to M7/U62



M8

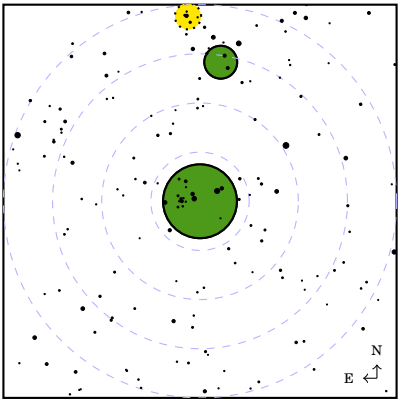
NGC 6523/6530 = U64 = BSC 45 = Lagoon Nebula

RA : 18.1^h	PSA: 67	Type: BN
Dec: $-24°$	OSA: E20	Mag : 3.0
Con: Sgr	CSA: 12/18	Size : 45.0′ × 30.0′

See:

Notes: M20 is 1.5° N. M21 is 2.0° N.

γ Sgr Alnasl (mag 3.0)
→ 6° N to M8/U64

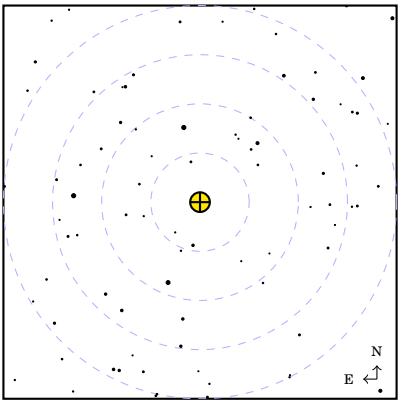


M9

NGC 6333 = BSC 123

RA : 17.3^h	PSA: 56	Type: GC
Dec: $-19°$	OSA: E17	Mag : 7.8
Con: Oph	CSA: 12/18	Size : 12.0′

See:

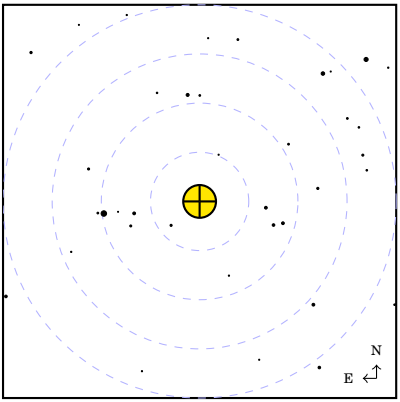


M10

NGC 6254 = U57 = BSC 61

RA : 17.0^h	PSA: 54/56	Type: GC
Dec: −04°	OSA: E17	Mag : 6.6
Con: Oph	CSA: 12	Size : 20.0′

See:

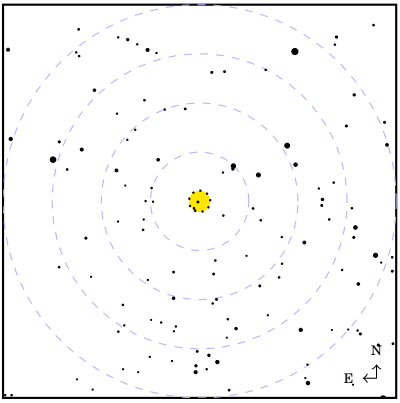


M11

NGC 6705 = U69 = Mel 213 = Cr 391 = Wild Duck Cluster

RA : 18.9^h	PSA: 65/67	Type: OC
Dec: −06°	OSA: E19	Mag : 5.3
Con: Sct	CSA: 12	Size : 13.0′

See:

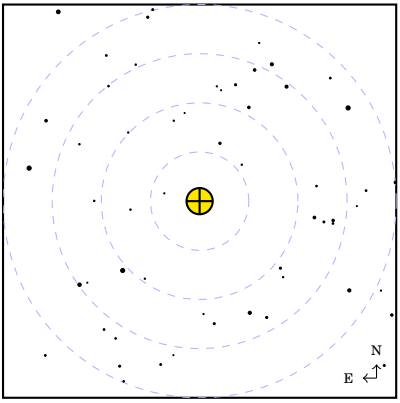


M12

NGC 6218 = U56 = BSC 94

RA : 16.8^h	PSA: 54/56	Type: GC
Dec: −02°	OSA: E17	Mag : 6.1
Con: Oph	CSA: 12	Size : 16.0′

See:



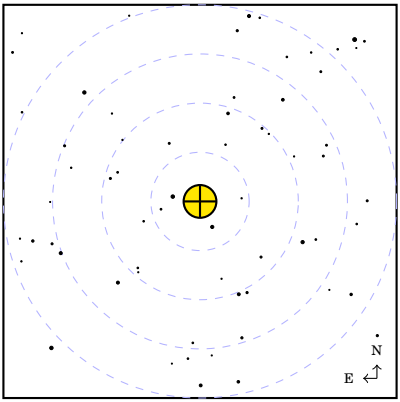
M13

NGC 6205 = U54 = BSC 40 = Hercules Cluster

RA : 16.7^h	PSA: 52	Type: GC
Dec: $+36°$	OSA: N14	Mag : 5.3
Con: Her	CSA: 5/6	Size : 20.0′

See:

ζ Her (mag 2.8)
→ 5° N (2/3 distance to η Her) to M13/U54

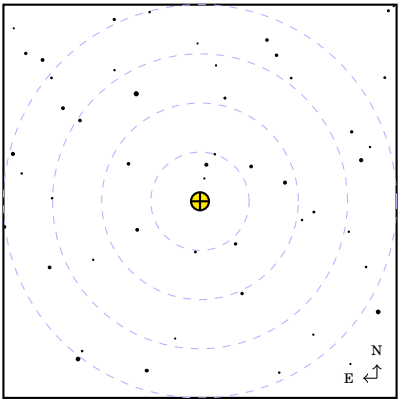


M14

NGC 6402 = BSC 116

RA : 17.6^h	PSA: 54	Type: GC
Dec: $-03°$	OSA: E17	Mag : 7.6
Con: Oph	CSA: 12	Size : 11.0′

See:

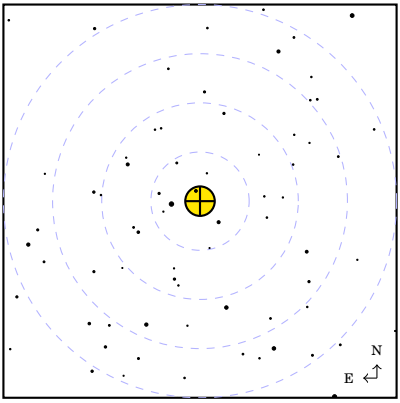


M15

NGC 7078 = U80 = BSC 54

RA : 21.5^h	PSA: 75	Type: GC
Dec: $+12°$	OSA: E23	Mag : 6.0
Con: Peg	CSA: 13	Size : 18.0′

See:



M16

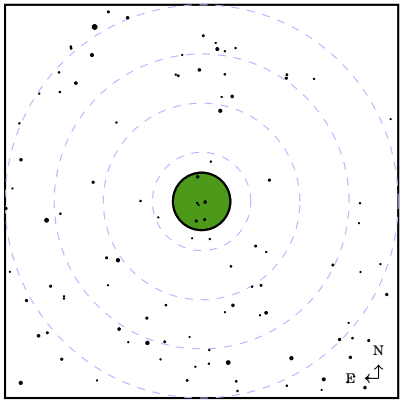
NGC 6611 = IC 4703 = BSC 46 = Eagle Nebula

RA : 18.3^h PSA: 67 Type: BN
Dec: $-14°$ OSA: E20 Mag : 6.0
Con: Ser CSA: 12 Size : 35.0′ × 28.0′

See:

ν Oph (mag 3.3)
→ 4° E
→ 4° S to M16

μ Sgr (mag 3.8)
→ 7° N
→ 1° E to M16



M17

NGC 6618 = U65 = BSC 47 = Omega Nebula

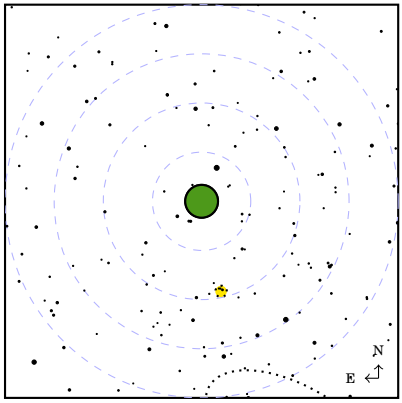
RA : 18.3^h PSA: 67 Type: BN
Dec: $-16°$ OSA: E20 Mag : 6.0
Con: Sgr CSA: 12 Size : 20.0′ × 15.0′

See:

Notes: M18 is 1° S. M24 is 3° SW.

ν Oph (mag 3.3)
→ 5° E
→ 6° S to M17/U65

μ Sgr (mag 3.8)
→ 5° N passing M24 and M18
→ 2° E to M17



M18

NGC 6613 = BSC 87

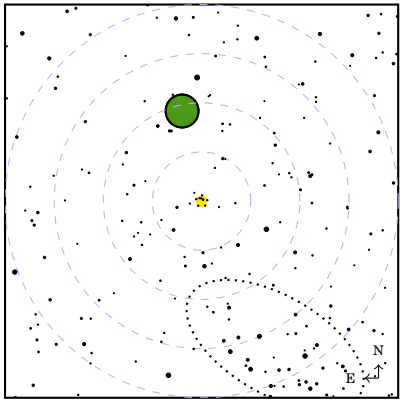
RA : 18.3^h PSA: 67 Type: OC
Dec: $-17°$ OSA: E20 Mag : 6.9
Con: Sgr CSA: 12/18 Size : 7.0′

See:

Notes: M17/U65 is 1° N. M24 is 2° SW.

ν Oph (mag 3.3)
→ 5° E
→ 7° S passing M17/U65 to M18

μ Sgr (mag 3.8)
→ 4° N passing M24
→ 2° E to M18

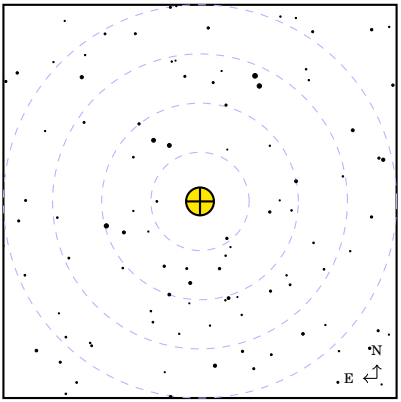


M19

NGC 6273 = BSC 86

RA : 17.0^h	PSA: 56	Type: GC
Dec: −26°	OSA: E18	Mag : 6.8
Con: Oph	CSA: 12/18	Size : 17.0′

See:



M20

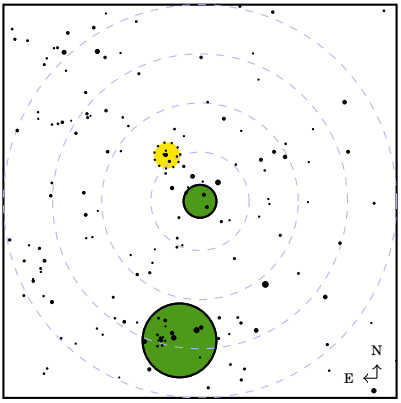
NGC 6514 = BSC 57 = Cr 360 = Trifid Nebula

RA : 18.0^h	PSA: 67	Type: BN
Dec: −23°	OSA: E20	Mag : 6.3
Con: Sgr	CSA: 12/18	Size : 20.0′

See:

Notes: M8/U64 is 1.5° S. M21 is 0.5° NE.

γ Sgr Alnasl (mag 3.0)
→ 7.5° N passing M8/U64 to M20



M21

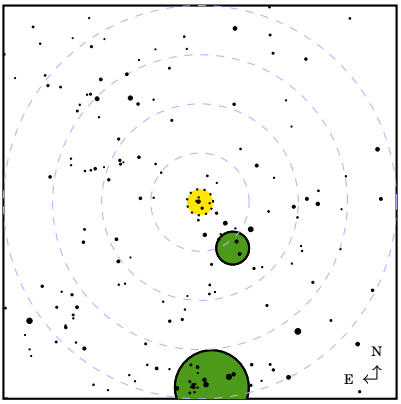
NGC 6531 = BSC 65

RA : 18.1^h	PSA: 67	Type: OC
Dec: −23°	OSA: E20	Mag : 5.9
Con: Sgr	CSA: 12/18	Size : 16.0′

See:

Notes: M8/U64 is 2.0° S. M20 is 0.5° SW.

γ Sgr Alnasl (mag 3.0)
→ 8.0° N passing M8/U64 and M20 to M21

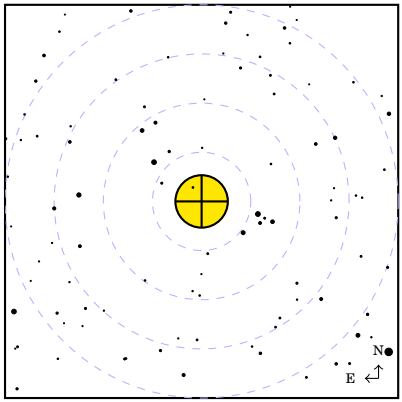


M22

NGC 6656 = U67 = BSC 28

RA : 18.6^h	PSA: 67	Type: GC
Dec: $-24°$	OSA: E20	Mag : 5.2
Con: Sgr	CSA: 12/18	Size : 32.0′

See:

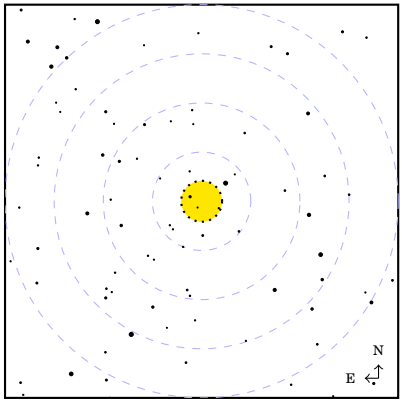


M23

NGC 6494 = BSC 69

RA : 17.9^h	PSA: 67	Type: OC
Dec: $-19°$	OSA: E20	Mag : 5.5
Con: Sgr	CSA: 12/18	Size : 25.0′

See:



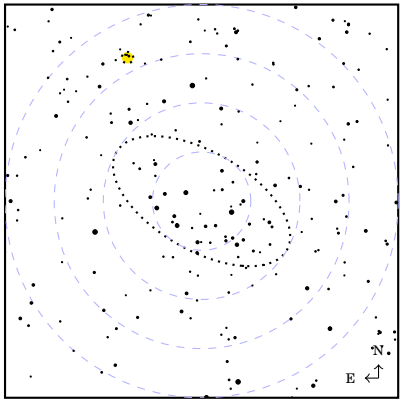
M24

IC 4715 = BSC 23

RA : 18.3^h	PSA: 67	Type: SC
Dec: $-19°$	OSA: E20	Mag : 2.5
Con: Sgr	CSA: 12/18	Size : 120.0′ × 60.0′

See:

Notes: M18 is 1.5° NE. M17/U65 is 3° NE.

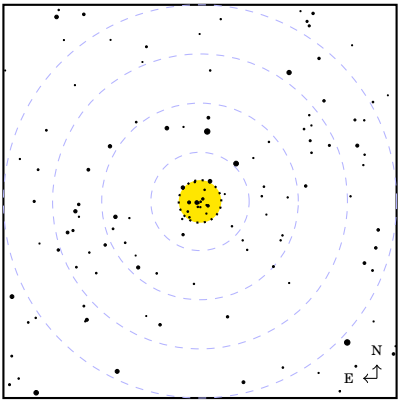


M25

IC 4725 = BSC 24

RA : 18.5^h	PSA: 67	Type: OC
Dec: $-19°$	OSA: E20	Mag : 4.6
Con: Sgr	CSA: 12/18	Size : 26.0′

See:

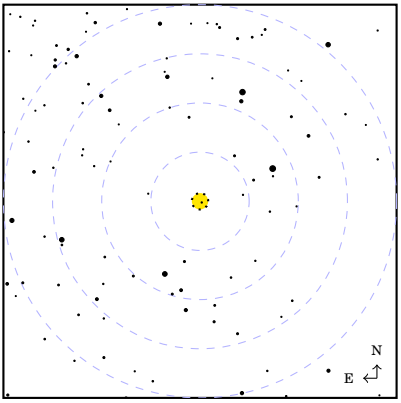


M26

NGC 6694 = BSC 104

RA : 18.8^h	PSA: 67	Type: OC
Dec: $-09°$	OSA: E19	Mag : 8.0
Con: Sct	CSA: 12	Size : 10.0′

See:

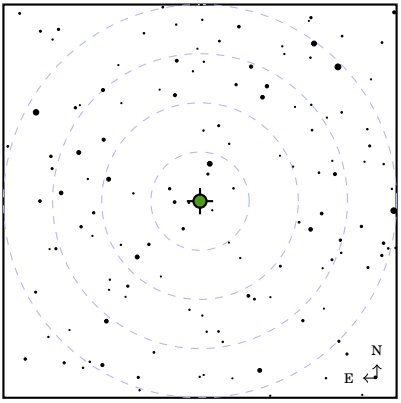


M27

NGC 6853 = U75 = BSC 88 = Dumbell Nebula

RA : 20.0^h	PSA: 64	Type: PN
Dec: $+23°$	OSA: E21	Mag : 7.3
Con: Vul	CSA: 6/7/12/13	Size : 8.0′×6.0′

See:

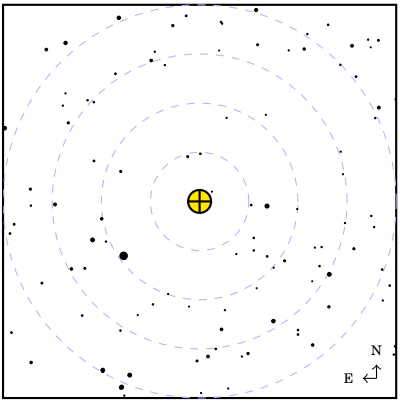


M28

NGC 6626 = BSC 93

RA : 18.4^h	PSA: 67/I	Type: GC
Dec: $-25°$	OSA: E20	Mag : 6.9
Con: Sgr	CSA: 12/18	Size : 13.8′

See:



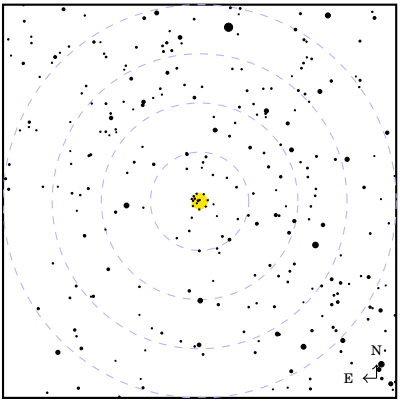
M29

NGC 6913 = BSC 75

RA : 20.4^h	PSA: 62	Type: OC
Dec: $+38°$	OSA: N20	Mag : 6.6
Con: Cyg	CSA: 6/7	Size : 10.0′

See:

γ Cyg Sadr (mag 2.2)
→ 1.8° S to M29

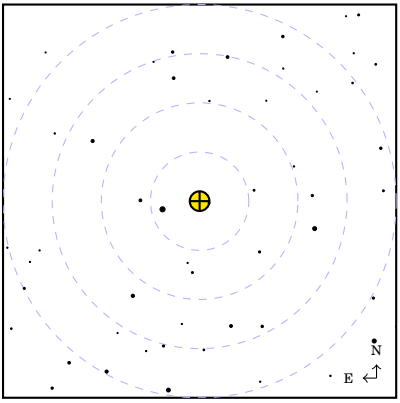


M30

NGC 7099 = BSC 95

RA : 21.7^h	PSA: 77	Type: GC
Dec: $-23°$	OSA: E22	Mag : 6.9
Con: Cap	CSA: 13/19	Size : 12.0′

See:



M31

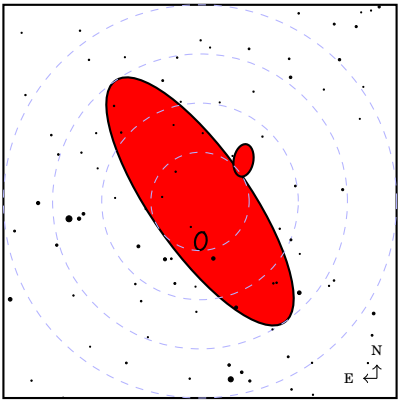
NGC 224 = U3 = BSC 11 = Andromeda Galaxy

RA : 00.7^h PSA: 3 Type: Gal
Dec: +41° OSA: N0 Mag : 3.4
Con: And CSA: 2/7 Size : 3.0° × 1.0°

See:

Notes: M32/U2 is 0.5° S. M100 is 0.5° NW.

β And Mirach (mag 2.1)
→ 2.5° W
→ 3.0° N to μ And (mag 3.9)
→ 2.5° W
→ 3.0° N to M31/U3



M32

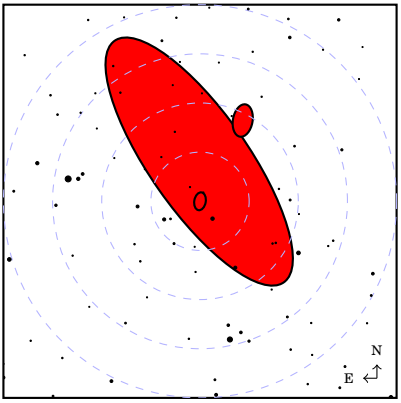
NGC 221 = U2 = BSC 110

RA : 00.7^h PSA: 3 Type: Gal
Dec: +41° OSA: N0 Mag : 8.2
Con: And CSA: 2/7 Size : 11.0′ × 7.3′

See:

Notes: M31/U3 is 0.5° N. M110 is 1° NW.

β And Mirach (mag 2.1)
→ 2.5° W
→ 3.0° N to μ And (mag 3.9)
→ 2.5° W
→ 2.5° N to M32/U2

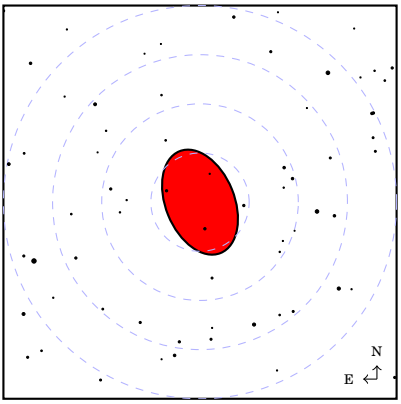


M33

NGC 598 = BSC 39

RA : 01.6^h PSA: 3 Type: Gal
Dec: +31° OSA: N0 Mag : 5.7
Con: Tri CSA: 2 Size : 67.0′ × 41.5′

See:



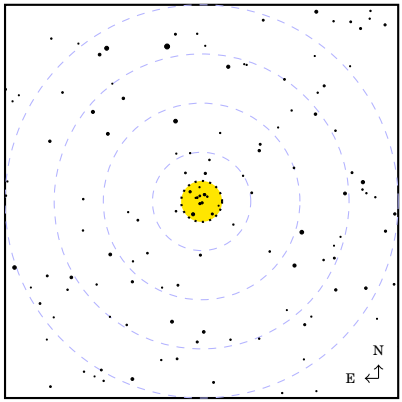
M34

NGC 1039 = BSC 35

RA : 02.7^h	PSA: 2	Type: OC
Dec: +43°	OSA: N4	Mag : 5.2
Con: Per	CSA: 2/3	Size : 25.0′

See:

β Per Algol (mag 2.1–3.4)
→ 2.0° N (1/2 distance to κ Per)
→ 4.5° W (2/5 distance to γ And) to M34

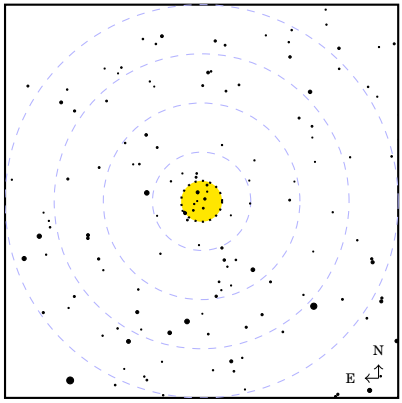


M35

NGC 2168 = U27 = BSC 31

RA : 06.2^h	PSA: 14	Type: OC
Dec: +24°	OSA: E7	Mag : 5.1
Con: Gem	CSA: 3/9	Size : 25.0′

See:

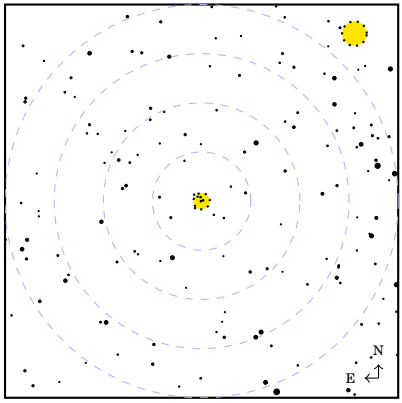


M36

NGC 1960 = U23 = BSC 58

RA : 05.6^h	PSA: 12	Type: OC
Dec: +34°	OSA: N6	Mag : 6.0
Con: Aur	CSA: 3	Size : 10.0′

See:



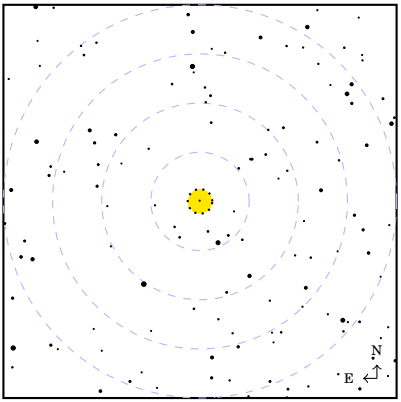
M37

NGC 2099 = U26 = BSC 55

RA : 05.9^h	PSA: 12	Type: OC
Dec: +33°	OSA: N6	Mag : 5.6
Con: Aur	CSA: 3	Size : 15.0′

See:

β Tau Elnath (mag 1.7)
→ 6° W to point 1° N of 136 Tau
→ 4° N to M37/U26



M38

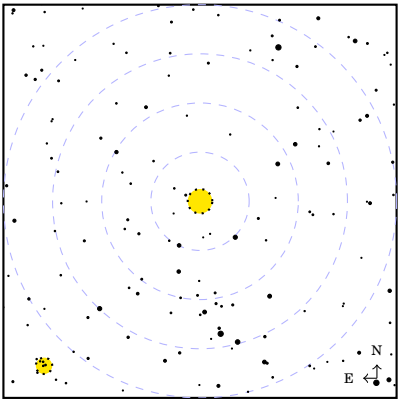
NGC 1912 = U22 = BSC 82

RA : 05.5^h	PSA: 12	Type: OC
Dec: +36°	OSA: N6	Mag : 6.4
Con: Aur	CSA: 3	Size : 15.0′

See:

Notes: M36/U23 is 2.5° SE.

β Tau Elnath (mag 1.7)
→ 7° N to M38/U23

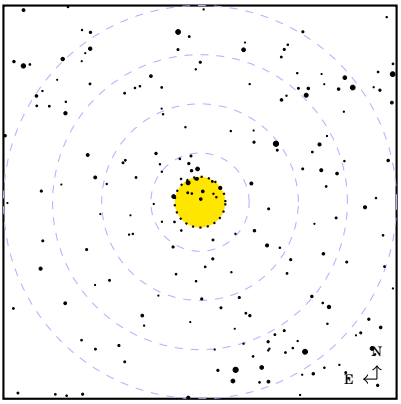


M39

NGC 7092 = U82 = BSC 36

RA : 21.5^h	PSA: 73	Type: OC
Dec: +48°	OSA: N24	Mag : 4.6
Con: Cyg	CSA: 1/6/7	Size : 31.0′

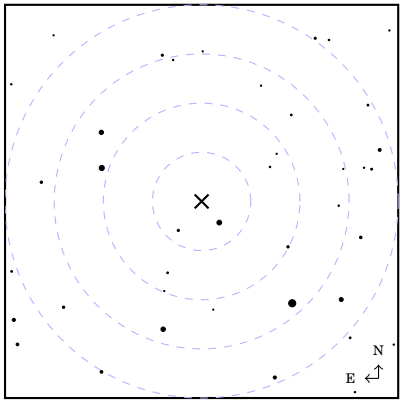
See:



M40

RA : 12.4^h PSA: 32 Type: DS
Dec: +58° OSA: N10 Mag : 9.0/9.6
Con: UMa CSA: 1/4/5 Size : 0.9′

See:



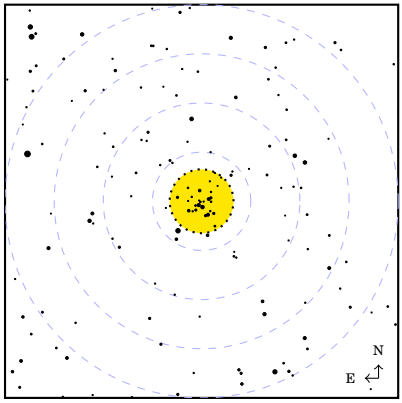
M41

NGC 2287 = U33 = BSC 21

RA : 06.8^h PSA: 27 Type: OC
Dec: −21° OSA: E6 Mag : 4.5
Con: CMa CSA: 9/15 Size : 39.0′

See:

α CMa Sirius (mag −1.5)
→ 4° S to M41/U33



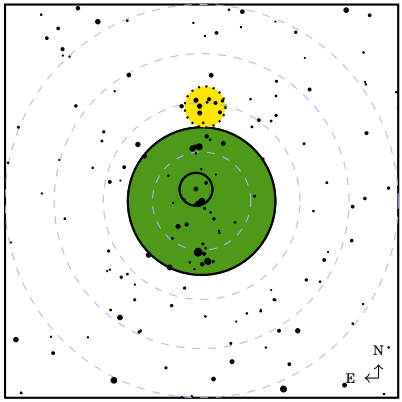
M42

NGC 1976 = U24 = BSC 15 = Orion Nebula

RA : 05.6^h PSA: 16/B Type: BN
Dec: −05° OSA: E4 Mag : 3.7
Con: Ori CSA: 9 Size : 90.0′ ×
60.0′

See:

Notes: M42/U24 surrounds M43. U25 is 1° N.



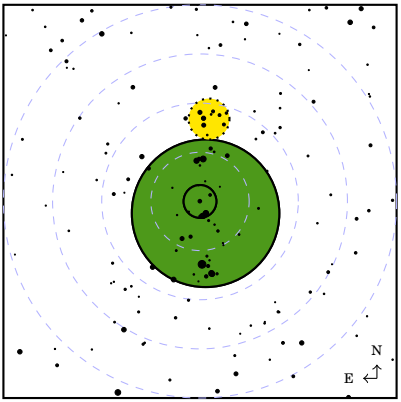
M43

NGC 1982 = BSC 136

RA : 05.6^h	PSA: 16/B	Type: BN
Dec: $-05°$	OSA: E4	Mag : 6.8
Con: Ori	CSA: 9	Size : 20.0′ × 15.0′

See:

Notes: M42/U24 surrounds M43. U25 is 1° N.

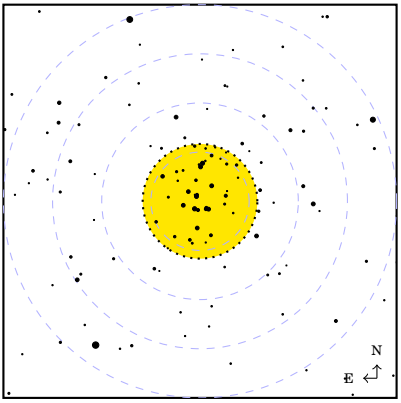


M44

NGC 2632 = U39 = BSC 12 = Mel 88 = Beehive Cluster

RA : 08.7^h	PSA: 24	Type: OC
Dec: $+20°$	OSA: E9	Mag : 3.1
Con: Cnc	CSA: 3/4/9/10	Size : 70.0′

See:

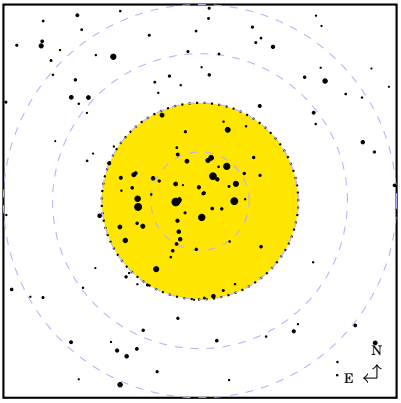


M45

U17 = BSC 4 = Mel 22 = Pleiades

RA : 03.8^h	PSA: 15/A	Type: OC
Dec: $+24°$	OSA: E3	Mag : 1.5
Con: Tau	CSA: 2/3/8/9	Size : 120.0′

See:



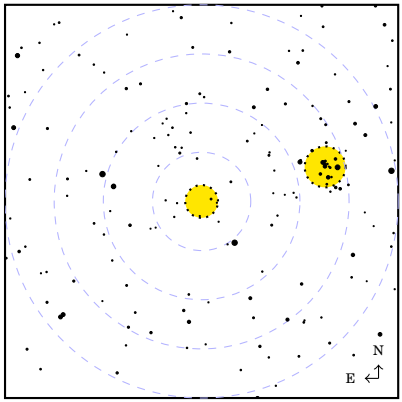
M46

NGC 2437 = BSC 50

RA : 07.7^h	PSA: 26	Type: OC
Dec: $-15°$	OSA: E8	Mag : 6.1
Con: Pup	CSA: 9/10/15/16	Size : 20.0′

See:

Notes: M47 is 1.3° W.



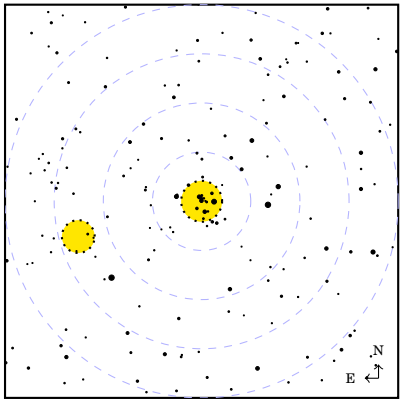
M47

NGC 2422 = BSC 19

RA : 07.6^h	PSA: 27	Type: OC
Dec: $-14°$	OSA: E8	Mag : 4.4
Con: Pup	CSA: 9/10/15/16	Size : 25.0′

See:

Notes: M46 is 1.3° E.

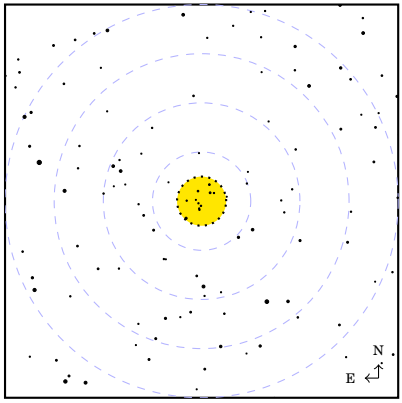


M48

NGC 2548 = U38 = BSC 37

RA : 08.2^h	PSA: 26	Type: OC
Dec: $-06°$	OSA: E10	Mag : 5.8
Con: Hya	CSA: 9/10	Size : 30.0′

See:



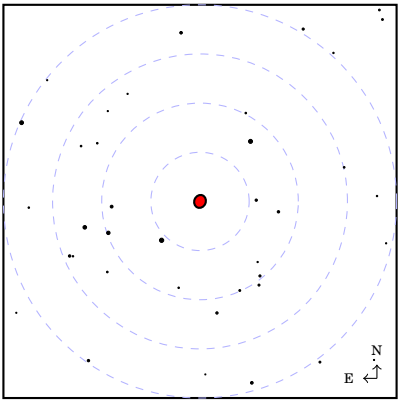
M49

NGC 4472 = BSC 150

RA : 12.5^h	PSA: 45/C	Type: Gal
Dec: $+08°$	OSA: E14	Mag : 8.4
Con: Vir	CSA: 10/11	Size : $8.1' \times 7.1'$

See:

Notes: In the Virgo Cluster.



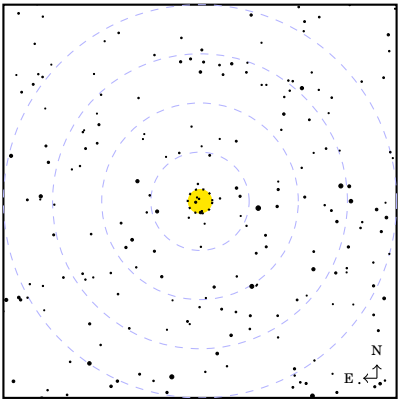
M50

NGC 2323 = U35 = BSC 44

RA : 07.0^h	PSA: 27	Type: OC
Dec: $-08°$	OSA: E8	Mag : 5.9
Con: Mon	CSA: 9	Size : 15.0′

See:

α CMa Sirius (mag −1.5)
→ 3° E to ι CMa
→ 5° N passing μ CMa to θ CMa
→ 4° N to asterism of three stars
→ 2° E to M50/U35

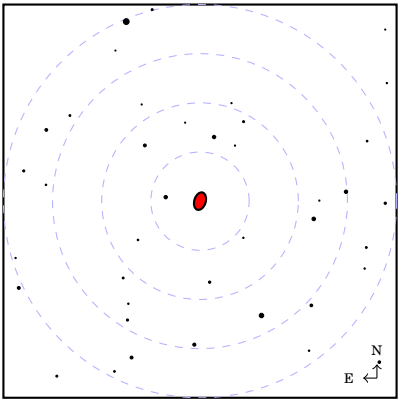


M51

NGC 5194 = BSC 120 = Whirlpool Galaxy

RA : 13.5^h	PSA: 43	Type: Gal
Dec: $+47°$	OSA: N12	Mag : 8.4
Con: CVn	CSA: 4/5	Size : 11.2′ × 6.9′

See:



M52

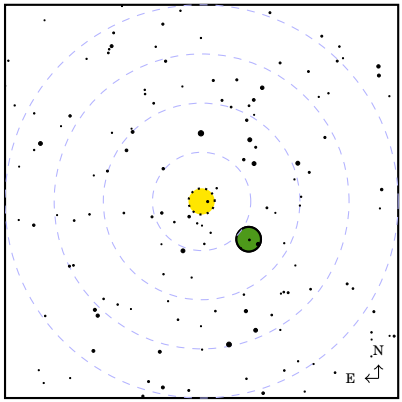
NGC 7654 = BSC 27

RA : 23.4^h PSA: 72 Type: OC
Dec: +62° OSA: N22 Mag : 6.9
Con: Cas CSA: 2/7 Size : 16.0′

See:

Notes: C11 is 0.6° to SW.

β Cas Chaph (mag 2.3)
→ 2° N
→ 6° W to M52

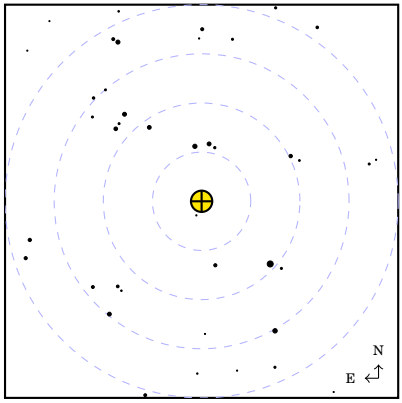


M53

NGC 5024 = BSC 117

RA : 13.2^h PSA: 45 Type: GC
Dec: +18° OSA: E13 Mag : 7.7
Con: Com CSA: 5/11 Size : 13.0′

See:

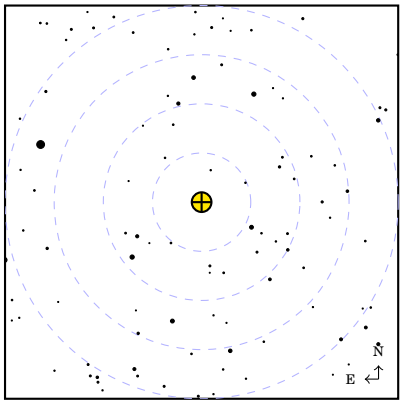


M54

NGC 6715 = BSC 118

RA : 18.9^h PSA: 69 Type: GC
Dec: −30° OSA: E20 Mag : 7.2
Con: Sgr CSA: 18/19 Size : 12.0′

See:

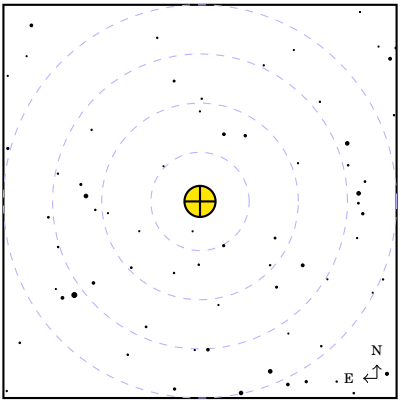


M55

NGC 6809 = BSC 85

RA : 19.7^h	PSA: 68	Type: GC
Dec: $-31°$	OSA: E22	Mag : 6.3
Con: Sgr	CSA: 18/19	Size : 19.0′

See:



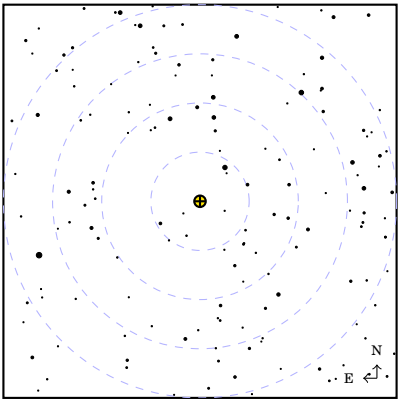
M56

NGC 6779 = BSC 113

RA : 19.3^h	PSA: 63	Type: GC
Dec: $+30°$	OSA: N18	Mag : 8.3
Con: Lyr	CSA: 6/7	Size : 7.0′

See:

γ Lyr Sulaphat (mag 3.2)
→ 3.5° E
→ 2.5° S to M56



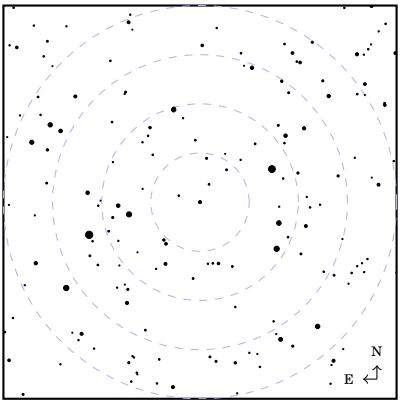
M57

NGC 6720 = U71 = BSC 127 = Ring Nebula

RA : 18.9^h	PSA: 63	Type: PN
Dec: $+33°$	OSA: N18	Mag : 8.8
Con: Lyr	CSA: 6/7	Size : 1.1′

See:

γ Lyr Sulaphat (mag 3.2)
→ 1.2° W
→ 0.4° N to M57/U71



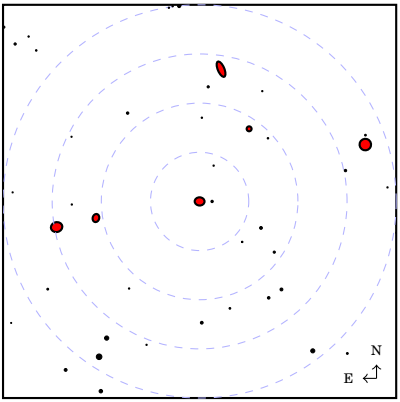
M58

NGC 4579

RA : 12.6^h	PSA: 45/C	Type: Gal
Dec: +12°	OSA: E14	Mag : 9.6
Con: Vir	CSA: 10/11	Size : 5.9′×4.7′

See:

Notes: In the Virgo Cluster.



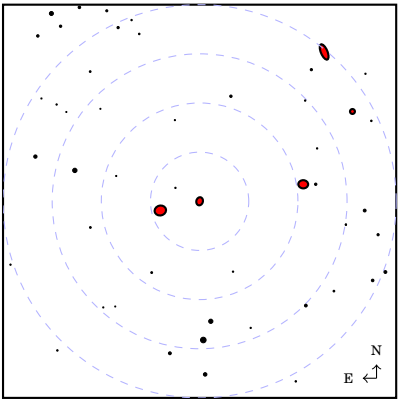
M59

NGC 4621

RA : 12.7^h	PSA: 45/C	Type: Gal
Dec: +12°	OSA: E14	Mag : 9.6
Con: Vir	CSA: 11	Size : 5.4′×3.7′

See:

Notes: In the Virgo Cluster.



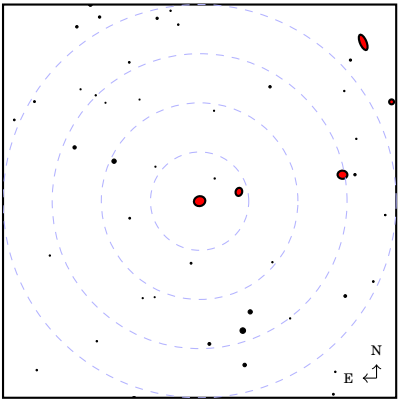
M60

NGC 4649

RA : 12.7^h	PSA: 45/C	Type: Gal
Dec: +12°	OSA: E14	Mag : 8.8
Con: Vir	CSA: 11	Size : 7.1′×6.1′

See:

Notes: In the Virgo Cluster.



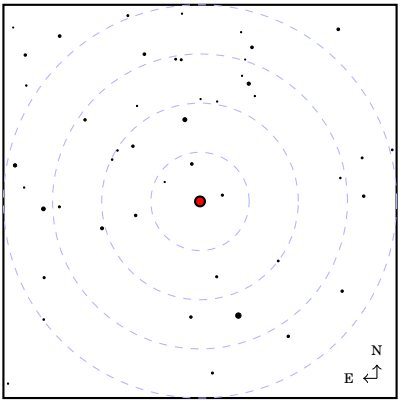
M61

NGC 4303

RA : 12.4^h	PSA: 45	Type: Gal
Dec: $+04°$	OSA: E14	Mag : 9.7
Con: Vir	CSA: 10/11	Size : $6.0' \times 5.9'$

See:

Notes: In the Virgo Cluster.



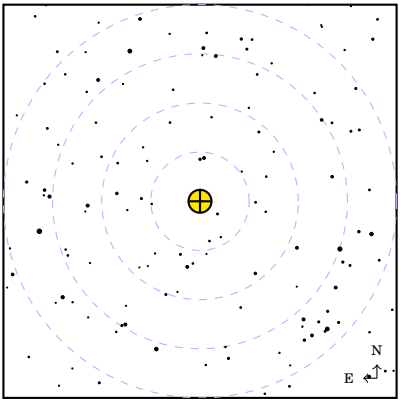
M62

NGC 6266 = U58 = BSC 81

RA : 17.0^h	PSA: 58	Type: GC
Dec: $-30°$	OSA: E18	Mag : 6.7
Con: Oph	CSA: 17/18	Size : $14.0'$

See:

ε Sco (mag 2.3)
→ 2° E
→ 4° N to M63

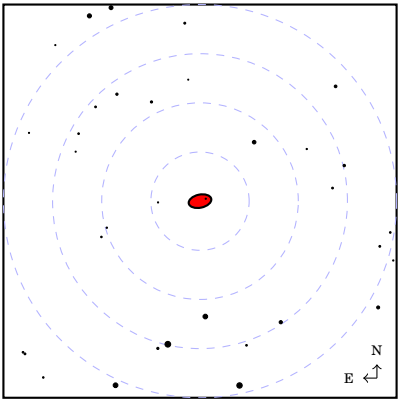


M63

NGC 5055 = BSC 145 = Sunflower Galaxy

RA : 13.3^h	PSA: 43	Type: Gal
Dec: $+42°$	OSA: N12	Mag : 8.6
Con: CVn	CSA: 4/5	Size : $13.5' \times 8.3'$

See:

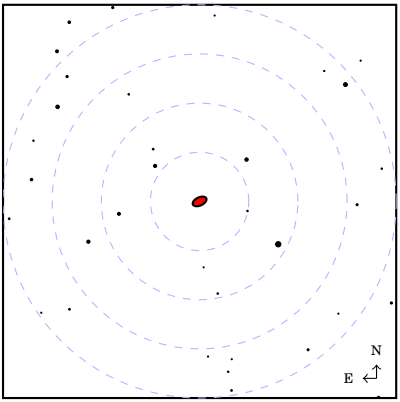


M64

NGC 4826 = U50 = BSC 149 = Black-Eye Galaxy

RA : 12.9^h	PSA: 45	Type: Gal
Dec: +22°	OSA: E13	Mag : 8.5
Con: Com	CSA: 5/11	Size : 9.2′×4.6′

See:



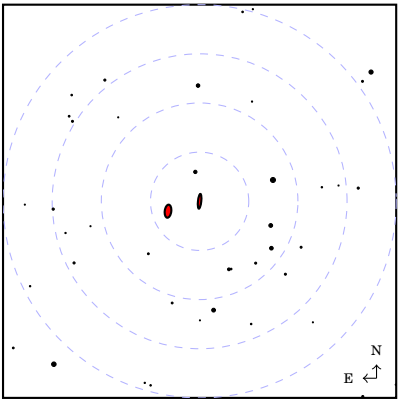
M65

NGC 3623

RA : 11.3^h	PSA: 34	Type: Gal
Dec: +13°	OSA: E11	Mag : 8.8
Con: Leo	CSA: 10/11	Size : 8.7′×2.2′

See:

Notes: M66 is 20′ E.



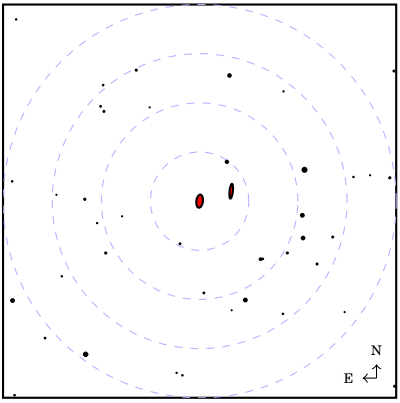
M66

NGC 3627 = BSC 130

RA : 11.3^h	PSA: 34	Type: Gal
Dec: +13°	OSA: E11	Mag : 8.9
Con: Leo	CSA: 10/11	Size : 8.2′×3.9′

See:

Notes: M65 is 20′ W.

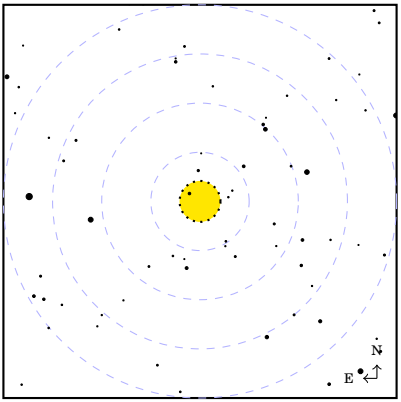


M67

NGC 2682 = U40 = BSC 51

RA : 08.9^h	PSA: 24	Type: OC
Dec: +12°	OSA: E9	Mag : 6.0
Con: Cnc	CSA: 10	Size : 25.0′

See:

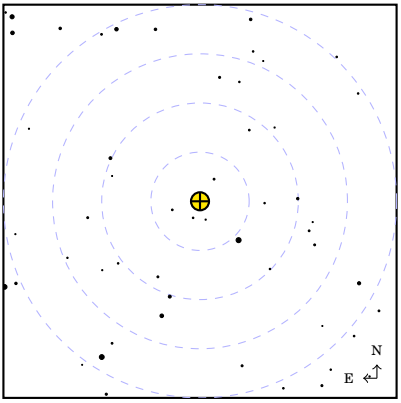


M68

NGC 4590

RA : 12.7^h	PSA: 47	Type: GC
Dec: −27°	OSA: E12	Mag : 7.6
Con: Hya	CSA: 10/11/16/17	Size : 11.0′

See:

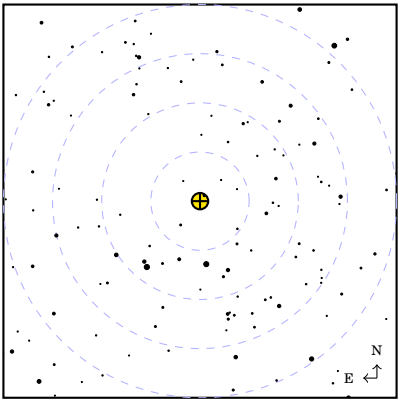


M69

NGC 6637

RA : 18.5^h	PSA: 67	Type: GC
Dec: −32°	OSA: E20	Mag : 7.4
Con: Sgr	CSA: 18	Size : 10.0′

See:

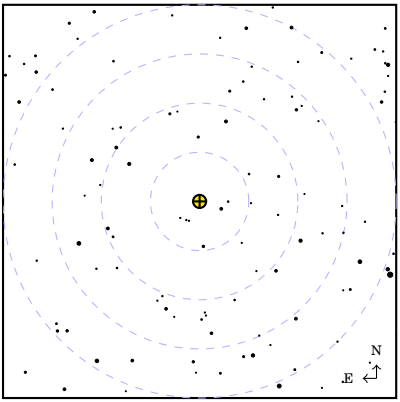


M70

NGC 6681 = BSC 105

RA : 18.7^h	PSA: 67	Type: GC
Dec: $-32°$	OSA: E20	Mag : 7.8
Con: Sgr	CSA: 18	Size : 8.0′

See:

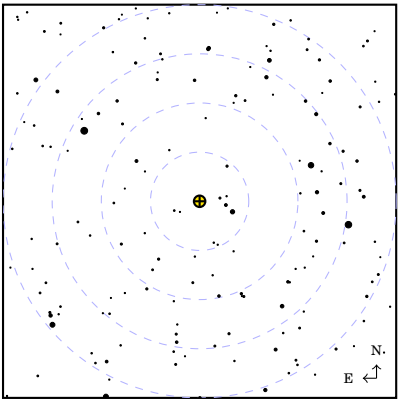


M71

NGC 6838 = BSC 52

RA : 19.9^h	PSA: 64	Type: GC
Dec: $+19°$	OSA: E21	Mag : 8.0
Con: Sge	CSA: 6/7/12/13	Size : 7.0′

See:



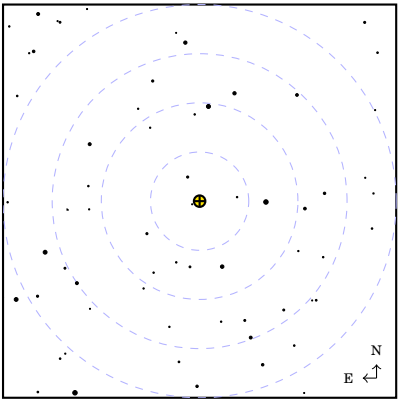
M72

NGC 6981 = BSC 148

RA : 20.9^h	PSA: 66/77	Type: GC
Dec: $-13°$	OSA: E24	Mag : 9.2
Con: Aqr	CSA: 13	Size : 7.0′

See:

Notes: M73 is 1.4° W.



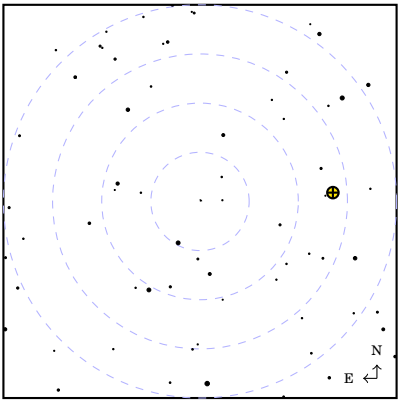
M73

NGC 6994 = Cr 426

RA : 21.0^h	PSA: 66/77	Type: AST
Dec: $-13°$	OSA: E24	Mag : 9.7
Con: Aqr	CSA: 13	Size : 1.4′

See:

Notes: Y-shaped asterism of four stars. M72 is 1.4° E.

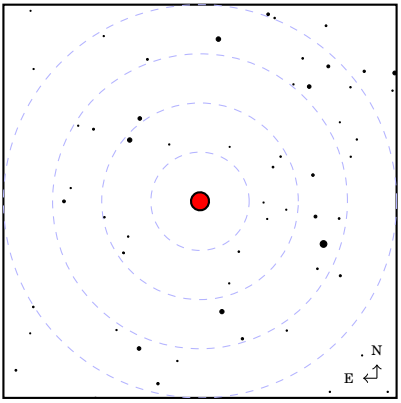


M74

NGC 628

RA : 01.6^h	PSA: 4/5	Type: Gal
Dec: $+16°$	OSA: E1	Mag : 8.5
Con: Psc	CSA: 8	Size : 11.0′

See:

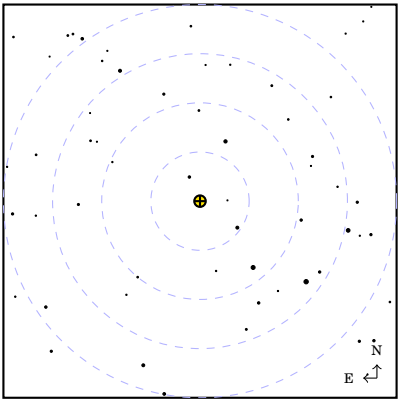


M75

NGC 6864 = BSC 144

RA : 20.1^h	PSA: 66	Type: GC
Dec: $-22°$	OSA: E22	Mag : 8.6
Con: Sgr	CSA: 12/13/18/19	Size : 7.0′

See:

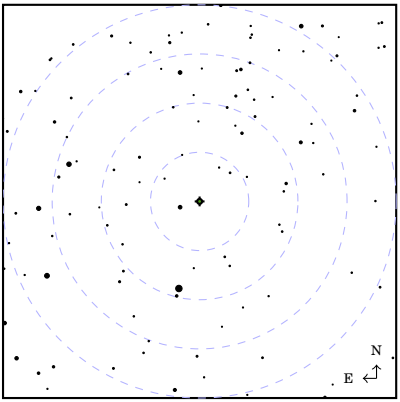


M76

NGC 651/650 = Little Dumbell

RA : 01.7^h	PSA: 2	Type: PN
Dec: $+52°$	OSA: N0	Mag : 10.4
Con: Per	CSA: 1/2/3/7	Size : $2.7' \times 1.8'$

See:

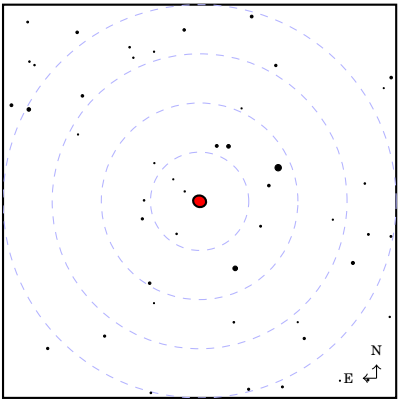


M77

NGC 1068 = U12

RA : 02.7^h	PSA: 4	Type: Gal
Dec: $-00°$	OSA: E0	Mag : 8.9
Con: Cet	CSA: 8	Size : $8.2' \times 7.3'$

See:

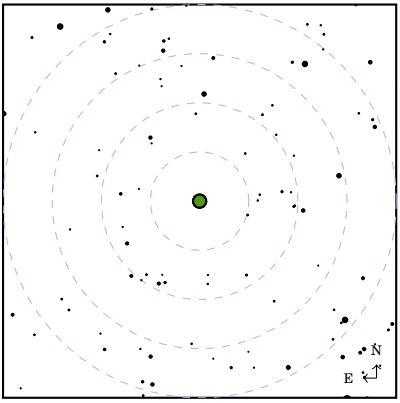


M78

NGC 2068 = BSC 114

RA : 05.8^h	PSA: 16	Type: BN
Dec: $+00°$	OSA: E5	Mag : 8.0
Con: Ori	CSA: 9	Size : $8.0' \times 6.0'$

See:

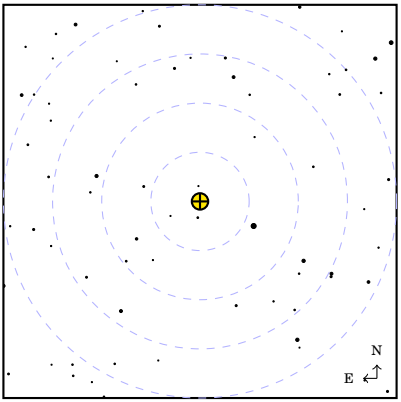


M79

NGC 1904 = BSC 126

RA : 05.4^h	PSA: 16	Type: GC
Dec: $-25°$	OSA: E4	Mag : 7.7
Con: Lep	CSA: 9/15	Size : 10.0′

See:



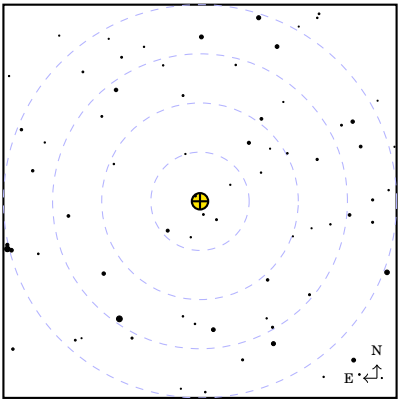
M80

NGC 6093 = BSC 101

RA : 16.3^h	PSA: 56	Type: GC
Dec: $-23°$	OSA: E18	Mag : 7.3
Con: Sco	CSA: 11/12/17/18	Size : 10.0′

See:

α Sco Antares (mag 0.9–1.1)
→ 3.0° W
→ 3.5° N to M80



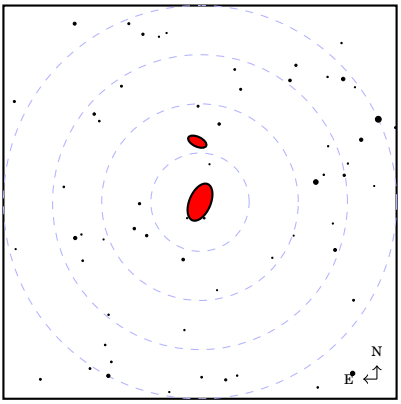
M81

NGC 3031 = U41 = BSC 72

RA : 09.9^h	PSA: 31	Type: Gal
Dec: $+69°$	OSA: N8	Mag : 6.9
Con: UMa	CSA: 1/4	Size : 24.0′ × 13.0′

See:

Notes: M82/U42 is 0.6° N.



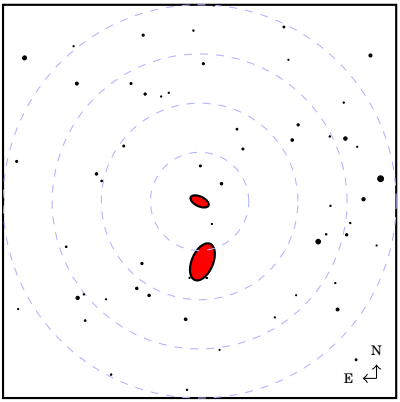
M82

NGC 3034 = U42 = BSC 122

RA : 09.9^h PSA: 31 Type: Gal
Dec: +70° OSA: N8 Mag : 8.4
Con: UMa CSA: 1/4 Size : 12.0′ × 5.6′

See:

Notes: M81/U41 is 0.6° S.

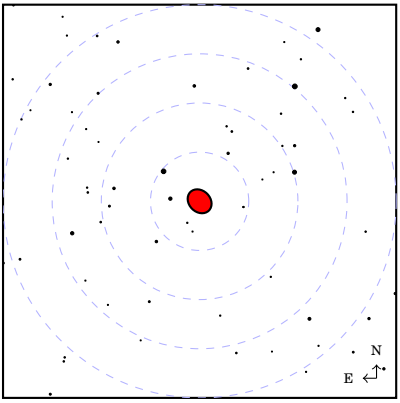


M83

NGC 5236 = BSC 89

RA : 13.6^h PSA: 47/48 Type: Gal
Dec: −30° OSA: E16 Mag : 7.6
Con: Hya CSA: 17 Size : 15.5′ × 13.0′

See:



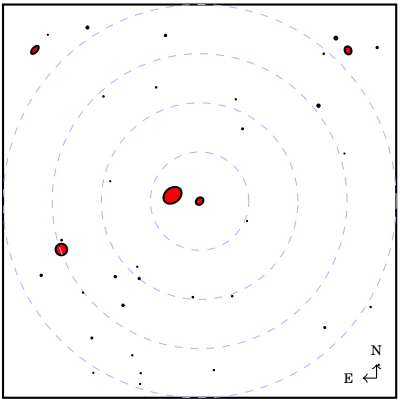
M84

NGC 4374 = U45

RA : 12.4^h PSA: 45/C Type: Gal
Dec: +13° OSA: E14 Mag : 9.1
Con: Vir CSA: 4/10/11 Size : 5.1′×4.1′

See:

Notes: In the Virgo Cluster.



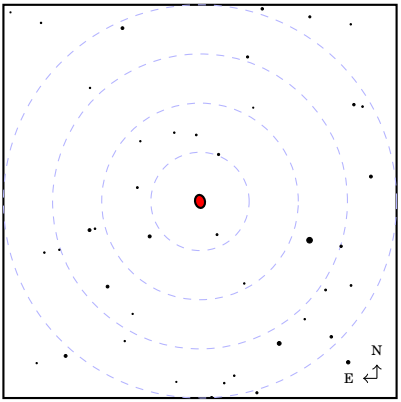
M85

NGC 4382

RA : 12.4^h	PSA: 45/C	Type: Gal
Dec: +18°	OSA: E14	Mag : 9.1
Con: Com	CSA: 4/5/10/11	Size : 7.5′×5.7′

See:

Notes: In the Virgo Cluster.



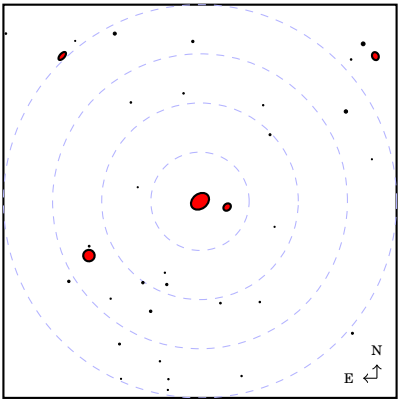
M86

NGC 4406 = U46

RA : 12.4^h	PSA: 45/C	Type: Gal
Dec: +13°	OSA: E14	Mag : 8.9
Con: Vir	CSA: 4/10/11	Size : 12.0′ × 9.3′

See:

Notes: In the Virgo Cluster.



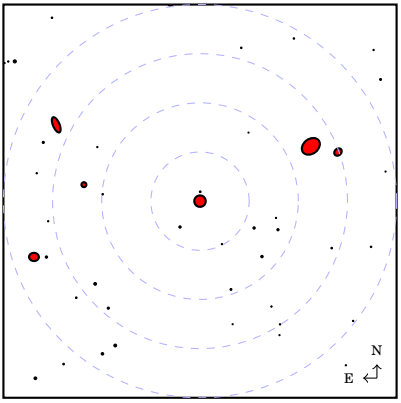
M87

NGC 4486 = U47

RA : 12.5^h	PSA: 45/C	Type: Gal
Dec: +12°	OSA: E14	Mag : 8.6
Con: Vir	CSA: 10/11	Size : 7.1′

See:

Notes: In the Virgo Cluster.



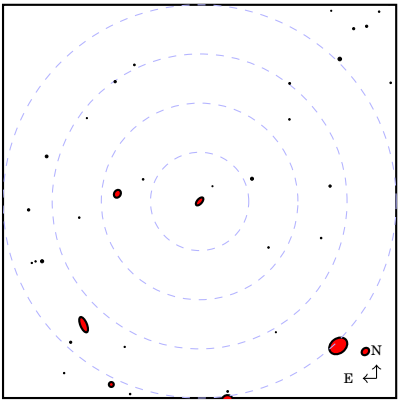
M88

NGC 4501

RA : 12.5^h PSA: 45/C Type: Gal
Dec: +14° OSA: E14 Mag : 9.6
Con: Com CSA: 5/10/11 Size : 6.1′×2.8′

See:

Notes: In the Virgo Cluster.



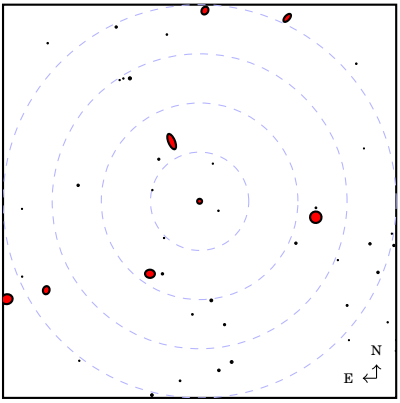
M89

NGC 4552

RA : 12.6^h PSA: 45/C Type: Gal
Dec: +13° OSA: E14 Mag : 9.8
Con: Vir CSA: 10/11 Size : 3.4′

See:

Notes: In the Virgo Cluster.



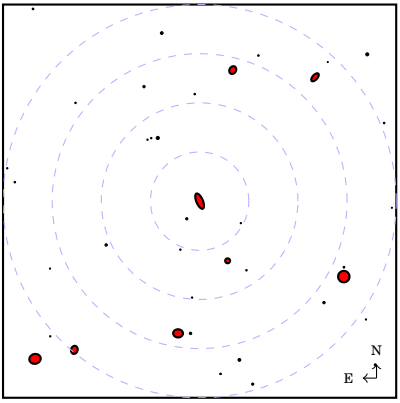
M90

NGC 4569

RA : 12.6^h PSA: 45/C Type: Gal
Dec: +13° OSA: E14 Mag : 9.5
Con: Vir CSA: 10/11 Size : 10.5′ × 4.4′

See:

Notes: In the Virgo Cluster.



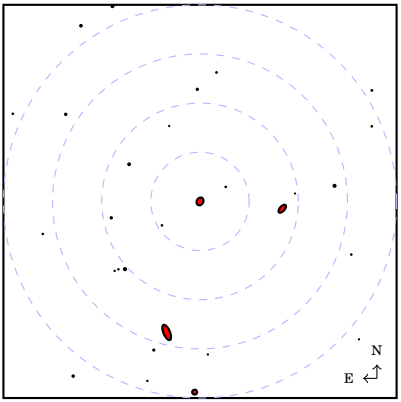
M91

NGC 4548

RA : 12.6^h	PSA: 45/C	Type: Gal
Dec: +14°	OSA: E14	Mag : 10.2
Con: Com	CSA: 5/10/11	Size : 5.0′×4.1′

See:

Notes: In the Virgo Cluster.



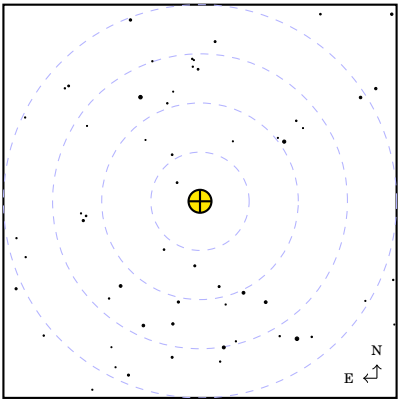
M92

NGC 6341 = U59 = BSC 59

RA : 17.3^h	PSA: 52	Type: GC
Dec: +43°	OSA: N14	Mag : 6.5
Con: Her	CSA: 5/6	Size : 14.0′

See:

π Her (mag 3.2)
→ 6.5° N to M92/U59

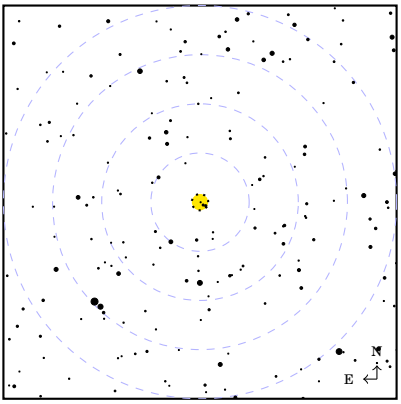


M93

NGC 2447 = BSC 48 = Cr 160

RA : 07.7^h	PSA: 26	Type: OC
Dec: −24°	OSA: E6	Mag : 6.2
Con: Pup	CSA: 9/10/15/16	Size : 10.0′

See:

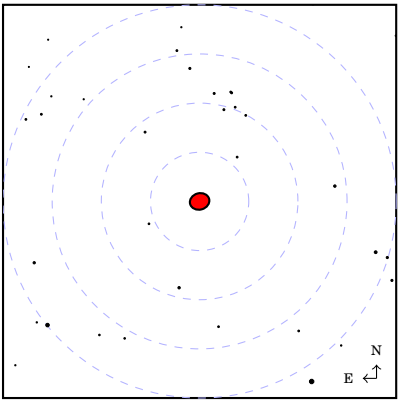


M94

NGC 4736 = U49 = BSC 134

RA : 12.8^h	PSA: 43	Type: Gal
Dec: +41°	OSA: N12	Mag : 8.2
Con: CVn	CSA: 4/5	Size : 11.5′ × 10.0′

See:



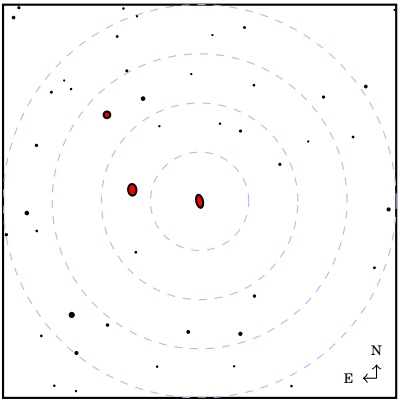
M95

NGC 3351

RA : 10.7^h	PSA: 34	Type: Gal
Dec: +12°	OSA: E11	Mag : 9.7
Con: Leo	CSA: 10	Size : 7.8′×4.5′

See:

Notes: M96 is 0.7° E. M105 is 1.3° NE.



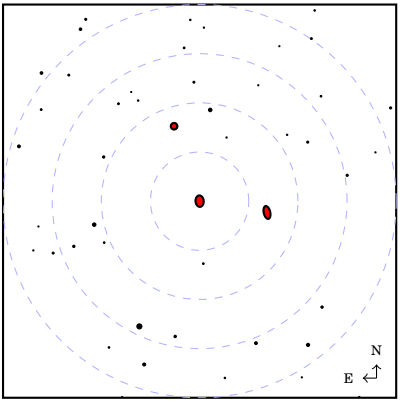
M96

NGC 3368

RA : 10.8^h	PSA: 34	Type: Gal
Dec: +12°	OSA: E11	Mag : 9.2
Con: Leo	CSA: 10	Size : 6.9′×4.6′

See:

Notes: M96 is 0.7° W. M105 is 0.8° NNE.



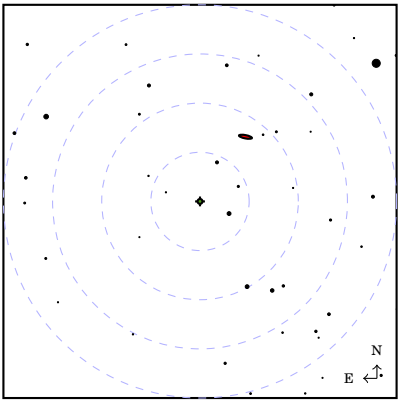
M97

NGC 3587

RA : 11.2^h	PSA: 32	Type: PN
Dec: +55°	OSA: N10	Mag : 9.8
Con: UMa	CSA: 1/4/5	Size : 3.4′

See:

Notes: M108 is 0.8° to NW.



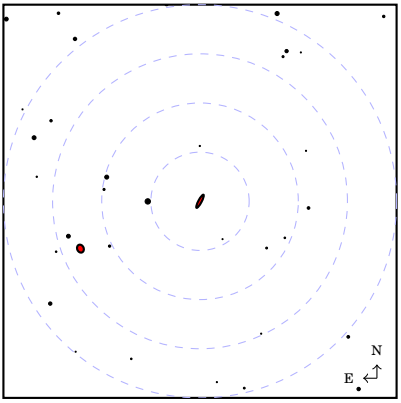
M98

NGC 4192

RA : 12.2^h	PSA: 45/C	Type: Gal
Dec: +15°	OSA: E14	Mag : 10.1
Con: Com	CSA: 4/5/10/11	Size : 9.1′×2.1′

See:

Notes: In the Virgo Cluster.



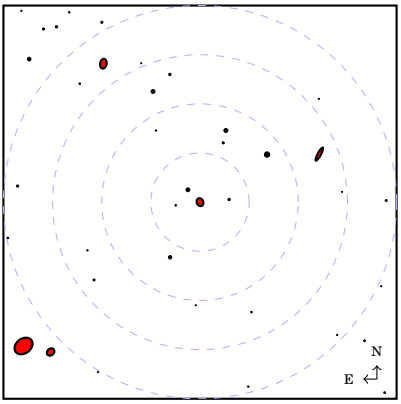
M99

NGC 4254

RA : 12.3^h	PSA: 45/C	Type: Gal
Dec: +14°	OSA: E14	Mag : 9.9
Con: Com	CSA: 4/5/10/11	Size : 4.6′×4.3′

See:

Notes: In the Virgo Cluster.



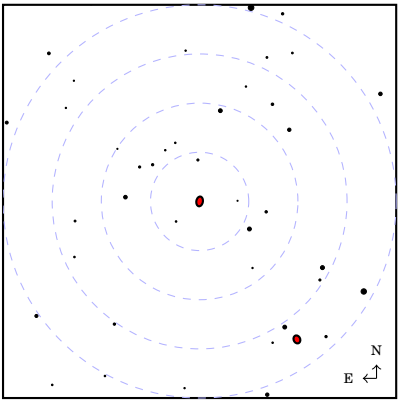
M100

NGC 4321

RA : 12.4^h PSA: 45/C Type: Gal
Dec: +16° OSA: E14 Mag : 9.3
Con: Com CSA: 4/5/10/11 Size : 6.2′×4.3′

See:

Notes: In the Virgo Cluster.

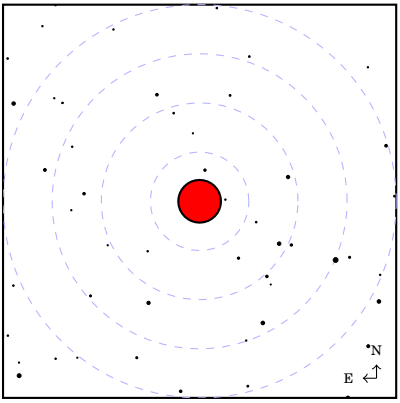


M101

NGC 5457 = BSC 100

RA : 14.1^h PSA: 42 Type: Gal
Dec: +54° OSA: N10 Mag : 7.9
Con: UMa CSA: 1/4/5/6 Size : 26.0′

See:



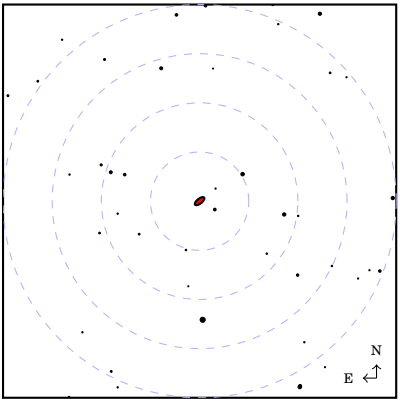
M102

NGC 5866

RA : 15.1^h PSA: 42 Type: Gal
Dec: +56° OSA: N16 Mag : 9.9
Con: Dra CSA: 1/5/6 Size : 6.6′×3.2′

See:

Notes: Labelled 5866 in PSA



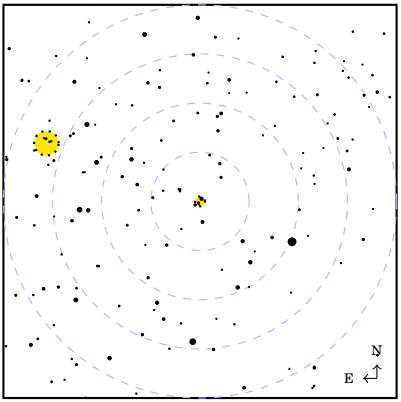
M103

NGC 581 = BSC 83

RA : 01.6^h	PSA: 3	Type: OC
Dec: +61°	OSA: N2	Mag : 7.4
Con: Cas	CSA: 1/2/3/7	Size : 6.0′

See:

Notes: C10 is 1.7° ENE.

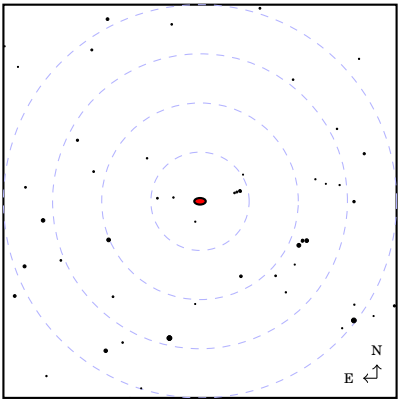


M104

NGC 4594 = U48 = BSC 133 = Sombrero Galaxy

RA : 12.7^h	PSA: 47	Type: Gal
Dec: −12°	OSA: E12	Mag : 8.0
Con: Vir	CSA: 10/11	Size : 7.1′×4.4′

See:



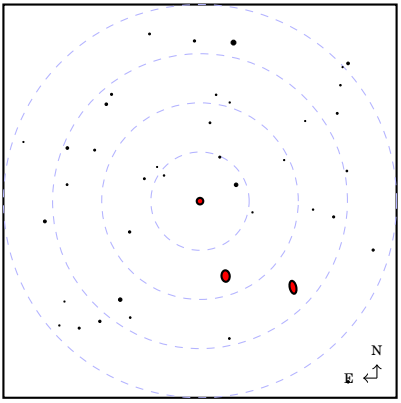
M105

NGC 3379

RA : 10.8^h	PSA: 34	Type: Gal
Dec: +13°	OSA: E11	Mag : 9.3
Con: Leo	CSA: 10	Size : 3.9′

See:

Notes: M95 is 1.3° SW. M96 is 0.8° SSW.

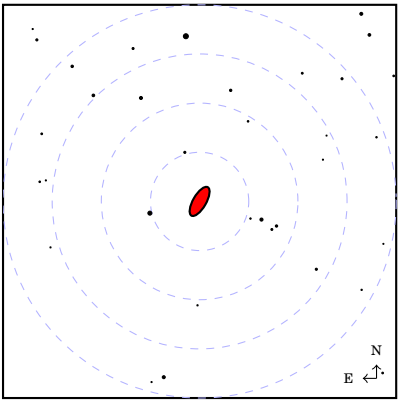


M106

NGC 4258 = BSC 141

RA : 12.3^h PSA: 43 Type: Gal
Dec: +47° OSA: N12 Mag : 8.4
Con: CVn CSA: 4/5 Size : 20.0′ × 8.4′

See:

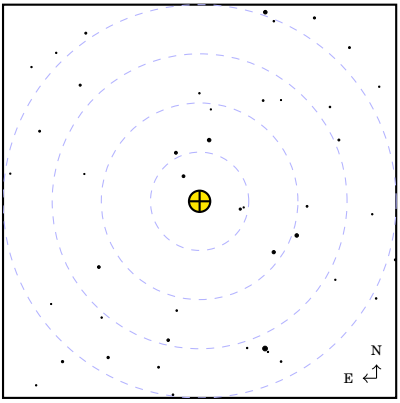


M107

NGC 6171 = BSC 129

RA : 16.5^h PSA: 56 Type: GC
Dec: −13° OSA: E17 Mag : 7.8
Con: Oph CSA: 11/12 Size : 13.0′

See:



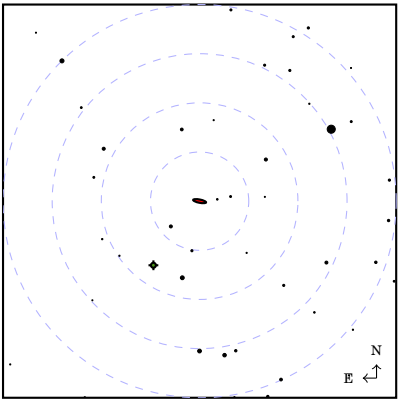
M108

NGC 3556

RA : 11.2^h PSA: 32/33 Type: Gal
Dec: +56° OSA: N10 Mag : 10.0
Con: UMa CSA: 1/4/5 Size : 8.2′×2.1′

See:

Notes: M97 is 0.8° to SE.

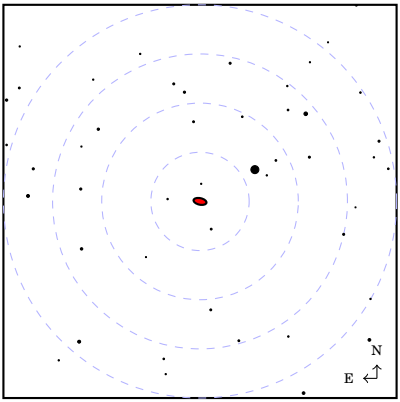


M109

NGC 3992

RA : 12.0^h	PSA: 32	Type: Gal
Dec: +53°	OSA: N10	Mag : 9.8
Con: UMa	CSA: 1/4/5	Size : 7.6′ ×4.3′

See:



M110

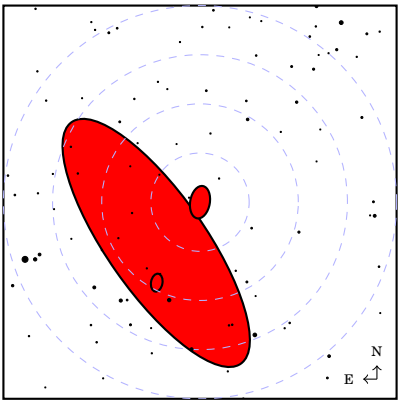
NGC 205 = BSC 132

RA : 00.7^h	PSA: 3	Type: Gal
Dec: +42°	OSA: N0	Mag : 8.1
Con: And	CSA: 2/7	Size : 19.5′ × 12.5′

See:

Notes: M31/U3 is 0.5° SE. M32/U2 is 1° SE.

β And Mirach (mag 2.1)
→ 2.5° W
→ 3.0° N to μ And (mag 3.9)
→ 3.0° W
→ 3.5° N to M31/U3



The Caldwell Objects

The Caldwell objects were selected by Patrick Moore as a complement to the Messier objects. Again, I find O'Meara's *Deep-Sky Companion: The Caldwell Objects* to be excellent on the origin of the catalog, the appearance of the objects, and their nature.

I use C1 to C109 to label the Caldwell objects. This is common practice, but does not confirm to the IAU's recommendation on nomenclature, as O'Meara rightly notes (pp. 14–15).

The following table lists the objects with their J2000 positions (decimal hours of right ascension and degrees of declination), the charts on which they appear in the *Pocket Sky Atlas*, their types, and other names. I follow the corrections given by O'Meara (p. 15) for C37, C49, C89, and C100.

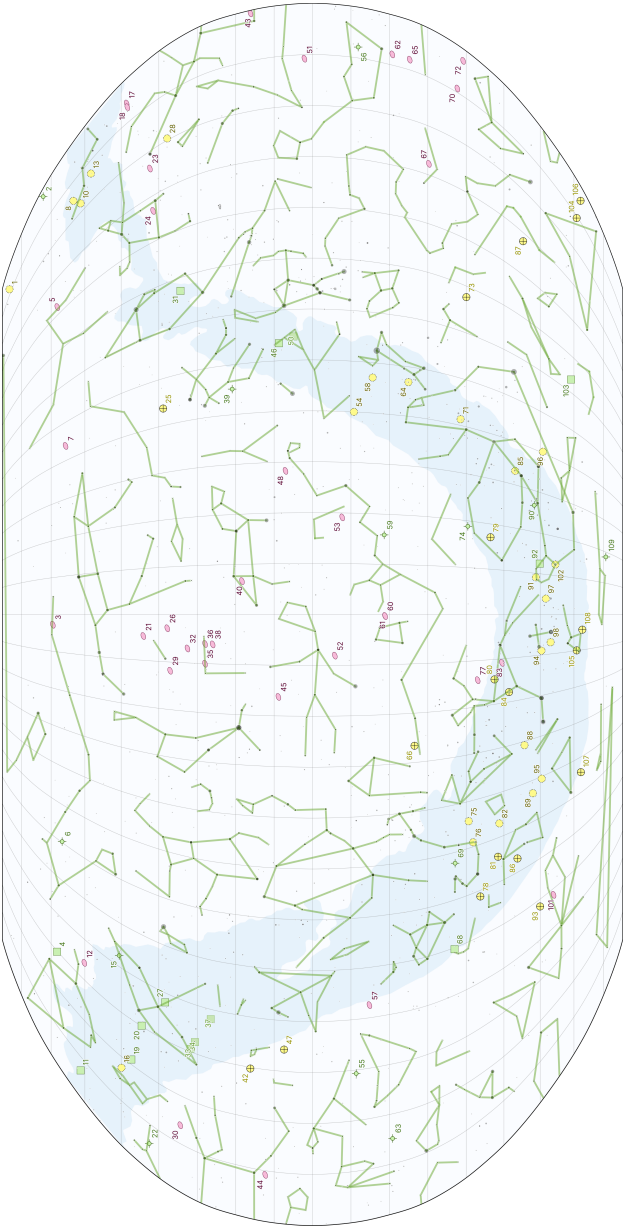
For completeness, I include finder charts for all the Caldwell objects, even bright ones like C41 (the Hyades).

I note that C41 (the Hyades) and C99 (the Coalsack Nebula) are much bigger than the finder charts. They are really binocular objects and are better located using a small-scale all-sky atlas.

Name	Position	Type	Other Names
C1	00.8 +85	OC	NGC 188 = BSC 111 = Mel 2
C2	00.2 +73	PN	NGC 40 = Bow-Tie Nebula
C3	12.3 +69	Gal	NGC 4236
C4	21.0 +68	BN	NGC 7023 = BSC 68 = Iris Nebula
C5	03.8 +68	Gal	IC 342 = BSC 142 = Hidden Galaxy
C6	18.0 +67	PN	NGC 6543 = Cat's Eye Nebula
C7	07.6 +66	Gal	NGC 2403 = BSC 131 = Fireworks Galaxy
C8	01.5 +63	OC	NGC 559 = Ghost's Goblet
C9	23.0 +63	BN	BSC 96 = Sh2-155 = Cave Nebula
C10	01.8 +61	OC	NGC 663 = U5 = BSC 76 = Horseshoe Cluster
C11	23.3 +61	BN	NGC 7635 = Bubble Nebula
C12	20.6 +60	Gal	NGC 6946 = Firecracker Galaxy
C13	01.3 +58	OC	NGC 457 = U4 = BSC 62 = E.T. Cluster = Owl Cluster
C14	02.3 +57	OC	NGC 869/884 = U9/10 = Double Cluster = h/χ Per Clusters
C15	19.7 +51	PN	NGC 6826 = BSC 128 = Blinking Planetary
C16	22.3 +50	OC	NGC 7243 = U85 = BSC 63
C17	00.6 +49	Gal	NGC 147
C18	00.6 +48	Gal	NGC 185
C19	21.9 +47	BN	IC 5146 = BSC 78 = Cr 470 = Cocoon Nebula
C20	21.0 +44	BN	NGC 7000 = BSC 16 = North America Nebula
C21	12.5 +44	Gal	NGC 4449 = Box Galaxy
C22	23.4 +43	PN	NGC 7662 = BSC 124 = Light Blue Snowball
C23	02.4 +42	Gal	NGC 891
C24	03.3 +42	Gal	NGC 1275 = Perseus A
C25	07.6 +39	GC	NGC 2419 = BSC 140
C26	12.3 +38	Gal	NGC 4244
C27	20.2 +38	BN	NGC 6888 = BSC 84 = Crescent Nebula
C28	01.9 +38	OC	NGC 752 = U7 = Mel 12
C29	13.2 +37	Gal	NGC 5005
C30	22.6 +34	Gal	NGC 7331
C31	05.3 +34	BN	IC 405 = BSC 49 = Flaming Star Nebula
C32	12.7 +33	Gal	NGC 4631 = Whale Galaxy
C33	20.9 +32	BN	NGC 6992 = BSC 73 = Eastern Veil Nebula
C34	20.8 +31	BN	NGC 6960 = BSC 74 = Western Veil Nebula
C35	13.0 +28	Gal	NGC 4889
C36	12.6 +28	Gal	NGC 4559
C37	20.2 +26	OC	NGC 6885
C38	12.6 +26	Gal	NGC 4565 = BSC 38 = Needle Galaxy
C39	07.5 +21	PN	NGC 2392 = U36 = Eskimo Nebula
C40	11.3 +18	Gal	NGC 3626
C41	04.4 +16	OC	U18 = BSC 1 = Mel 25 = Cr 50 = Hyades
C42	21.0 +16	GC	NGC 7006
C43	00.1 +16	Gal	NGC 7814
C44	23.1 +12	Gal	NGC 7479
C45	13.6 +09	Gal	NGC 5248
C46	06.7 +09	BN	NGC 2261 = BSC 137 = Hubble's Variable Nebula
C47	20.6 +07	GC	NGC 6934 = BSC 121
C48	09.2 +07	Gal	NGC 2775
C49	06.5 +05	BN	NGC 2237/38/46 = BSC 135 = Rosette Nebula
C50	06.5 +05	OC	NGC 2244 = U30 = BSC 25

Name	Position	Type	Other Names
C51	01.1 +02	Gal	IC 1613
C52	12.8 −06	Gal	NGC 4697
C53	10.1 −08	Gal	NGC 3115 = The Spindle
C54	08.0 −11	OC	NGC 2506 = BSC 90 = Mel 80 = Cr 170
C55	21.1 −11	PN	NGC 7009 = BSC 106 = Saturn Nebula
C56	00.8 −12	PN	NGC 246 = BSC 107
C57	19.7 −15	Gal	NGC 6822 = IC 4895 = BSC 146 = Barnard's Galaxy
C58	07.3 −16	OC	NGC 2360 = BSC 79 = Mel 64 = Caroline's Cluster
C59	10.4 −19	PN	NGC 3242 = U43 = BSC 125 = Ghost of Jupiter
C60	12.0 −19	Gal	NGC 4038 = NW Antennae Galaxy
C61	12.0 −19	Gal	NGC 4039 = SE Antennae Galaxy
C62	00.8 −21	Gal	NGC 247
C63	22.5 −21	PN	NGC 7293 = BSC 91 = Helix Nebula
C64	07.3 −25	OC	NGC 2362 = BSC 17 = τ CMa Cluster
C65	00.8 −25	Gal	NGC 253 = BSC 108 = Sculpter Galaxy
C66	14.7 −27	GC	NGC 5694
C67	02.8 −30	Gal	NGC 1097
C68	19.0 −37	BN	NGC 6729 = R CrA Nebula
C69	17.2 −37	PN	NGC 6302 = BSC 77 = Bug Nebula
C70	00.9 −38	Gal	NGC 300 = BSC 138
C71	07.9 −39	OC	NGC 2477 = BSC 41 = Mel 78
C72	00.2 −39	Gal	NGC 55 = BSC 102
C73	05.2 −40	GC	NGC 1851 = BSC 80
C74	10.1 −40	PN	NGC 3132
C75	16.4 −41	OC	NGC 6124 = BSC 42
C76	16.9 −42	OC	NGC 6231 = BSC 7 = Cr 315
C77	13.4 −43	Gal	NGC 5128 = Cen A
C78	18.1 −44	GC	NGC 6541 = BSC 60
C79	10.3 −46	GC	NGC 3201 = BSC 112
C80	13.4 −47	GC	NGC 5139 = BSC 14 = ω Cen
C81	17.4 −48	GC	NGC 6352 = BSC 98
C82	16.7 −49	OC	NGC 6193 = BSC 30
C83	13.1 −49	Gal	NGC 4945 = BSC 147
C84	13.8 −51	GC	NGC 5286 = BSC 92
C85	08.7 −53	OC	IC 2391 = BSC 6 = o Vel Cluster
C86	17.7 −54	GC	NGC 6397 = BSC 67
C87	03.2 −55	GC	NGC 1261 = BSC 115
C88	15.1 −56	OC	NGC 5823 = BSC 103 = Mel 131
C89	16.3 −58	OC	NGC 6087 = BSC 33 = S Nor Cluster
C90	09.4 −58	PN	NGC 2867
C91	11.1 −59	OC	NGC 3532 = BSC 9
C92	10.8 −60	BN	NGC 3372 = BSC 3 = η Car Nebula
C93	19.2 −60	GC	NGC 6752 = BSC 34
C94	12.9 −60	OC	NGC 4755 = BSC 20 = Mel 114 = Jewel Box
C95	16.1 −60	OC	NGC 6025 = BSC 29
C96	08.0 −61	OC	NGC 2516 = BSC 13 = Southern Beehive Cluster
C97	11.6 −62	OC	NGC 3766 = BSC 32 = Cr 248
C98	12.7 −63	OC	NGC 4609 = BSC 70 = Coalsack Cluster
C99	12.5 −64	DN	Coalsack
C100	11.6 −63	OC	BSC 22 = Cr 249 = λ Cen Cluster

Name	Position	Type	Other Names
C101	19.2 −64	Gal	NGC 6744 = BSC 143
C102	10.7 −64	OC	IC 2602 = BSC 5 = Mel 102 = Cr 229 = θ Car Cluster
C103	05.6 −69	BN	NGC 2070 = BSC 109 = Tarantula Nebula
C104	01.1 −71	GC	NGC 362 = BSC 64 = Mel 4
C105	13.0 −71	GC	NGC 4833 = BSC 97
C106	00.4 −72	GC	NGC 104 = BSC 26 = Mel 1 = 47 Tuc
C107	16.4 −72	GC	NGC 6101 = BSC 139
C108	12.4 −73	GC	NGC 4372 = BSC 99
C109	10.2 −81	PN	NGC 3195



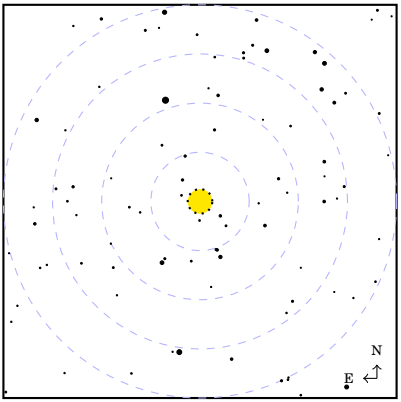
The distribution of Caldwell objects over the whole sky

C1

NGC 188 = BSC 111 = Mel 2

RA : 00.8^h	PSA: 1	Type: OC
Dec: +85°	OSA:	Mag : 7.1
Con: Cep	CSA: 1	Size : 15.0′

See:



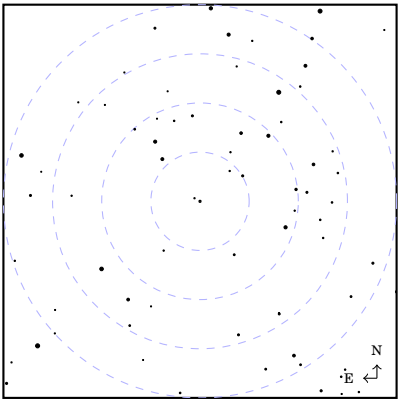
C2

NGC 40 = Bow-Tie Nebula

RA : 00.2^h	PSA: 1/71	Type: PN
Dec: +73°	OSA:	Mag :
Con: Cep	CSA: 1	Size : 0.6′

See:

Notes: Central star is magnitude 11.6 (DSC:TCO).

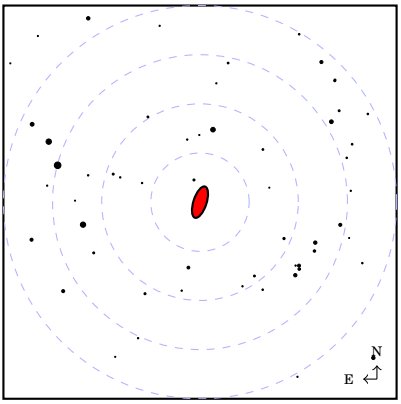


C3

NGC 4236

RA : 12.3^h	PSA: 31/41	Type: Gal
Dec: +69°	OSA:	Mag : 9.6
Con: Dra	CSA: 1/5	Size : 20.0′ × 8.0′

See:



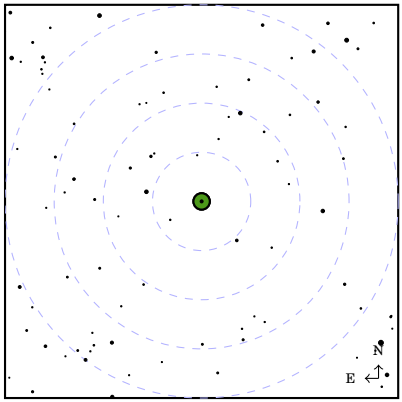
C4

NGC 7023 = BSC 68 = Iris Nebula

RA : 21.0^h PSA: 61/71 Type: BN
Dec: +68° OSA: Mag : 7.7
Con: Cep CSA: 1/6/7 Size : 10.0′ × 8.0′

See:

Notes: Central star is magnitude 6.8 (DSC:TCO).

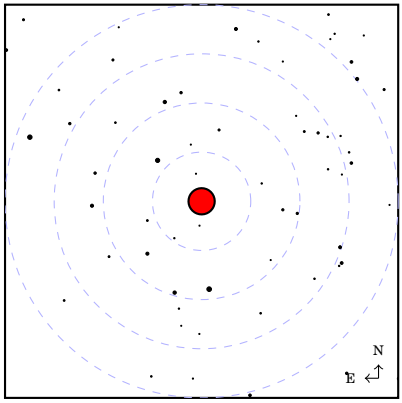


C5

IC 342 = BSC 142 = Hidden Galaxy

RA : 03.8^h PSA: 1/11 Type: Gal
Dec: +68° OSA: Mag : 7.9
Con: Cam CSA: 1/2/3 Size : 16.0′

See:

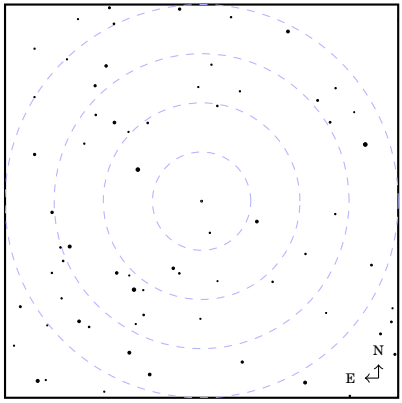


C6

NGC 6543 = Cat's Eye Nebula

RA : 18.0^h PSA: 51/61 Type: PN
Dec: +67° OSA: N16 Mag : 8.1
Con: Dra CSA: 1/6 Size : 0.4′×0.3′

See:

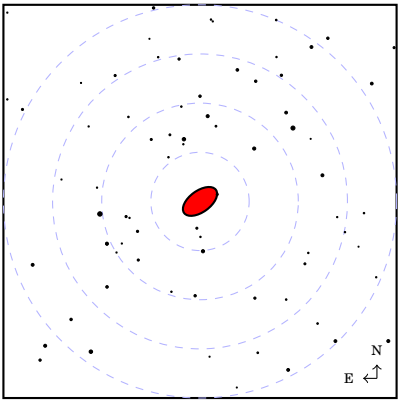


C7

NGC 2403 = BSC 131 = Fireworks Galaxy

RA : 07.6^h	PSA: 21	Type: Gal
Dec: +66°	OSA:	Mag : 7.3
Con: Cam	CSA: 1/3/4	Size : 24.0′ × 13.0′

See:



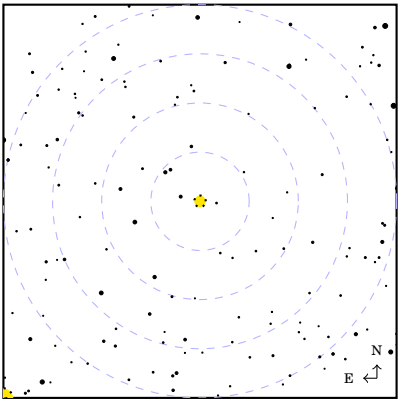
C8

NGC 559 = Ghost's Goblet

RA : 01.5^h	PSA: 1/3	Type: OC
Dec: +63°	OSA: N2	Mag : 9.5
Con: Cas	CSA: 1/2	Size : 7.0′

See:

Notes: C10 is 3° to SE.



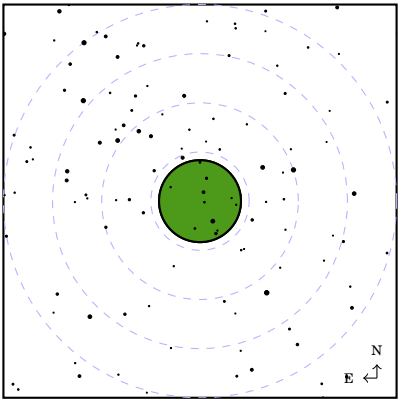
C9

BSC 96 = Sh2-155 = Cave Nebula

RA : 23.0^h	PSA: 71/72	Type: BN
Dec: +63°	OSA:	Mag : 10.0
Con: Cep	CSA: 1/7	Size : 50.0′ × 30.0′

See:

Notes: Not shown in the CSA.



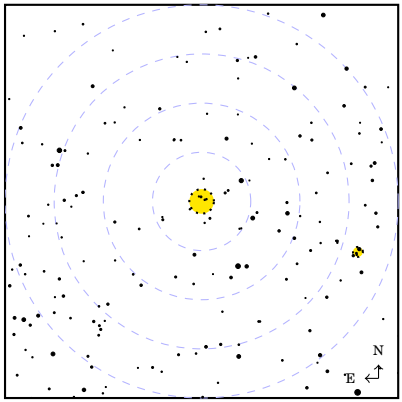
C10

NGC 663 = U5 = BSC 76 = Horseshoe Cluster

RA : 01.8^h	PSA: 1/2	Type: OC
Dec: +61°	OSA: N2	Mag : 6.7
Con: Cas	CSA: 1/2	Size : 15.0′

See:

Notes: M103 is 1.7° WSW. C8 is 3° NW.



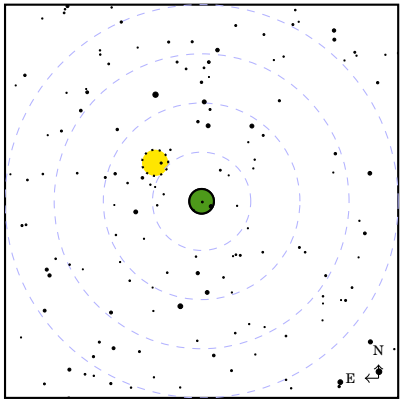
C11

NGC 7635 = Bubble Nebula

RA : 23.3^h	PSA: 71/72	Type: BN
Dec: +61°	OSA:	Mag : 10
Con: Cas	CSA: 1/2/7	Size : 15.0′ × 8.0′

See:

Notes: M52 is 0.6° to NE.

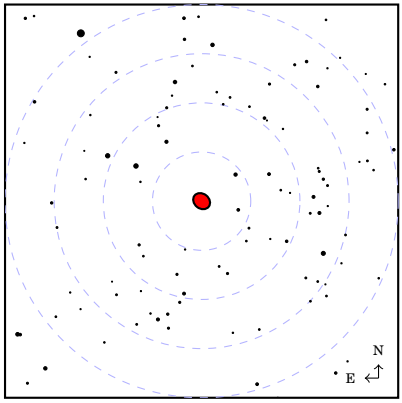


C12

NGC 6946 = Firecracker Galaxy

RA : 20.6^h	PSA: 61/62	Type: Gal
Dec: +60°	OSA: N22	Mag : 8.8
Con: Cep/Cyg	CSA: 1/6/7	Size : 11.0′ × 9.0′

See:

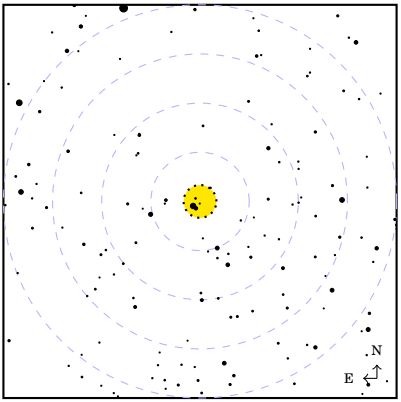


C13

NGC 457 = U4 = BSC 62 = E.T. Cluster = Owl Cluster

RA : 01.3^h	PSA: 1/3	Type: OC
Dec: +58°	OSA: N2	Mag : 6.4
Con:	CSA:	Size : 20.0′

See:



C14

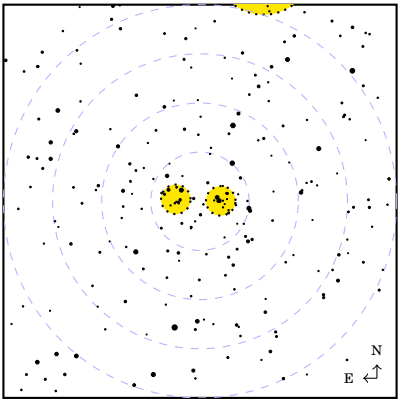
NGC 869/884 = U9/10 = Double Cluster = h/χ Per Clusters

RA : 02.3^h	PSA: 1/2	Type: OC
Dec: +57°	OSA: N2	Mag : 4.5/5.7
Con:	CSA:	Size : 18.0′

See:

Notes: NGC 869 (h Per) is to W and brighter; NGC 884 (χ Per) is to E and fainter. Both are about 18′ in diameter (DSC:TCO). U8 is 2° N. U11 is 2.5° SE.

α Per Mirfak (mag 1.8)
→ 9° E passing ι Per and θ Per to 64/65 And
→ 7° N to C14/U9/U10

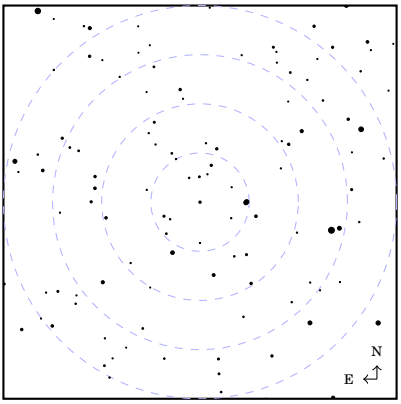


C15

NGC 6826 = BSC 128 = Blinking Planetary

RA : 19.7^h	PSA: 62	Type: PN
Dec: +51°	OSA: N18	Mag : 8.5
Con:	CSA:	Size : 0.5′×0.4′

See:

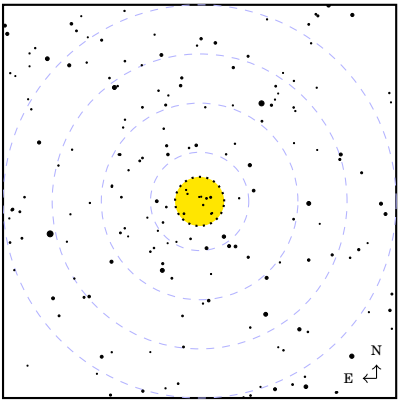


C16

NGC 7243 = U85 = BSC 63

RA : 22.3^h PSA: 73 Type: OC
Dec: +50° OSA: N24 Mag : 6.4
Con: CSA: Size : 30.0′

See:



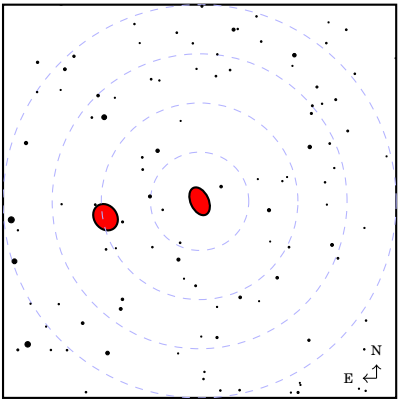
C17

NGC 147

RA : 00.6^h PSA: 3 Type: Gal
Dec: +49° OSA: Mag : 9.5
Con: CSA: Size : 18.0′ ×
11.0′

See:

Notes: C18 is 1° E.



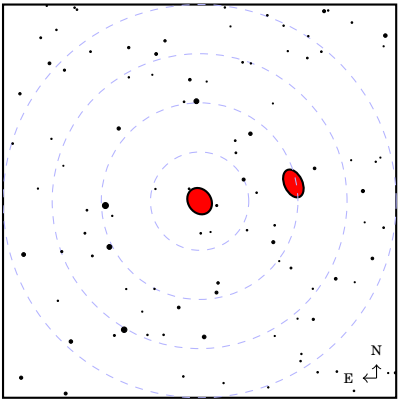
C18

NGC 185

RA : 00.6^h PSA: 3 Type: Gal
Dec: +48° OSA: Mag : 9.2
Con: CSA: Size : 17.0′ ×
14.0′

See:

Notes: C17 is 1° W.



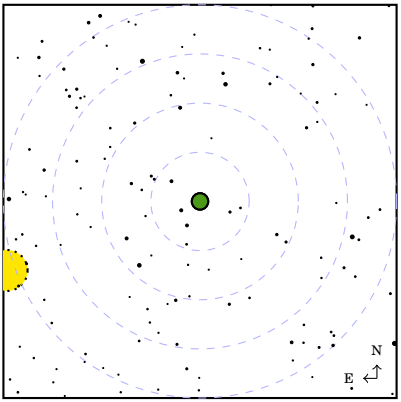
C19

IC 5146 = BSC 78 = Cr 470 = Cocoon Nebula

RA : 21.9^h	PSA: 73	Type: BN
Dec: +47°	OSA:	Mag : 9.3
Con:	CSA:	Size : 10.0′

See:

Notes: Cluster is magitude 7.2 (DSC:TCO). U84 is 2° ESE.



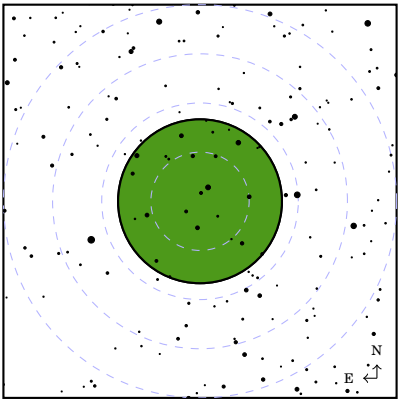
C20

NGC 7000 = BSC 16 = North America Nebula

RA : 21.0^h	PSA: 62	Type: BN
Dec: +44°	OSA: N20	Mag : 3.8
Con:	CSA:	Size : 100.0′ × 60.0′

See:

α Cyg Deneb (mag 1.3)
→ 1° S
→ 3° E to C20

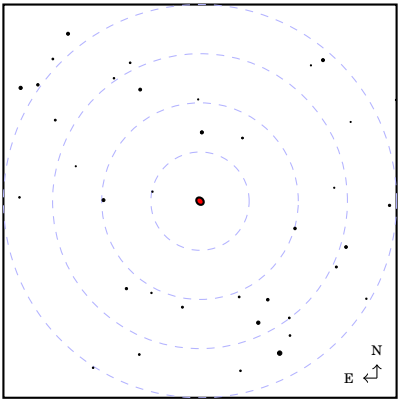


C21

NGC 4449 = Box Galaxy

RA : 12.5^h	PSA: 32/43	Type: Gal
Dec: +44°	OSA: N12	Mag : 9.6
Con:	CSA:	Size : 5.0′×4.0′

See:

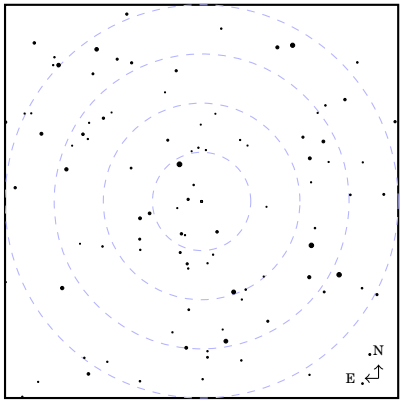


C22

NGC 7662 = BSC 124 = Light Blue Snowball

RA : 23.4^h	PSA: 72	Type: PN
Dec: +43°	OSA: N24	Mag : 8.3
Con:	CSA:	Size : 0.5′

See:

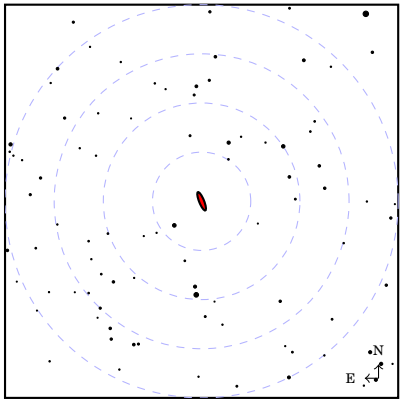


C23

NGC 891

RA : 02.4^h	PSA: 2	Type: Gal
Dec: +42°	OSA: N0	Mag : 9.9
Con:	CSA:	Size : 12.0′ × 3.0′

See:

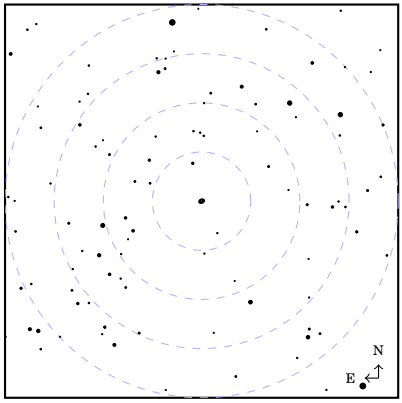


C24

NGC 1275 = Perseus A

RA : 03.3^h	PSA: 13	Type: Gal
Dec: +42°	OSA:	Mag : 11.9
Con:	CSA:	Size : 3.0′×2.0′

See:

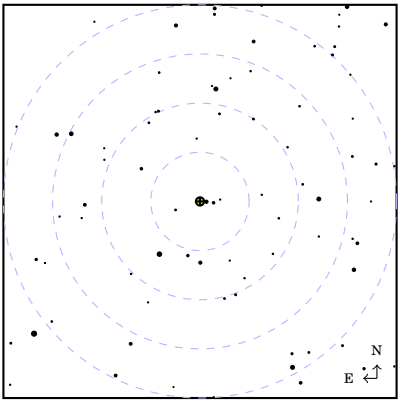


C25

NGC 2419 = BSC 140

RA : 07.6^h	PSA: 23	Type: GC
Dec: +39°	OSA:	Mag : 10.4
Con:	CSA:	Size : 4.6′

See:

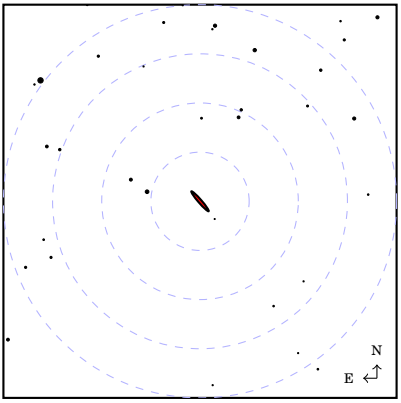


C26

NGC 4244

RA : 12.3^h	PSA: 32/43	Type: Gal
Dec: +38°	OSA: N12	Mag : 10.4
Con:	CSA:	Size : 16.0′ × 2.0′

See:

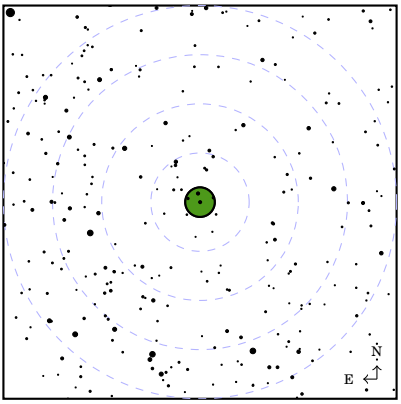


C27

NGC 6888 = BSC 84 = Crescent Nebula

RA : 20.2^h	PSA: 62	Type: BN
Dec: +38°	OSA:	Mag : 8.8
Con:	CSA:	Size : 18.0′ × 13.0′

See:

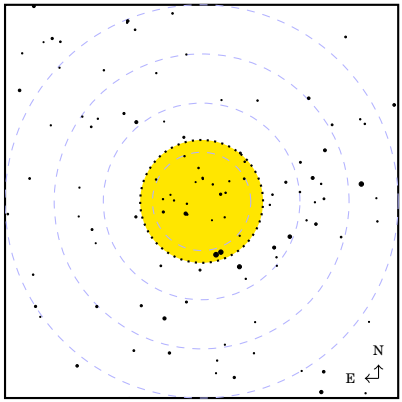


C28

NGC 752 = U7 = Mel 12

RA : 01.9^h PSA: 2 Type: OC
Dec: +38° OSA: N0 Mag : 5.7
Con: CSA: Size : 75.0′

See:

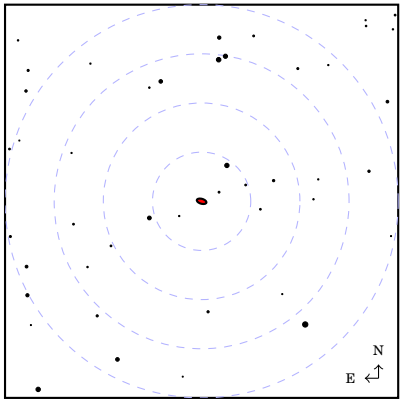


C29

NGC 5005

RA : 13.2^h PSA: 43 Type: Gal
Dec: +37° OSA: Mag : 9.8
Con: CSA: Size : 6.0′×3.0′

See:

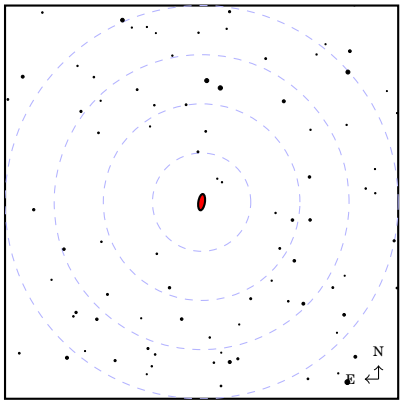


C30

NGC 7331

RA : 22.6^h PSA: 72/74 Type: Gal
Dec: +34° OSA: E23 Mag : 9.5
Con: CSA: Size : 9.7′×4.5′

See:



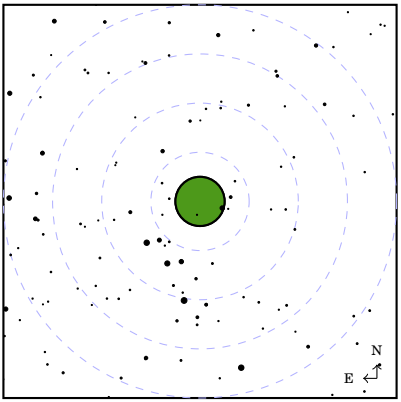
C31

IC 405 = BSC 49 = Flaming Star Nebula

RA : 05.3^h	PSA: 12	Type: BN
Dec: +34°	OSA:	Mag : 9.2
Con:	CSA:	Size : 30.0′ × 20.0′

See:

Notes: AE Aur (towards the SW edge) is magnitude 6 and variable (DSC:TCO).

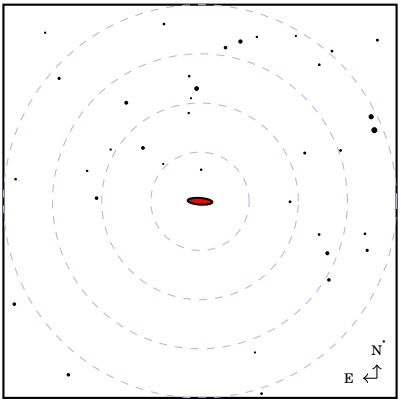


C32

NGC 4631 = Whale Galaxy

RA : 12.7^h	PSA: 43/45	Type: Gal
Dec: +33°	OSA: E13	Mag : 9.2
Con:	CSA:	Size : 15.0′ × 4.0′

See:



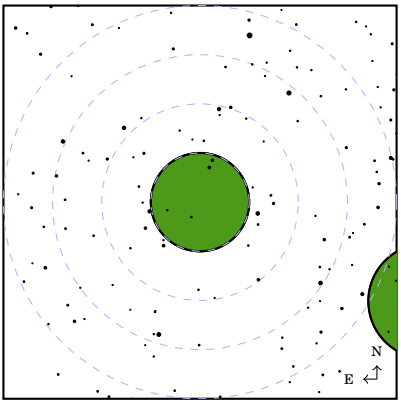
C33

NGC 6992 = BSC 73 = Eastern Veil Nebula

RA : 20.9^h	PSA: 62	Type: BN
Dec: +32°	OSA: N20	Mag : 7.5
Con:	CSA:	Size : 60.0′ × 8.0′

See:

Notes: C34 is 2.5° SW.



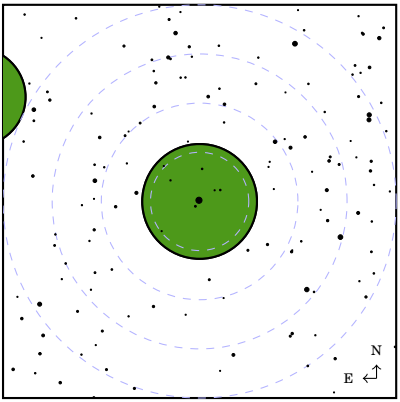
C34

NGC 6960 = BSC 74 = Western Veil Nebula

RA : 20.8^h PSA: 62 Type: BN
Dec: +31° OSA: N20 Mag : 7.9
Con: CSA: Size : 70.0′ × 6.0′

See:

Notes: C33 is 2.5° NE.

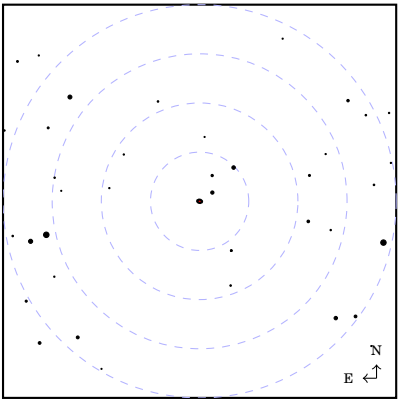


C35

NGC 4889

RA : 13.0^h PSA: 43/45 Type: Gal
Dec: +28° OSA: Mag : 11.5
Con: CSA: Size : 3.0′×2.0′

See:



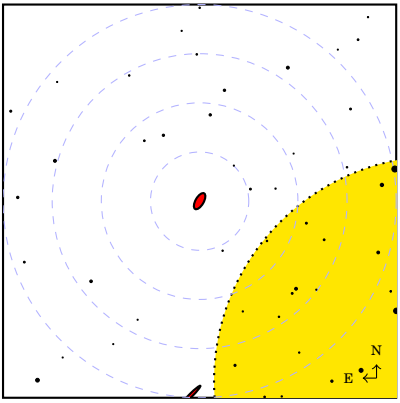
C36

NGC 4559

RA : 12.6^h PSA: 43/45 Type: Gal
Dec: +28° OSA: E13 Mag : 10.0
Con: CSA: Size : 11.0′ × 5.0′

See:

Notes: C38 is 2° S. U44 is 4° to SW.

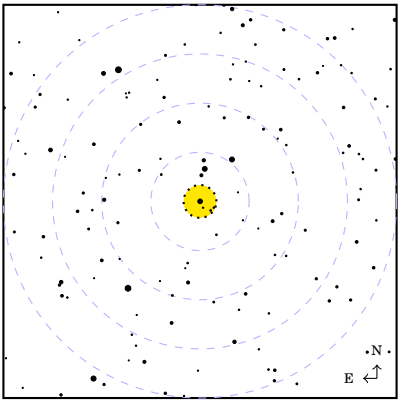


C37

NGC 6885

RA : 20.2^h	PSA: 62/64	Type: OC
Dec: +26°	OSA:	Mag : 8.1
Con:	CSA:	Size : 20.0′

See:



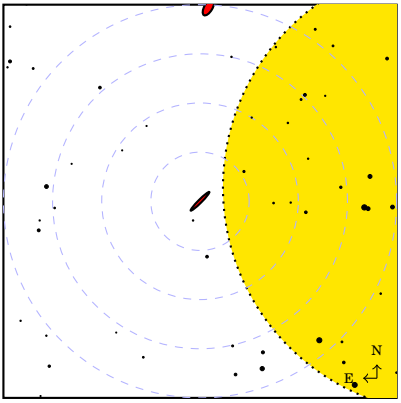
C38

NGC 4565 = BSC 38 = Needle Galaxy

RA : 12.6^h	PSA: 43/45	Type: Gal
Dec: +26°	OSA: E13	Mag : 9.6
Con:	CSA:	Size : 16.0′ × 2.0′

See:

Notes: C36 is 2° N. U44 is 3° to W.



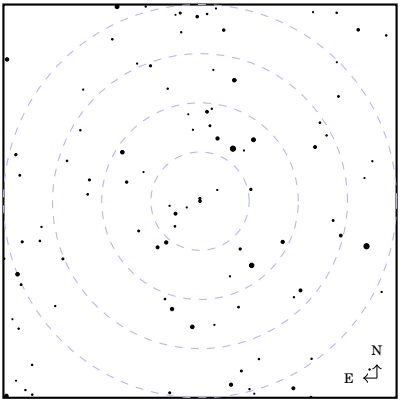
C39

NGC 2392 = U36 = Eskimo Nebula

RA : 07.5^h	PSA: 25	Type: PN
Dec: +21°	OSA: E7	Mag : 9.2
Con:	CSA:	Size : 0.8′×0.7′

See:

Notes: The open cluster NGC 2420 is 2° N (see DSC:TCO).

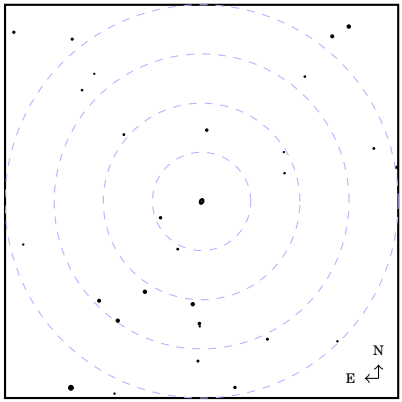


C40

NGC 3626

RA : 11.3^h	PSA: 34	Type: Gal
Dec: +18°	OSA:	Mag : 11.0
Con:	CSA:	Size : 3.0′×2.2′

See:



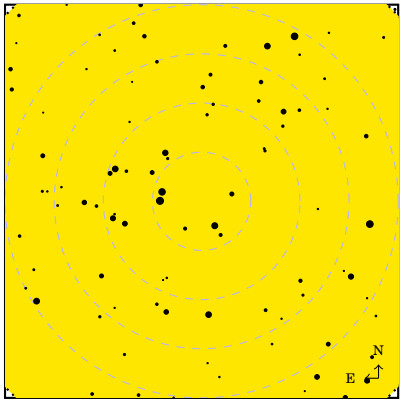
C41

U18 = BSC 1 = Mel 25 = Cr 50 = Hyades

RA : 04.4^h	PSA: 15	Type: OC
Dec: +16°	OSA: E3	Mag : 0.5
Con:	CSA:	Size : 5.5°

See:

Notes: Much bigger than the finder chart.

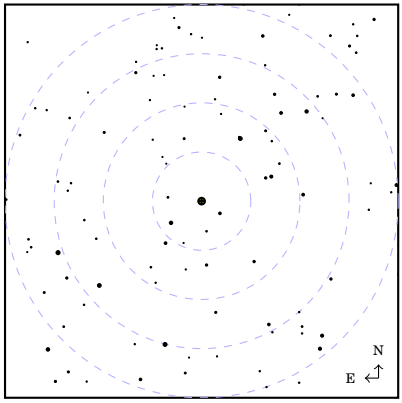


C42

NGC 7006

RA : 21.0^h	PSA: 64/75	Type: GC
Dec: +16°	OSA:	Mag : 10.6
Con:	CSA:	Size : 3.6′

See:

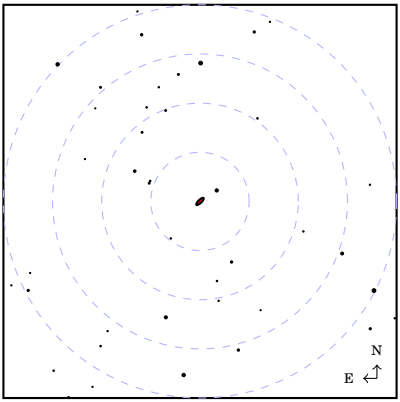


C43

NGC 7814

RA : 00.1^h	PSA: 5/74	Type: Gal
Dec: +16°	OSA:	Mag : 10.6
Con:	CSA:	Size : 6.0′×2.0′

See:

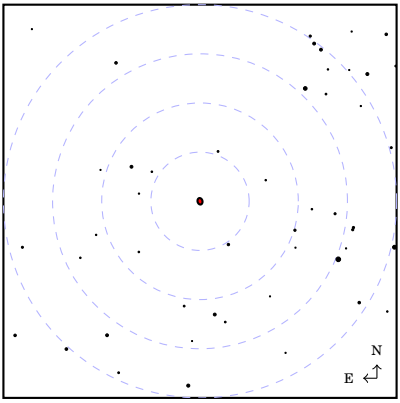


C44

NGC 7479

RA : 23.1^h	PSA: 74	Type: Gal
Dec: +12°	OSA:	Mag : 10.8
Con:	CSA:	Size : 4.0′×3.0′

See:

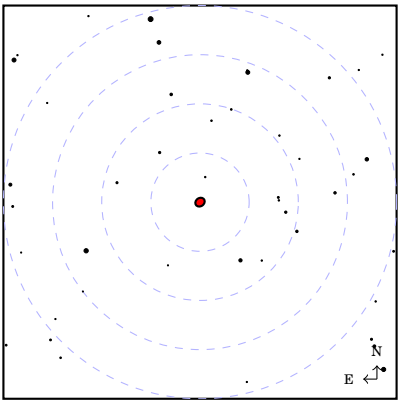


C45

NGC 5248

RA : 13.6^h	PSA: 44	Type: Gal
Dec: +09°	OSA:	Mag : 10.3
Con:	CSA:	Size : 6.0′×5.0′

See:



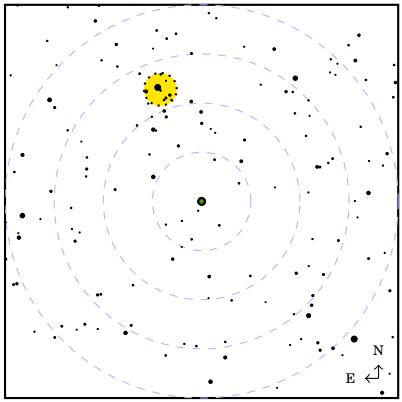
C46

NGC 2261 = BSC 137 = Hubble's Variable Nebula

RA : 06.7^h	PSA: 25/E	Type: BN
Dec: +09°	OSA: E7	Mag : 10
Con:	CSA:	Size : 4.0′×2.0′

See:

Notes: The magnitude of the nebula is variable (DSC:TCO). U31 is 1.25° NNE.

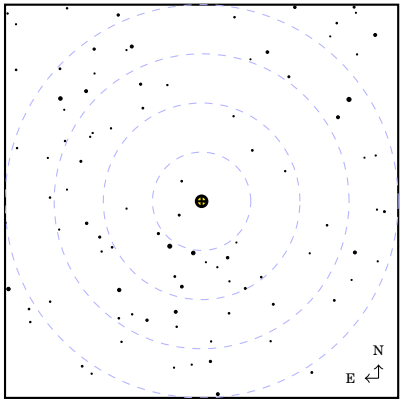


C47

NGC 6934 = BSC 121

RA : 20.6^h	PSA: 64	Type: GC
Dec: +07°	OSA:	Mag : 8.8
Con:	CSA:	Size : 7.1′

See:

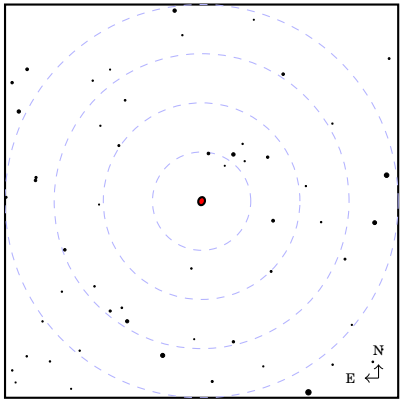


C48

NGC 2775

RA : 09.2^h	PSA: 24/35	Type: Gal
Dec: +07°	OSA:	Mag : 10.1
Con:	CSA:	Size : 5.0′×4.0′

See:



C49

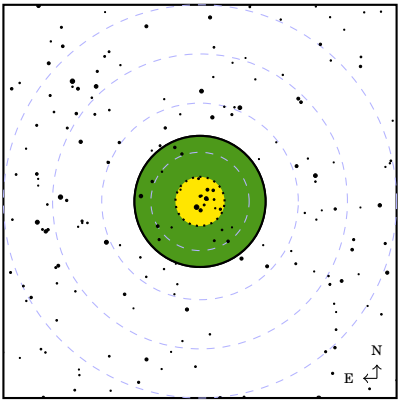
NGC 2237/38/46 = BSC 135 = Rosette Nebula

RA : 06.5^h	PSA: 25/E	Type: BN	
Dec: +05°	OSA: E9	Mag : 5	
Con:	CSA:	Size : 80.0′	×
		60.0′	

See:

Notes: Surrounding C50.

α Ori Betelgeuse (mag 0.3–0.9)
→ 3° S
→ 7° E to 8 Mon
→ 2° E to C49



C50

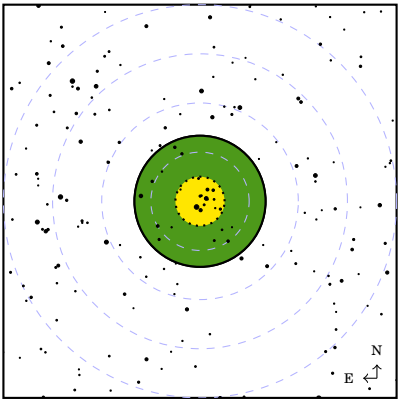
NGC 2244 = U30 = BSC 25

RA : 06.5^h	PSA: 25/E	Type: OC	
Dec: +05°	OSA: E9	Mag : 4.4	
Con:	CSA:	Size : 30.0′	

See:

Notes: Surrounded by C49.

α Ori Betelgeuse (mag 0.3–0.9)
→ 3° S
→ 7° E to 8 Mon
→ 2° E to C50/U30

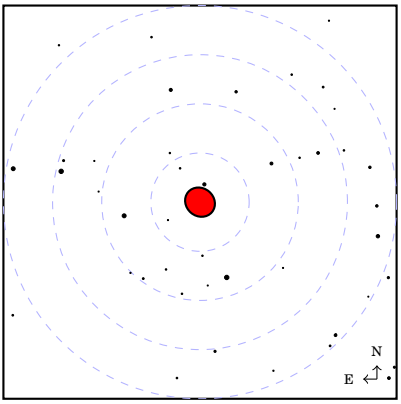


C51

IC 1613

RA : 01.1^h	PSA: 5/7	Type: Gal	
Dec: +02°	OSA:	Mag : 9.2	
Con:	CSA:	Size : 19.0′	×
		17.0′	

See:

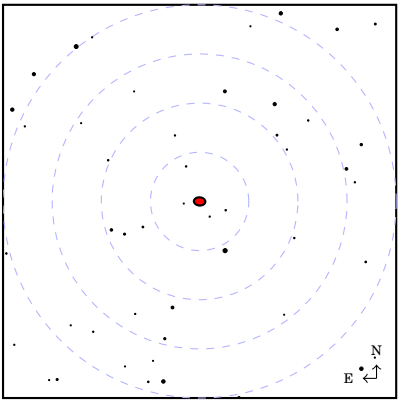


C52

NGC 4697

RA : 12.8^h	PSA: 47	Type: Gal
Dec: $-06°$	OSA: E12	Mag : 9.0
Con:	CSA:	Size : $7.0' \times 5.0'$

See:

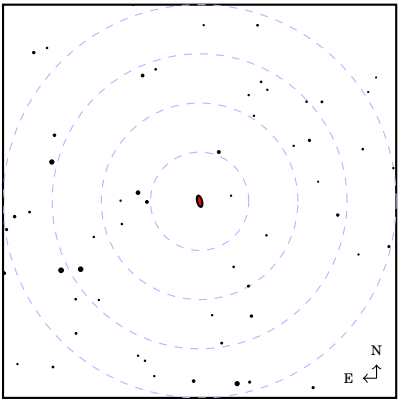


C53

NGC 3115 = The Spindle

RA : 10.1^h	PSA: 37	Type: Gal
Dec: $-08°$	OSA: E10	Mag : 8.9
Con:	CSA:	Size : $7.0' \times 3.0'$

See:

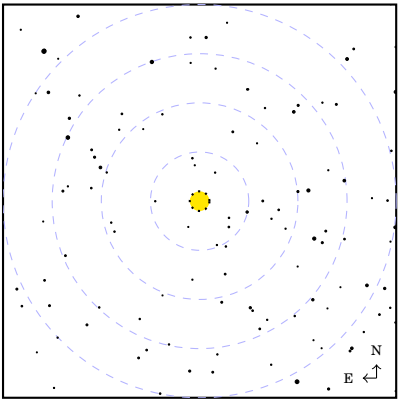


C54

NGC 2506 = BSC 90 = Mel 80 = Cr 170

RA : 08.0^h	PSA: 26	Type: OC
Dec: $-11°$	OSA:	Mag : 7.6
Con:	CSA:	Size : $12.0'$

See:

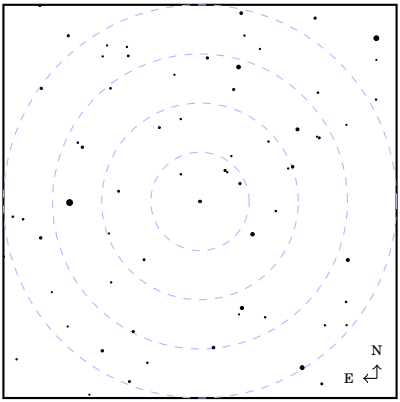


C55

NGC 7009 = BSC 106 = Saturn Nebula

RA : 21.1^h	PSA: 77	Type: PN
Dec: $-11°$	OSA: E24	Mag : 8.0
Con:	CSA:	Size : $0.7' \times 0.4'$

See:

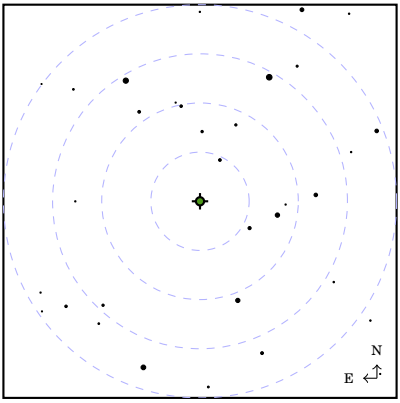


C56

NGC 246 = BSC 107

RA : 00.8^h	PSA: 7	Type: PN
Dec: $-12°$	OSA:	Mag : 10.9
Con:	CSA:	Size : $5.0' \times 4.0'$

See:



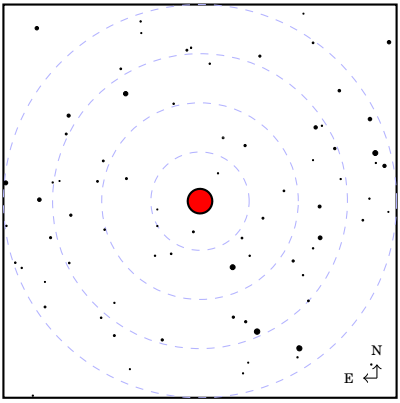
C57

NGC 6822 = IC 4895 = BSC 146 = Barnard's Galaxy

RA : 19.7^h	PSA: 66	Type: Gal
Dec: $-15°$	OSA: E22	Mag : 7.9
Con:	CSA:	Size : $15.0'$

See:

Notes: U73 is 0.7° NNW.

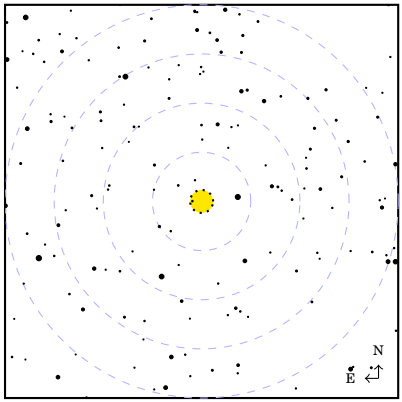


C58

NGC 2360 = BSC 79 = Mel 64 = Caroline's Cluster

RA : 07.3^h	PSA: 27	Type: OC
Dec: $-16°$	OSA: E8	Mag : 7.0
Con:	CSA:	Size : 14.0′

See:

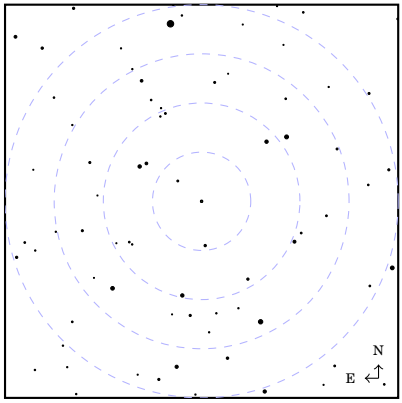


C59

NGC 3242 = U43 = BSC 125 = Ghost of Jupiter

RA : 10.4^h	PSA: 36/37	Type: PN
Dec: $-19°$	OSA: E10	Mag : 7.3
Con:	CSA:	Size : 0.8′×0.6′

See:



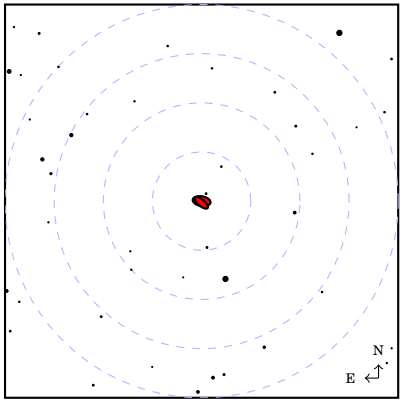
C60

NGC 4038 = NW Antennae Galaxy

RA : 12.0^h	PSA: 36/47	Type: Gal
Dec: $-19°$	OSA:	Mag : 10.5
Con:	CSA:	Size : 11.0′ × 6.0′

See:

Notes: C61 is 1.2′ SW.



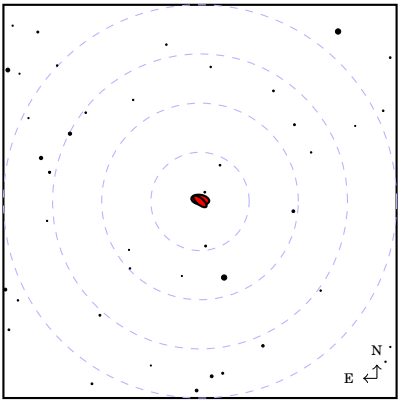
C61

NGC 4039 = SE Antennae Galaxy

RA : 12.0^h	PSA: 36/47	Type: Gal	
Dec: $-19°$	OSA:	Mag : 11.2	
Con:	CSA:	Size : 10.0′	×
		4.0′	

See:

Notes: C60 is 1.2′ NW

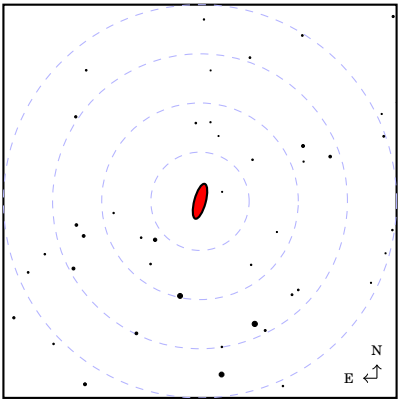


C62

NGC 247

RA : 00.8^h	PSA: 7	Type: Gal	
Dec: $-21°$	OSA: E0	Mag : 8.9	
Con:	CSA:	Size : 22.0′	×
		7.0′	

See:

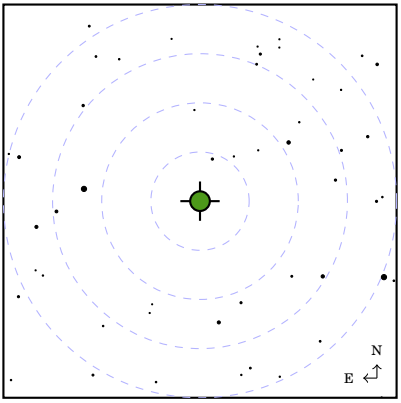


C63

NGC 7293 = BSC 91 = Helix Nebula

RA : 22.5^h	PSA: 76/77	Type: PN	
Dec: $-21°$	OSA: E24	Mag : 6.0	
Con:	CSA:	Size : 12.0′	×
		10.0′	

See:

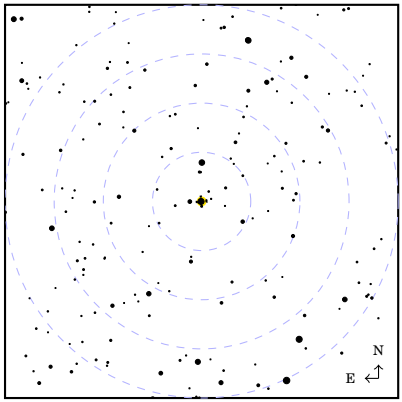


C64

NGC 2362 = BSC 17 = τ CMa Cluster

RA : 07.3^h	PSA: 27	Type: OC
Dec: $-25°$	OSA: E6	Mag : 4.1
Con:	CSA:	Size : 6.0′

See:



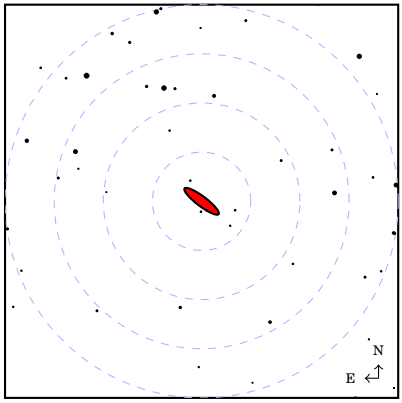
C65

NGC 253 = BSC 108 = Sculpter Galaxy

RA : 00.8^h	PSA: 7/9	Type: Gal
Dec: $-25°$	OSA: E0	Mag : 7.1
Con:	CSA:	Size : 26.0′ × 6.0′

See:

β Cet Deneb Kaitos (mag 2.0)
→ 7° S
→ 1° E to C65

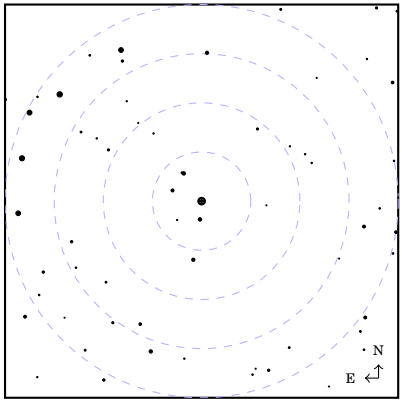


C66

NGC 5694

RA : 14.7^h	PSA: 46	Type: GC
Dec: $-27°$	OSA:	Mag : 10.2
Con:	CSA:	Size : 4.3′

See:

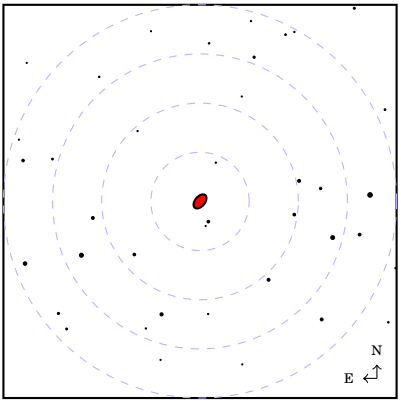


C67

NGC 1097

RA : 02.8^h	PSA: 6/8	Type: Gal
Dec: $-30°$	OSA:	Mag : 9.2
Con:	CSA:	Size : 10.0′ × 6.0′

See:



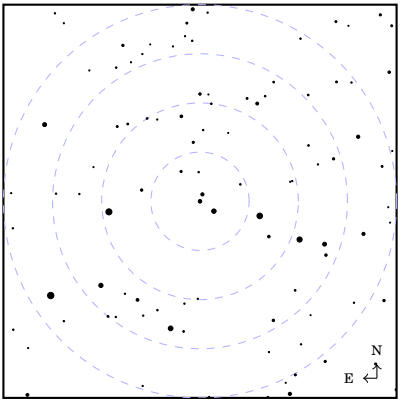
C68

NGC 6729 = R CrA Nebula

RA : 19.0^h	PSA: 69	Type: BN
Dec: $-37°$	OSA:	Mag : 9.5
Con:	CSA:	Size : 1.3′

See:

Notes: The morphology and magnitude of the nebula are variable (DSC:TCO).

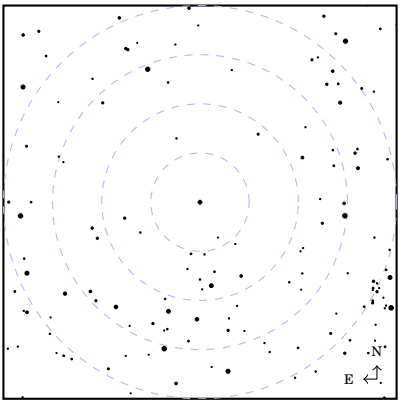


C69

NGC 6302 = BSC 77 = Bug Nebula

RA : 17.2^h	PSA: 58	Type: PN
Dec: $-37°$	OSA:	Mag : 9.6
Con:	CSA:	Size : 1.4′×0.4′

See:

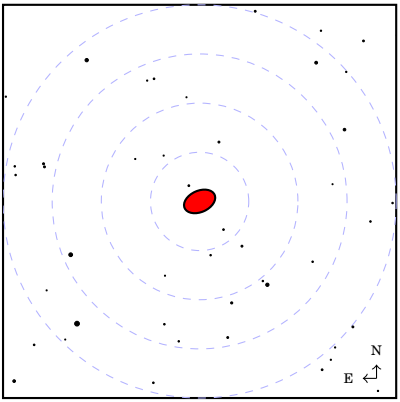


C70

NGC 300 = BSC 138

RA : 00.9^h PSA: 9 Type: Gal
Dec: −38° OSA: Mag : 7.2
Con: CSA: Size : 20.0′ ×
13.0′

See:

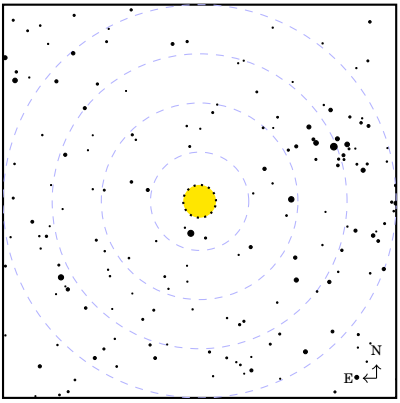


C71

NGC 2477 = BSC 41 = Mel 78

RA : 07.9^h PSA: 28 Type: OC
Dec: −39° OSA: S6 Mag : 5.8
Con: CSA: Size : 20.0′

See:

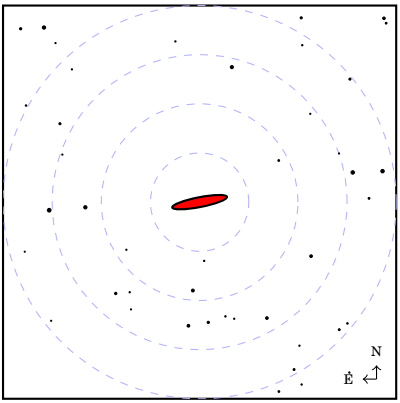


C72

NGC 55 = BSC 102

RA : 00.2^h PSA: 9 Type: Gal
Dec: −39° OSA: S0 Mag : 7.5
Con: CSA: Size : 34.0′ ×
6.0′

See:

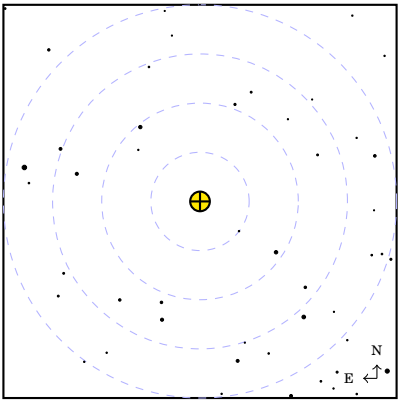


C73

NGC 1851 = BSC 80

RA : 05.2^h PSA: 18 Type: GC
Dec: −40° OSA: S0 Mag : 7.1
Con: CSA: Size : 12.0′

See:

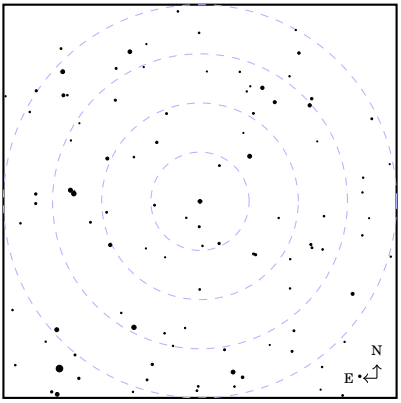


C74

NGC 3132

RA : 10.1^h PSA: 39 Type: PN
Dec: −40° OSA: S6 Mag : 9.2
Con: CSA: Size : 1.4′×0.9′

See:

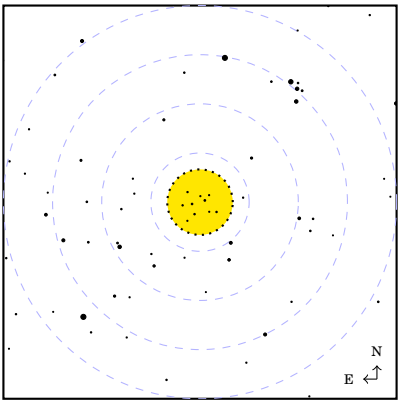


C75

NGC 6124 = BSC 42

RA : 16.4^h PSA: 58 Type: OC
Dec: −41° OSA: S21 Mag : 5.3
Con: CSA: Size : 40.0′

See:

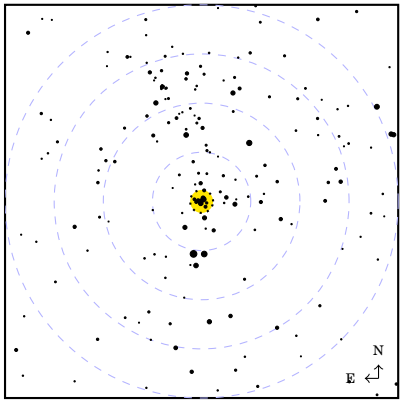


C76

NGC 6231 = BSC 7 = Cr 315

RA : 16.9^h PSA: 58 Type: OC
Dec: −42° OSA: S21 Mag : 2.6
Con: CSA: Size : 14.0′

See:

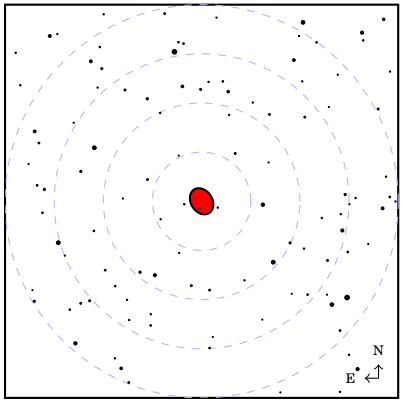


C77

NGC 5128 = Cen A

RA : 13.4^h PSA: 48/49 Type: Gal
Dec: −43° OSA: S15 Mag : 6.6
Con: CSA: Size : 17.0′ × 13.0′

See:

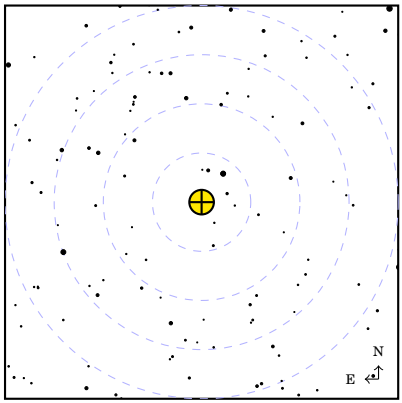


C78

NGC 6541 = BSC 60

RA : 18.1^h PSA: 69 Type: GC
Dec: −44° OSA: S21 Mag : 6.3
Con: CSA: Size : 15.0′

See:

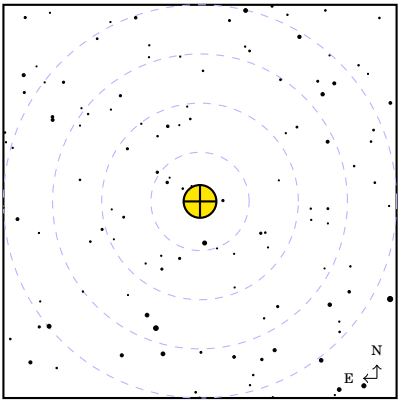


C79

NGC 3201 = BSC 112

RA : 10.3^h	PSA: 39	Type: GC
Dec: $-46°$	OSA: S6	Mag : 6.7
Con:	CSA:	Size : 20.0′

See:

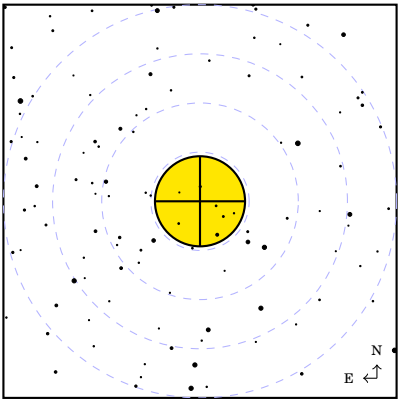


C80

NGC 5139 = BSC 14 = ω Cen

RA : 13.4^h	PSA: 48/49	Type: GC
Dec: $-47°$	OSA: S15	Mag : 3.9
Con:	CSA:	Size : 55.0′

See:

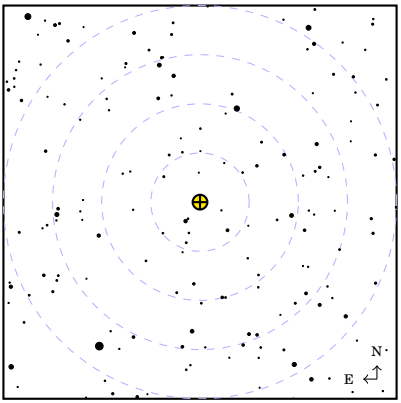


C81

NGC 6352 = BSC 98

RA : 17.4^h	PSA: 58	Type: GC
Dec: $-48°$	OSA:	Mag : 7.8
Con:	CSA:	Size : 9.0′

See:

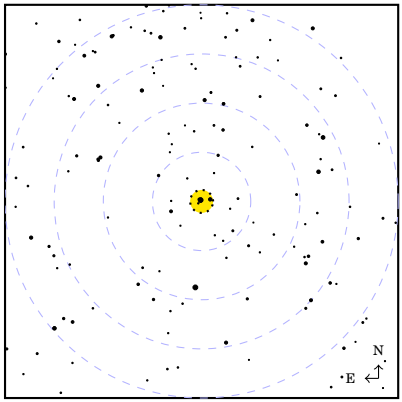


C82

NGC 6193 = BSC 30

RA : 16.7^h PSA: 58 Type: OC
Dec: −49° OSA: Mag : 4.8
Con: CSA: Size : 14.0′

See:

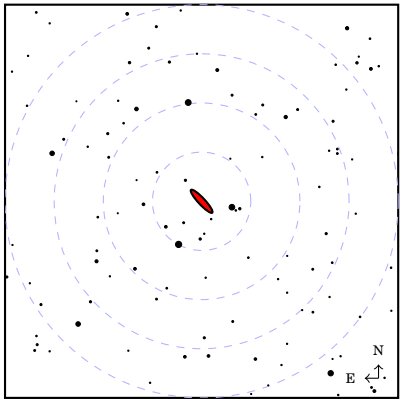


C83

NGC 4945 = BSC 147

RA : 13.1^h PSA: 49 Type: Gal
Dec: −49° OSA: Mag : 8.8
Con: CSA: Size : 19.0′ × 4.0′

See:

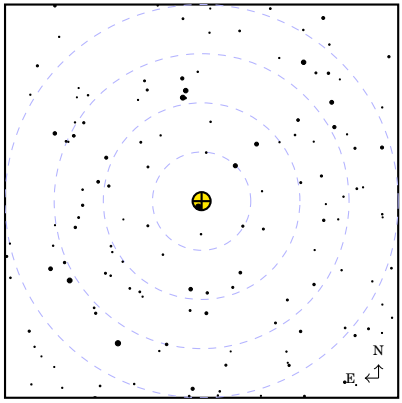


C84

NGC 5286 = BSC 92

RA : 13.8^h PSA: 48 Type: GC
Dec: −51° OSA: Mag : 7.2
Con: CSA: Size : 11.0′

See:

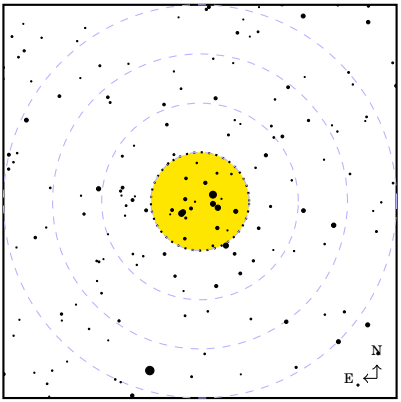


C85

IC 2391 = BSC 6 = *o* Vel Cluster

RA : 08.7^h	PSA: 28	Type: OC
Dec: −53°	OSA: S6	Mag : 2.8
Con:	CSA:	Size : 60.0′

See:

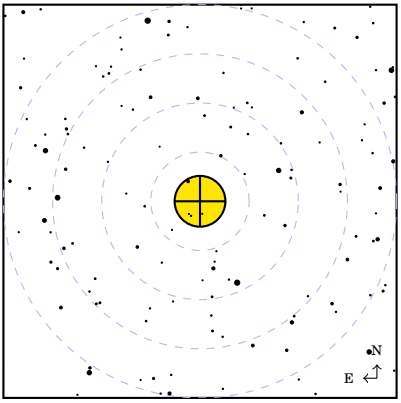


C86

NGC 6397 = BSC 67

RA : 17.7^h	PSA: 58	Type: GC
Dec: −54°	OSA: S18	Mag : 5.3
Con:	CSA:	Size : 31.0′

See:

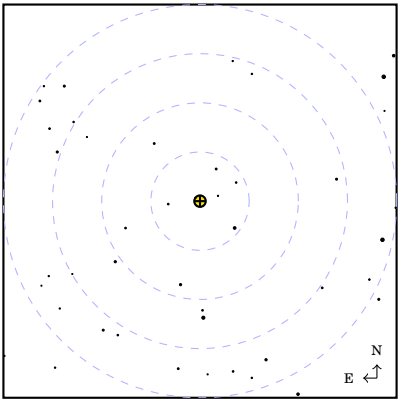


C87

NGC 1261 = BSC 115

RA : 03.2^h	PSA: 19	Type: GC
Dec: −55°	OSA:	Mag : 8.3
Con:	CSA:	Size : 6.8′

See:

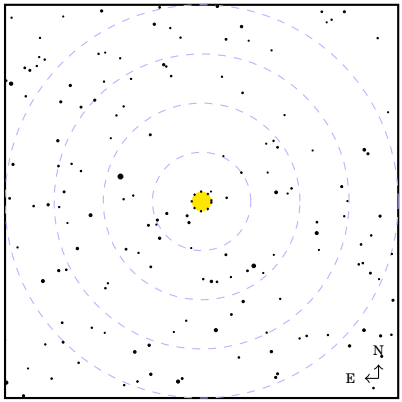


C88

NGC 5823 = BSC 103 = Mel 131

RA : 15.1^h	PSA: 59	Type: OC
Dec: −56°	OSA:	Mag : 7.9
Con:	CSA:	Size : 12.0′

See:

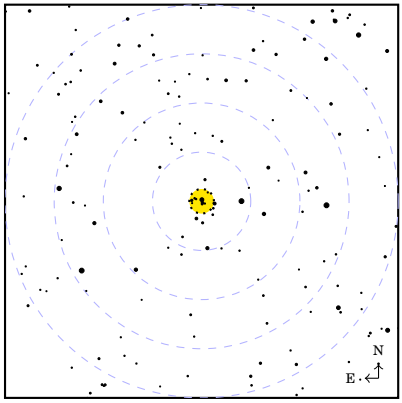


C89

NGC 6087 = BSC 33 = S Nor Cluster

RA : 16.3^h	PSA: 58/60	Type: OC
Dec: −58°	OSA: S18	Mag : 5.2
Con:	CSA:	Size : 15.0′

See:

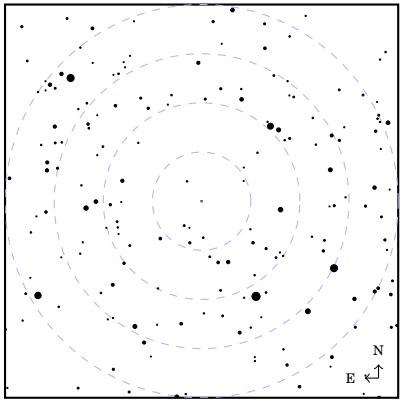


C90

NGC 2867

RA : 09.4^h	PSA: 39	Type: PN
Dec: −58°	OSA:	Mag : 9.7
Con:	CSA:	Size : 0.2′

See:



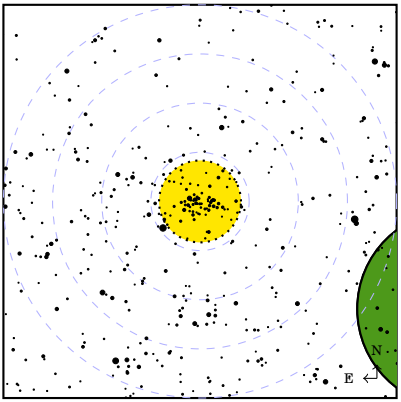
C91

NGC 3532 = BSC 9

RA : 11.1^h	PSA: 38/40	Type: OC
Dec: $-59°$	OSA: S9	Mag : 3.2
Con:	CSA:	Size : 50.0′

See:

Notes: C90 is 2.5° SW.



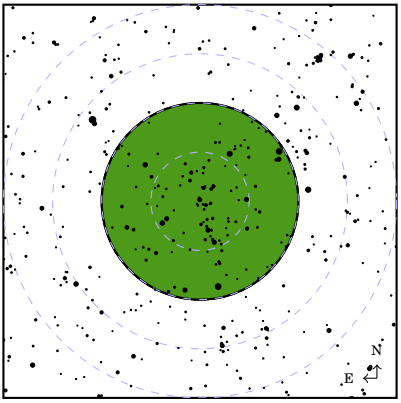
C92

NGC 3372 = BSC 3 = η Car Nebula

RA : 10.8^h	PSA: 38/40	Type: BN
Dec: $-60°$	OSA: S9	Mag : 4.5
Con:	CSA:	Size : 120.0′

See:

Notes: The nebula is variable in magnitude. C91 is 2.5° NE.

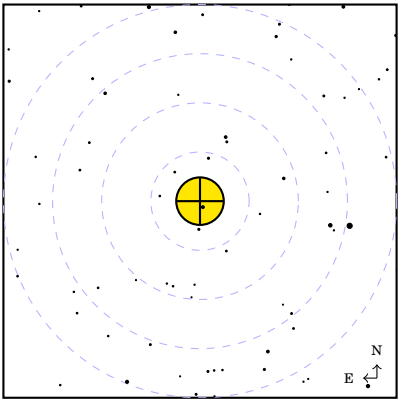


C93

NGC 6752 = BSC 34

RA : 19.2^h	PSA: 69/70	Type: GC
Dec: $-60°$	OSA: S24	Mag : 5.4
Con:	CSA:	Size : 29.0′

See:



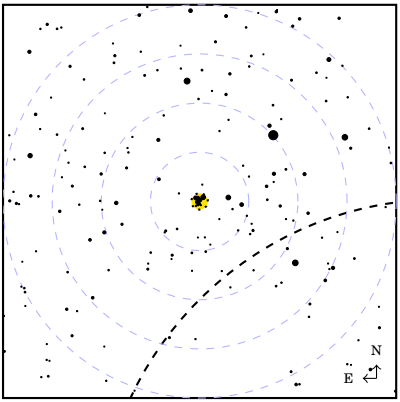
C94

NGC 4755 = BSC 20 = Mel 114 = Jewel Box

RA : 12.9^h	PSA: 49/50	Type: OC
Dec: −60°	OSA: S12	Mag : 4.2
Con:	CSA:	Size : 10.0′

See:

Notes: The northern edge of C99 is 1° S.

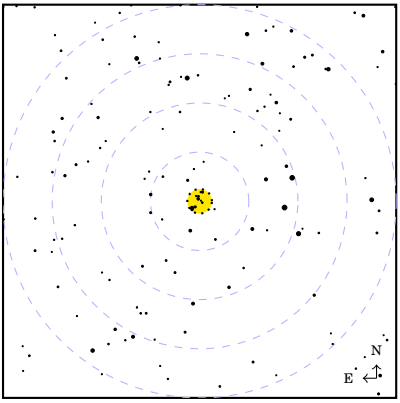


C95

NGC 6025 = BSC 29

RA : 16.1^h	PSA: 59/60	Type: OC
Dec: −60°	OSA:	Mag : 5.5
Con:	CSA:	Size : 15.0′

See:

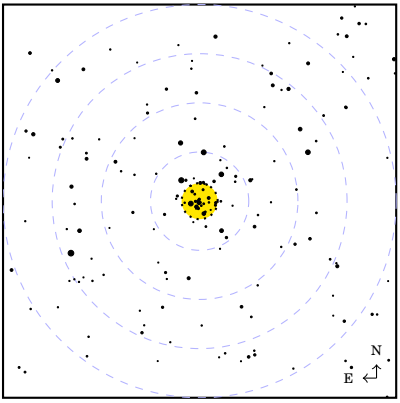


C96

NGC 2516 = BSC 13 = Southern Beehive Cluster

RA : 08.0^h	PSA: 28/30	Type: OC
Dec: −61°	OSA: S3	Mag : 3.5
Con:	CSA:	Size : 22.0′

See:



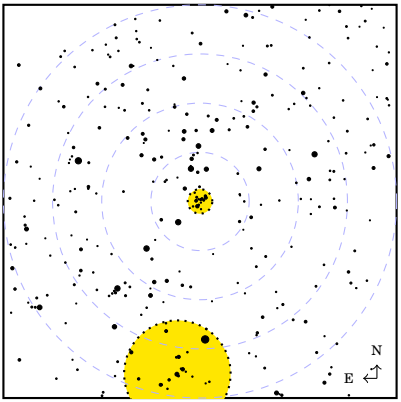
C97

NGC 3766 = BSC 32 = Cr 248

RA : 11.6^h	PSA: 38/40	Type: OC
Dec: −62°	OSA: S12	Mag : 5.3
Con:	CSA:	Size : 15.0′

See:

Notes: C100 is 1.8° S.



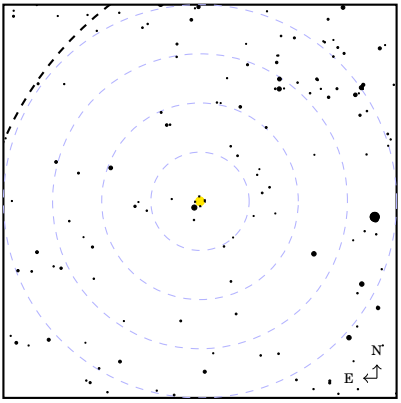
C98

NGC 4609 = BSC 70 = Coalsack Cluster

RA : 12.7^h	PSA: 49/50	Type: OC
Dec: −63°	OSA:	Mag : 6.9
Con:	CSA:	Size : 6.0′

See:

Notes: Surrounded by C99.



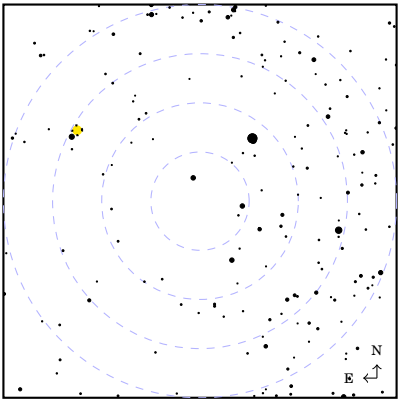
C99

Coalsack

RA : 12.5^h	PSA: 49/50	Type: DN
Dec: −64°	OSA: S12	Mag :
Con:	CSA:	Size : 7.0°×5.0°

See:

Notes: Much bigger than the finder chart.
Surrounds C98. C94 is 1° N of the northern edge.



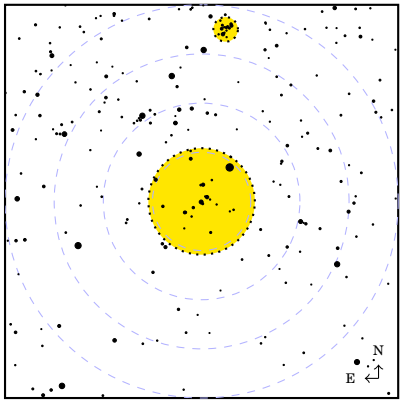
C100

BSC 22 = Cr 249 = λ Cen Cluster

RA : 11.6^h	PSA: 38/40	Type: OC
Dec: $-63°$	OSA:	Mag : 2.9
Con:	CSA:	Size : 65.0′ × 40.0′

See:

Notes: C97 is 1.8° N.

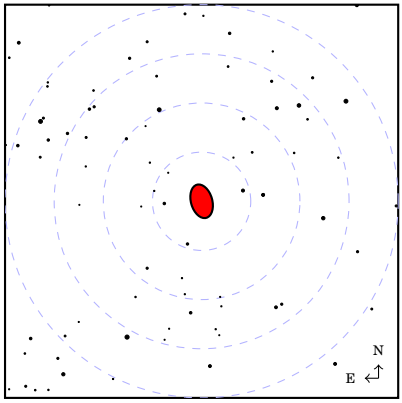


C101

NGC 6744 = BSC 143

RA : 19.2^h	PSA: 70	Type: Gal
Dec: $-64°$	OSA:	Mag : 8.6
Con:	CSA:	Size : 21.0′ × 13.0′

See:

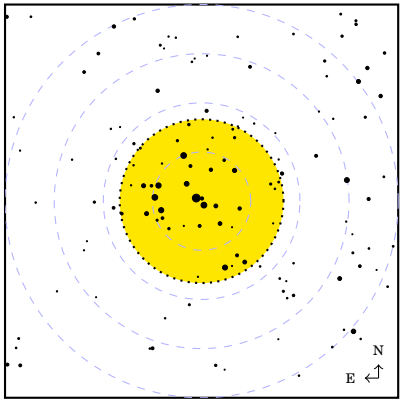


C102

IC 2602 = BSC 5 = Mel 102 = Cr 229 = θ Car Cluster

RA : 10.7^h	PSA: 38/40	Type: OC
Dec: $-64°$	OSA: S9	Mag : 1.9
Con:	CSA:	Size : 100.0′

See:



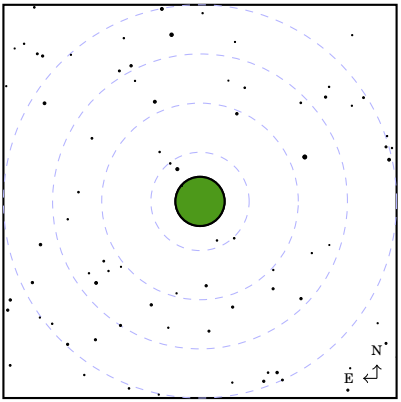
C103

NGC 2070 = BSC 109 = Tarantula Nebula

RA : 05.6^h	PSA: 20/30/D	Type: BN	
Dec: −69°	OSA: S3	Mag : 4.0/8.3	
Con:	CSA:	Size : 30.0′	× 20.0′

See:

Notes: The nebula is magnitude 4.0 (DSC:TCO). the cluster is magnitude 8.3 and has a diameter of 5′ (DSC:TCO).

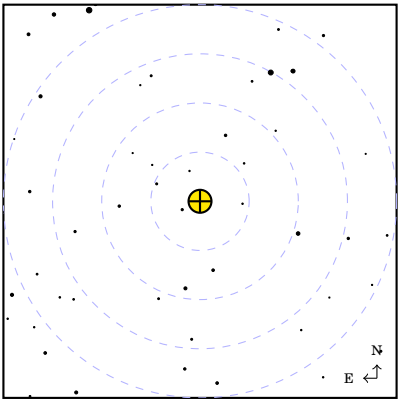


C104

NGC 362 = BSC 64 = Mel 4

RA : 01.1^h	PSA: 10/80	Type: GC
Dec: −71°	OSA: S0	Mag : 6.4
Con:	CSA:	Size : 14.0′

See:

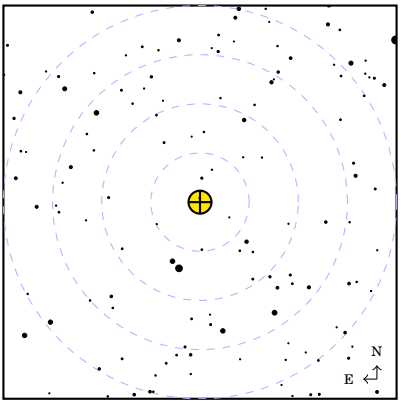


C105

NGC 4833 = BSC 97

RA : 13.0^h	PSA: 50	Type: GC
Dec: −71°	OSA:	Mag : 8.4
Con:	CSA:	Size : 14.0′

See:

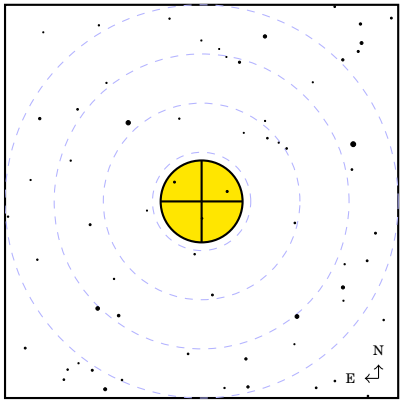


C106

NGC 104 = BSC 26 = Mel 1 = 47 Tuc

RA : 00.4^h	PSA: 10/80	Type: GC
Dec: $-72°$	OSA: S0	Mag : 4.0
Con:	CSA:	Size : 50.0′

See:

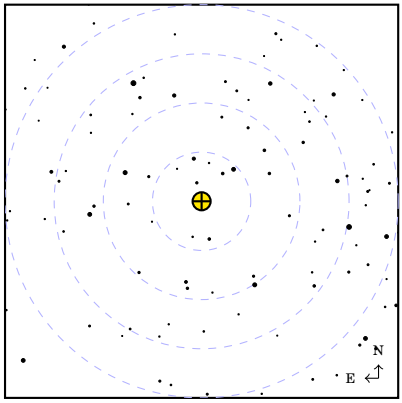


C107

NGC 6101 = BSC 139

RA : 16.4^h	PSA: 60	Type: GC
Dec: $-72°$	OSA:	Mag : 9.2
Con:	CSA:	Size : 11.0′

See:

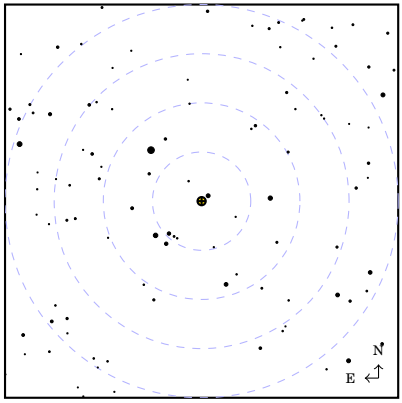


C108

NGC 4372 = BSC 99

RA : 12.4^h	PSA: 40/50	Type: GC
Dec: $-73°$	OSA:	Mag : 7.2
Con:	CSA:	Size : 5.0′

See:

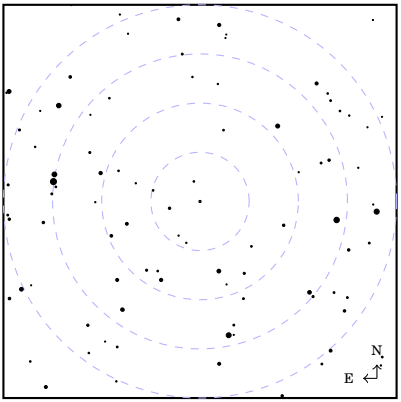


C109

NGC 3195

RA : 10.2^{h}	PSA : 40	Type: PN
Dec: -81°	OSA:	Mag : 11.6
Con:	CSA:	Size : 0.6′

See:



The Urban Observing Program Objects

The Urban Observing Program of the Astronomical League consists of bright objects selected by Terry Trees for northern observers under light-polluted skies. There are 87 deep-sky objects, 12 double stars, and 1 variable star. The deep-sky objects range in declination from −35 to +72 degrees, cover all seasons, and include 41 Messier objects and 14 Caldwell objects.

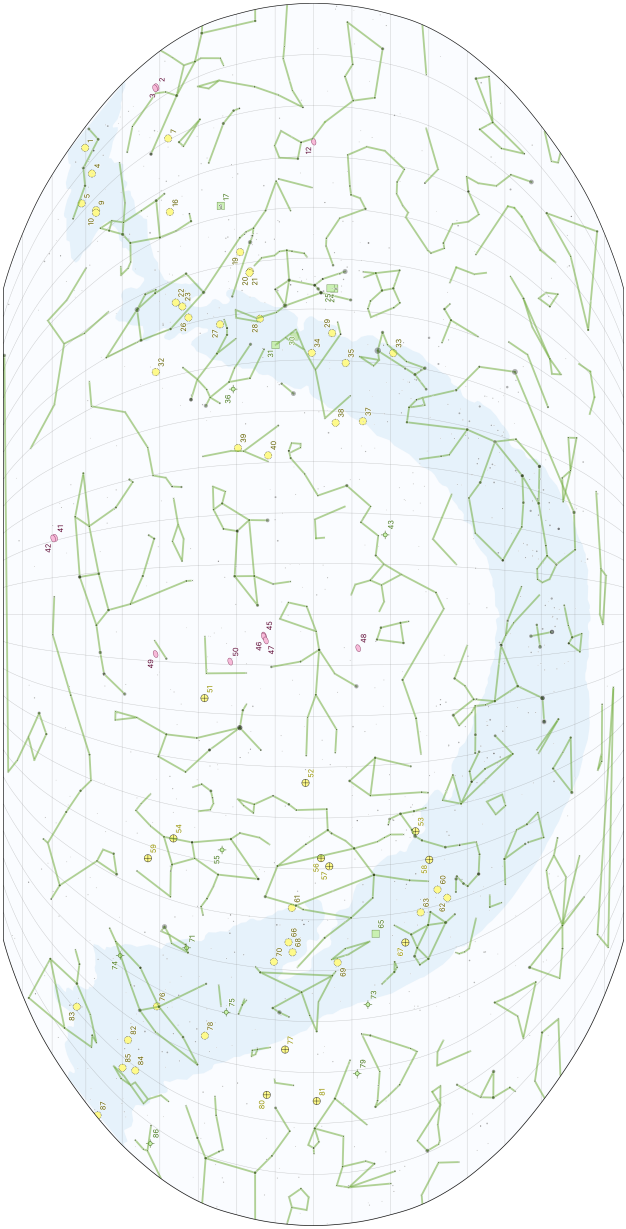
I give finder charts for the deep-sky objects only; the stars are all no fainter than magnitude 5.2 and have Bayer designations, and so a standard all-sky atlas is adequate. I use U1 to U87 to label the deep-sky objects, following the order in right ascension given by Trees, but this is not a standard designation. For completeness, I include all the deep-sky objects, even bright ones like the Hyades (U18).

The following table lists the deep-sky objects with their J2000 positions (decimal hours of right ascension and degrees of declination), the charts on which they appear in the *Pocket Sky Atlas*, their types, other names, and references to O'Meara's *Deep-Sky Companions* volumes (*The Messier Objects*, *The Caldwell Objects*, *The Secret Deep*, and *Hidden Treasures*).

I note that the Hyades (U18) and Coma Star Cluster (U44) are much bigger than the finder charts. They are really binocular objects and better located using a small-scale all-sky atlas. That said, be aware that the α Per Cluster (U15) and Coma Star Cluster (U44) are not labelled in the *Pocket Sky Atlas*.

Name	Position	Type	Other Names
U1	00.5 +60	OC	NGC 129
U2	00.7 +41	Gal	NGC 221 = M32 = BSC 110
U3	00.7 +41	Gal	NGC 224 = M31 = BSC 11 = Andromeda Galaxy
U4	01.3 +58	OC	NGC 457 = C13 = BSC 62 = E.T. Cluster = Owl Cluster
U5	01.8 +61	OC	NGC 663 = C10 = BSC 76 = Horseshoe Cluster
U6	01.8 +72	OC	Cr 463
U7	01.9 +38	OC	NGC 752 = C28 = Mel 12
U8	02.3 +60	OC	Stock 2
U9	02.3 +57	OC	NGC 869 = Mel 13 = Cr 24 = h Per Cluster = half of C14
U10	02.4 +57	OC	NGC 884 = Mel 14 = Cr 25 = χ Per Cluster = half of C14
U11	02.6 +56	OC	Tr 2
U12	02.7 −00	Gal	NGC 1068 = M77
U13	03.2 +63	OC	Tr 3
U14	03.3 +60	OC	Stock 23 = Pazmino's Cluster
U15	03.4 +49	OC	Mel 20 = Cr 39 = α Per Cluster
U16	03.5 +37	OC	NGC 1342
U17	03.8 +24	OC	M45 = BSC 4 = Mel 22 = Pleiades
U18	04.4 +16	OC	C41 = BSC 1 = Mel 25 = Cr 50 = Hyades
U19	04.8 +19	OC	NGC 1647
U20	05.2 +17	OC	NGC 1807 = Mel 29 = Cr 61
U21	05.2 +17	OC	NGC 1817 = Cr 60
U22	05.5 +36	OC	NGC 1912 = M38 = BSC 82
U23	05.6 +34	OC	NGC 1960 = M36 = BSC 58
U24	05.6 −05	BN	NGC 1976 = M42 = BSC 15 = Orion Nebula
U25	05.6 −04	OC	NGC 1981
U26	05.9 +33	OC	NGC 2099 = M37 = BSC 55
U27	06.2 +24	OC	NGC 2168 = M35 = BSC 31
U28	06.1 +14	OC	NGC 2169
U29	06.5 −05	OC	NGC 2232
U30	06.5 +05	OC	NGC 2244 = C50 = BSC 25
U31	06.7 +10	OC	NGC 2264
U32	06.8 +41	OC	NGC 2281
U33	06.8 −21	OC	NGC 2287 = M41 = BSC 21
U34	06.9 +00	OC	NGC 2301
U35	07.0 −08	OC	NGC 2323 = M50 = BSC 44
U36	07.5 +21	PN	NGC 2392 = C39 = Eskimo Nebula
U37	08.2 −13	OC	NGC 2539 = Mel 83
U38	08.2 −06	OC	NGC 2548 = M48 = BSC 37
U39	08.7 +20	OC	NGC 2632 = M44 = BSC 12 = Mel 88 = Beehive Cluster
U40	08.9 +12	OC	NGC 2682 = M67 = BSC 51
U41	09.9 +69	Gal	NGC 3031 = M81 = BSC 72
U42	09.9 +70	Gal	NGC 3034 = M82 = BSC 122
U43	10.4 −19	PN	NGC 3242 = C59 = BSC 125 = Ghost of Jupiter
U44	12.4 +26	OC	Mel 111 = Cr 256 = Coma Star Cluster
U45	12.4 +13	Gal	NGC 4374 = M84
U46	12.4 +13	Gal	NGC 4406 = M86
U47	12.5 +12	Gal	NGC 4486 = M87
U48	12.7 −12	Gal	NGC 4594 = M104 = BSC 133 = Sombrero Galaxy
U49	12.8 +41	Gal	NGC 4736 = M94 = BSC 134
U50	12.9 +22	Gal	NGC 4826 = M64 = BSC 149 = Black-Eye Galaxy

Name	Position	Type	Other Names
U51	13.7 +28	GC	NGC 5272 = M3 = BSC 53
U52	15.3 +02	GC	NGC 5904 = M5 = BSC 66
U53	16.4 −27	GC	NGC 6121 = M4 = BSC 43
U54	16.7 +36	GC	NGC 6205 = M13 = BSC 40 = Hercules Cluster
U55	16.7 +24	PN	NGC 6210
U56	16.8 −02	GC	NGC 6218 = M12 = BSC 94
U57	17.0 −04	GC	NGC 6254 = M10 = BSC 61
U58	17.0 −30	GC	NGC 6266 = M62 = BSC 81
U59	17.3 +43	GC	NGC 6341 = M92 = BSC 59
U60	17.7 −32	OC	NGC 6405 = M6 = BSC 18 = Butterfly Nebula
U61	17.8 +06	OC	IC 4665 = Mel 179 = Cr 349
U62	17.9 −35	OC	NGC 6475 = M7 = BSC 10
U63	18.1 −28	OC	NGC 6520
U64	18.1 −24	BN	NGC 6523/6530 = M8 = BSC 45 = Lagoon Nebula
U65	18.3 −16	BN	NGC 6618 = M17 = BSC 47 = Omega Nebula
U66	18.5 +07	OC	NGC 6633
U67	18.6 −24	GC	NGC 6656 = M22 = BSC 28
U68	18.6 +05	OC	IC 4756 = Mel 210
U69	18.9 −06	OC	NGC 6705 = M11 = Mel 213 = Cr 391 = Wild Duck Cluster
U70	18.9 +10	OC	NGC 6709
U71	18.9 +33	PN	NGC 6720 = M57 = BSC 127 = Ring Nebula
U72	19.4 +20	OC	Cr 399 = Brocchi's Cluster
U73	19.7 −14	PN	NGC 6818
U74	19.7 +51	PN	NGC 6826
U75	20.0 +23	PN	NGC 6853 = M27 = BSC 88 = Dumbell Nebula
U76	20.4 +41	OC	NGC 6910
U77	20.6 +07	GC	NGC 6934
U78	20.6 +28	OC	NGC 6940 = Mel 232
U79	21.1 −11	PN	NGC 7009
U80	21.5 +12	GC	NGC 7078 = M15 = BSC 54
U81	21.6 −01	GC	NGC 7089 = M2 = BSC 56
U82	21.5 +48	OC	NGC 7092 = M39 = BSC 36
U83	21.9 +63	OC	NGC 7160
U84	22.1 +47	OC	NGC 7209
U85	22.3 +50	OC	NGC 7243 = C16 = BSC 63
U86	23.4 +43	PN	NGC 7662
U87	23.9 +57	OC	NGC 7789 = Mel 245



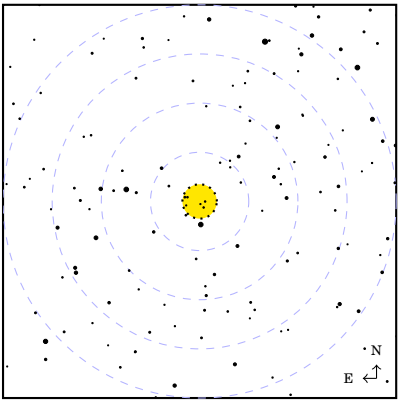
The distribution of Urban List objects over the whole sky

U1

NGC 129

RA : 00.5^h	PSA: 3	Type: OC
Dec: +60°	OSA:	Mag :
Con:	CSA:	Size : 21.0′

See:



U2

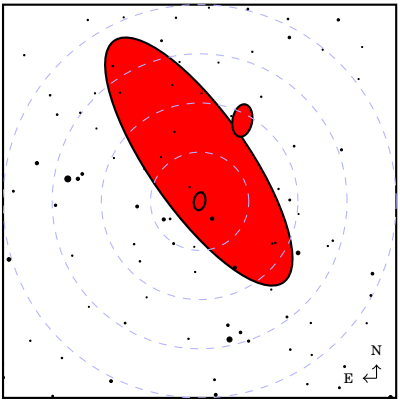
NGC 221 = M32 = BSC 110

RA : 00.7^h	PSA: 3	Type: Gal
Dec: +41°	OSA: N0	Mag : 8.2
Con: And	CSA: 2/7	Size : 11.0′ × 7.3′

See:

Notes: M31/U3 is 0.5° N. M110 is 1° NW.

β And Mirach (mag 2.1)
→ 2.5° W
→ 3.0° N to μ And (mag 3.9)
→ 2.5° W
→ 2.5° N to M32/U2



U3

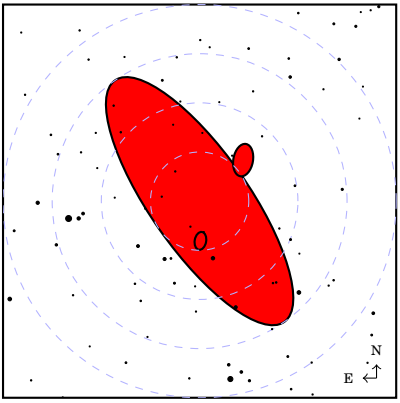
NGC 224 = M31 = BSC 11 = Andromeda Galaxy

RA : 00.7^h	PSA: 3	Type: Gal
Dec: +41°	OSA: N0	Mag : 3.4
Con: And	CSA: 2/7	Size : 3.0° × 1.0°

See:

Notes: M32/U2 is 0.5° S. M100 is 0.5° NW.

β And Mirach (mag 2.1)
→ 2.5° W
→ 3.0° N to μ And (mag 3.9)
→ 2.5° W
→ 3.0° N to M31/U3

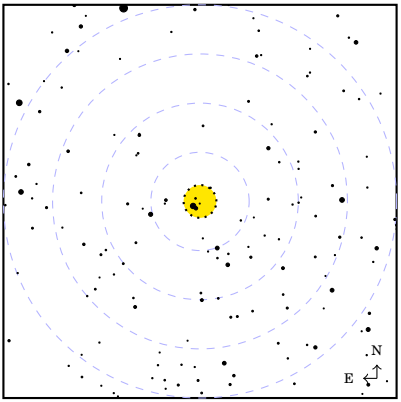


U4

NGC 457 = C13 = BSC 62 = E.T. Cluster = Owl Cluster

RA : 01.3^h	PSA: 1/3	Type: OC
Dec: +58°	OSA: N2	Mag : 6.4
Con:	CSA:	Size : 20.0′

See:



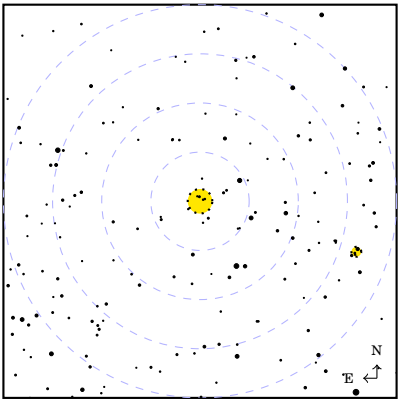
U5

NGC 663 = C10 = BSC 76 = Horseshoe Cluster

RA : 01.8^h	PSA: 1/2	Type: OC
Dec: +61°	OSA: N2	Mag : 6.7
Con: Cas	CSA: 1/2	Size : 15.0′

See:

Notes: M103 is 1.7° WSW. C8 is 3° NW.

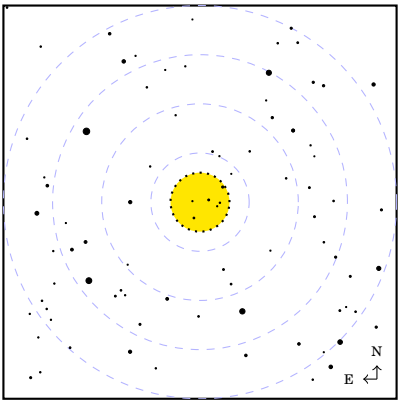


U6

Cr 463

RA : 01.8^h	PSA: 1	Type: OC
Dec: +72°	OSA:	Mag :
Con:	CSA:	Size : 36.0′

See:

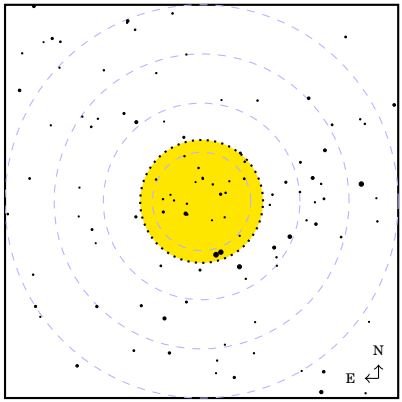


U7

NGC 752 = C28 = Mel 12

RA : 01.9^h	PSA: 2	Type: OC
Dec: +38°	OSA: N0	Mag : 5.7
Con:	CSA:	Size : 75.0′

See:



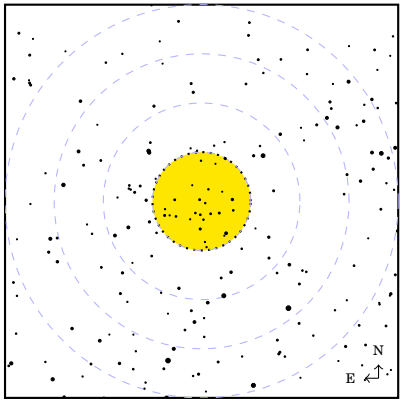
U8

Stock 2

RA : 02.3^h	PSA: 2	Type: OC
Dec: +60°	OSA:	Mag :
Con:	CSA:	Size : 60.0′

See:

Notes: C14/U9/U10 is 2° S.



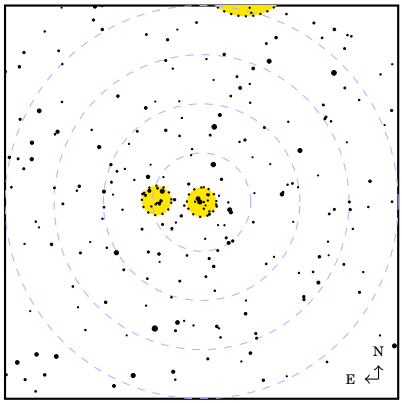
U9

NGC 869 = Mel 13 = Cr 24 = h Per Cluster = half of C14

RA : 02.3^h	PSA: 2	Type: OC
Dec: +57°	OSA: N2	Mag :
Con:	CSA:	Size : 18.0′

See:

Notes: U9 and U10 form C14. U10 is 0.5° W. U8 is 2° N. U11 is 3.0° SE.



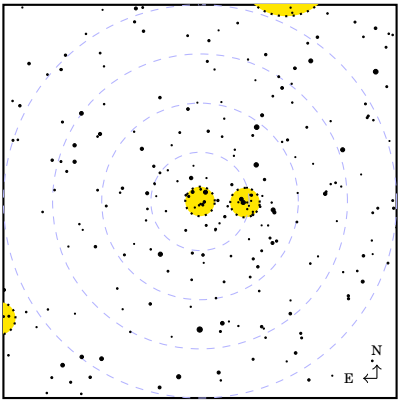
U10

NGC 884 = Mel 14 = Cr 25 = χ Per Cluster = half of C14

RA : 02.4^h	PSA: 2	Type: OC
Dec: $+57°$	OSA: N2	Mag :
Con:	CSA:	Size : $18.0'$

See:

Notes: U9 and U10 form C14. U9 is 0.5° E. U8 is 2° N. U11 is 2.5° SE.



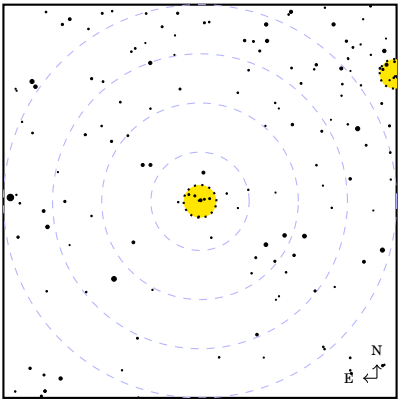
U11

Tr 2

RA : 02.6^h	PSA: 2	Type: OC
Dec: $+56°$	OSA:	Mag :
Con:	CSA:	Size : $20.0'$

See:

Notes: C14/U9/10 are 2.5° NW.

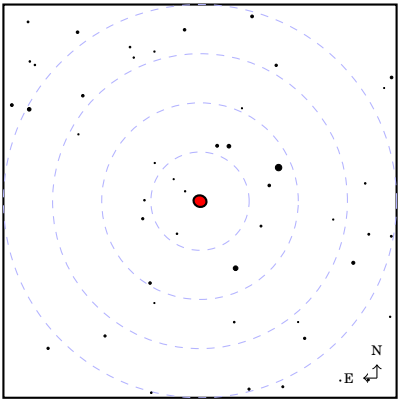


U12

NGC 1068 = M77

RA : 02.7^h	PSA: 4	Type: Gal
Dec: $-00°$	OSA: E0	Mag : 8.9
Con: Cet	CSA: 8	Size : $8.2' \times 7.3'$

See:

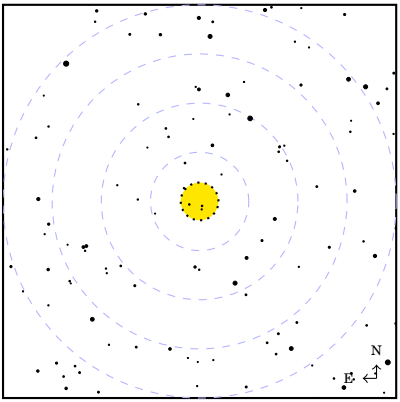


U13

Tr 3

RA : 03.2^h	PSA: 11/13	Type: OC
Dec: +63°	OSA:	Mag :
Con:	CSA:	Size : 23.0′

See:

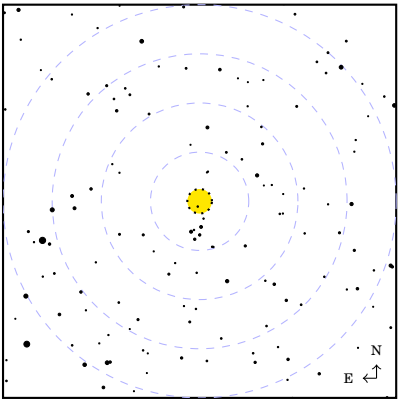


U14

Stock 23 = Pazmino's Cluster

RA : 03.3^h	PSA: 11/13	Type: OC
Dec: +60°	OSA:	Mag :
Con:	CSA:	Size : 15.0′

See:



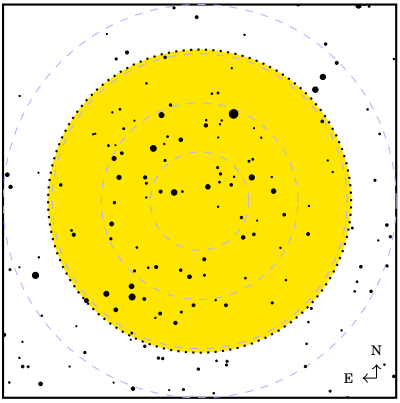
U15

Mel 20 = Cr 39 = α Per Cluster

RA : 03.4^h	PSA: 3	Type: OC
Dec: +49°	OSA:	Mag :
Con:	CSA:	Size : 3.1°

See:

Notes: Not shown in the PSA.

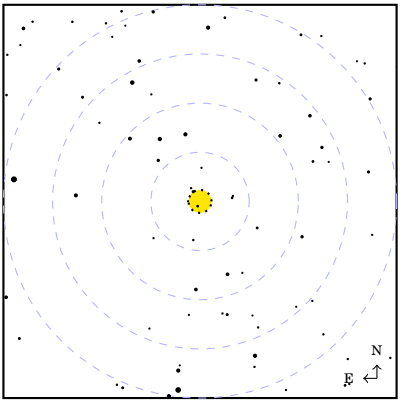


U16

NGC 1342

RA : 03.5^h	PSA: 13	Type: OC
Dec: +37°	OSA:	Mag :
Con:	CSA:	Size : 14.0′

See:

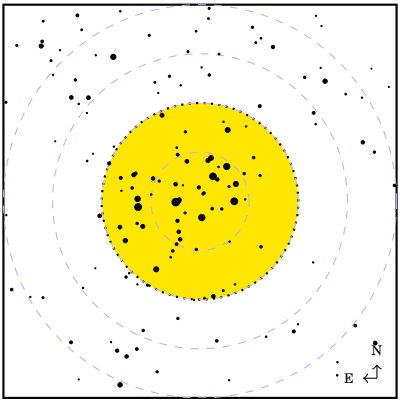


U17

M45 = BSC 4 = Mel 22 = Pleiades

RA : 03.8^h	PSA: 15/A	Type: OC
Dec: +24°	OSA: E3	Mag : 1.5
Con: Tau	CSA: 2/3/8/9	Size : 120.0′

See:



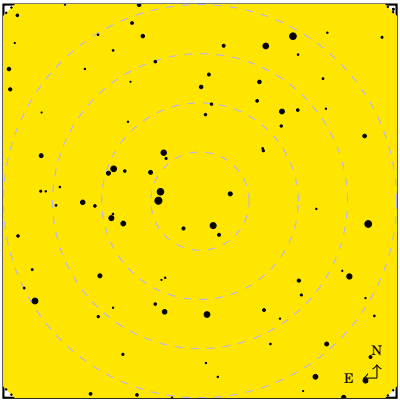
U18

C41 = BSC 1 = Mel 25 = Cr 50 = Hyades

RA : 04.4^h	PSA: 15	Type: OC
Dec: +16°	OSA: E3	Mag : 0.5
Con:	CSA:	Size : 5.5°

See:

Notes: Much bigger than the finder chart.



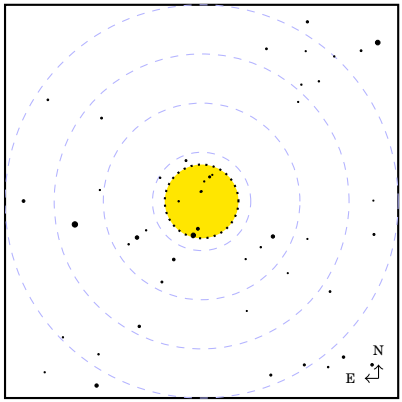
U19

NGC 1647

RA : 04.8^h	PSA: 15	Type: OC
Dec: +19°	OSA: E3	Mag :
Con:	CSA:	Size : 45.0′

See:

α Tau Betelgeuse (mag 0.3–0.9)
→ 2.5° N
→ 2.5° E to U19



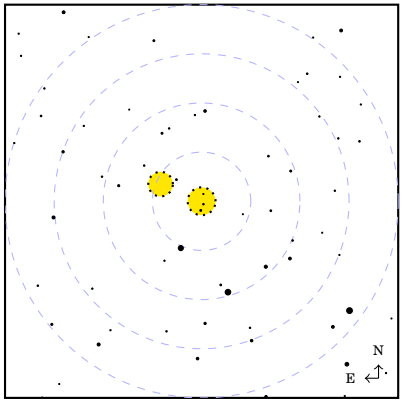
U20

NGC 1807 = Mel 29 = Cr 61

RA : 05.2^h	PSA: 14	Type: OC
Dec: +17°	OSA:	Mag :
Con:	CSA:	Size : 17.0′

See:

Notes: U21 is 0.5° WSW.



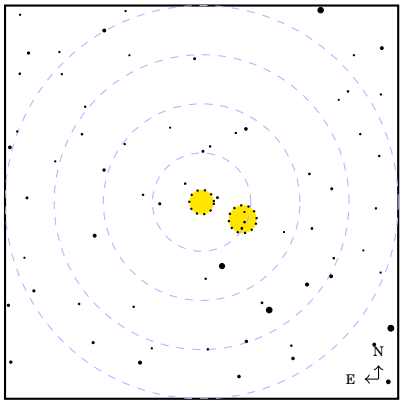
U21

NGC 1817 = Cr 60

RA : 05.2^h	PSA: 14	Type: OC
Dec: +17°	OSA:	Mag :
Con:	CSA:	Size : 15.0′

See:

Notes: U20 is 0.5° ENE.



U22

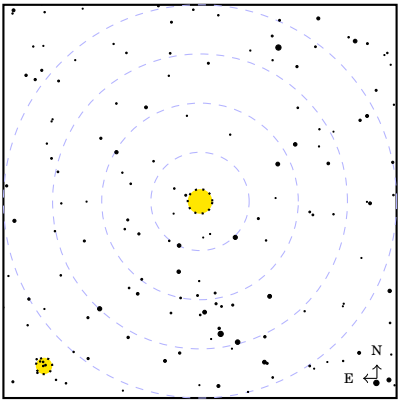
NGC 1912 = M38 = BSC 82

RA : 05.5^h	PSA: 12	Type: OC
Dec: +36°	OSA: N6	Mag : 6.4
Con: Aur	CSA: 3	Size : 15.0′

See:

Notes: M36/U23 is 2.5° SE.

β Tau Elnath (mag 1.7)
→ 7° N to M38/U23

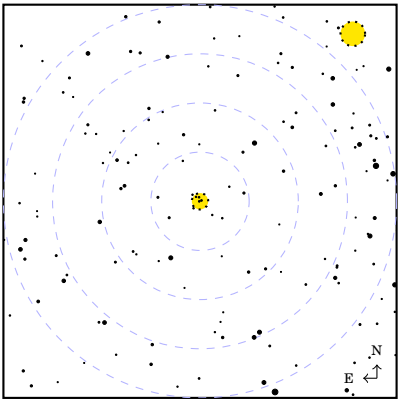


U23

NGC 1960 = M36 = BSC 58

RA : 05.6^h	PSA: 12	Type: OC
Dec: +34°	OSA: N6	Mag : 6.0
Con: Aur	CSA: 3	Size : 10.0′

See:



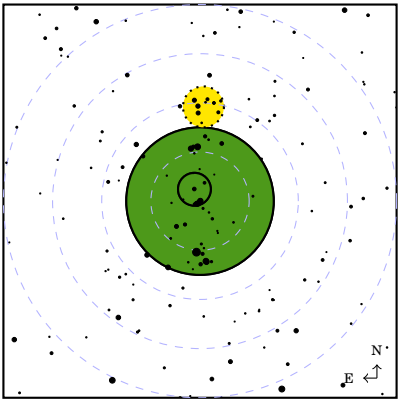
U24

NGC 1976 = M42 = BSC 15 = Orion Nebula

RA : 05.6^h	PSA: 16/B	Type: BN
Dec: −05°	OSA: E4	Mag : 3.7
Con: Ori	CSA: 9	Size : 90.0′ × 60.0′

See:

Notes: M42/U24 surrounds M43. U25 is 1° N.



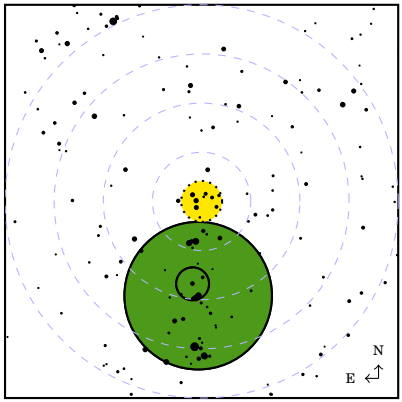
U25

NGC 1981

RA : 05.6^h	PSA: 16/B	Type: OC
Dec: $-04°$	OSA: E4	Mag :
Con:	CSA:	Size : 25.0′

See:

Notes: M42/U24 and M43 are 1° S.



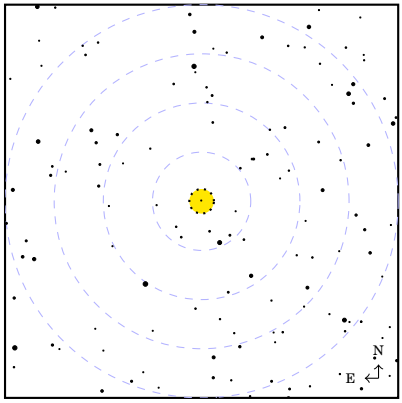
U26

NGC 2099 = M37 = BSC 55

RA : 05.9^h	PSA: 12	Type: OC
Dec: $+33°$	OSA: N6	Mag : 5.6
Con: Aur	CSA: 3	Size : 15.0′

See:

β Tau Elnath (mag 1.7)
→ 6° W to point 1° N of 136 Tau
→ 4° N to M37/U26

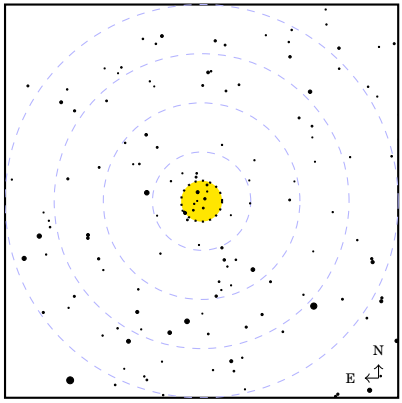


U27

NGC 2168 = M35 = BSC 31

RA : 06.2^h	PSA: 14	Type: OC
Dec: $+24°$	OSA: E7	Mag : 5.1
Con: Gem	CSA: 3/9	Size : 25.0′

See:

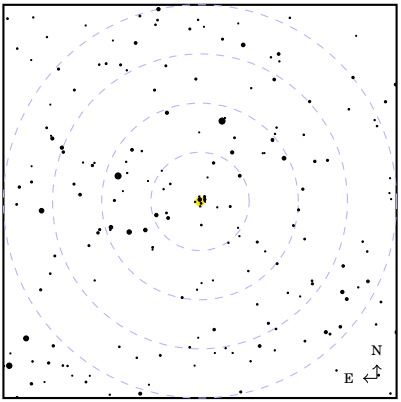


U28

NGC 2169

RA : 06.1^h	PSA: 14/25	Type: OC
Dec: +14°	OSA:	Mag :
Con:	CSA:	Size : 6.0′

See:



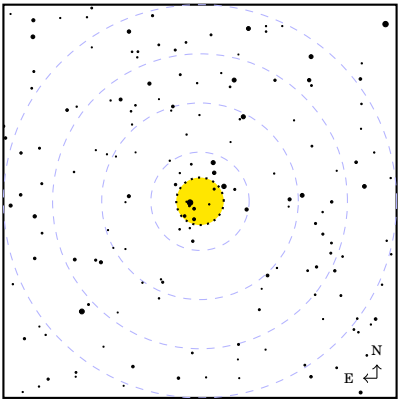
U29

NGC 2232

RA : 06.5^h	PSA: 25/27	Type: OC
Dec: −05°	OSA:	Mag :
Con:	CSA:	Size : 29.0′

See:

κ Ori Saiph (mag 2.1)
→ 3° N
→ 11° E passing γ Mon to β Mon
→ 2.5° N to U29



U30

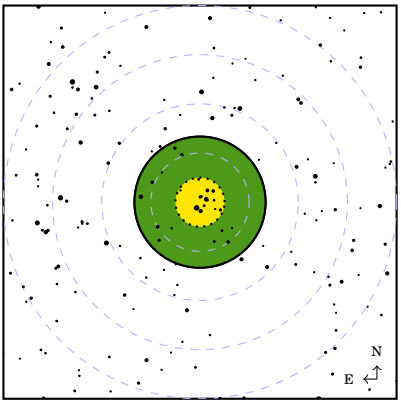
NGC 2244 = C50 = BSC 25

RA : 06.5^h	PSA: 25/E	Type: OC
Dec: +05°	OSA: E9	Mag : 4.4
Con:	CSA:	Size : 30.0′

See:

Notes: Surrounded by C49.

α Ori Betelgeuse (mag 0.3–0.9)
→ 3° S
→ 7° E to 8 Mon
→ 2° E to C50/U30



U31

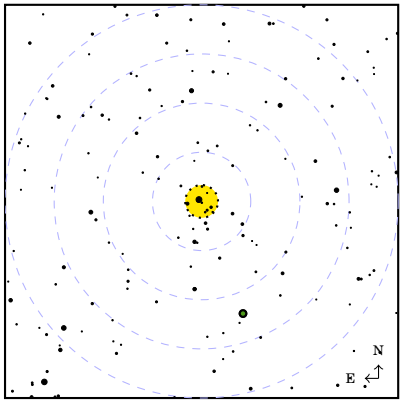
NGC 2264

RA : 06.7^h	PSA: 25/E	Type: OC
Dec: +10°	OSA: E7	Mag :
Con:	CSA:	Size : 20.0′

See:

Notes: C46 is 1.25° SSW.

γ Gem Alhena (mag 1.9)
→ 7° S passing ξ Gem and 30 Gem to U31

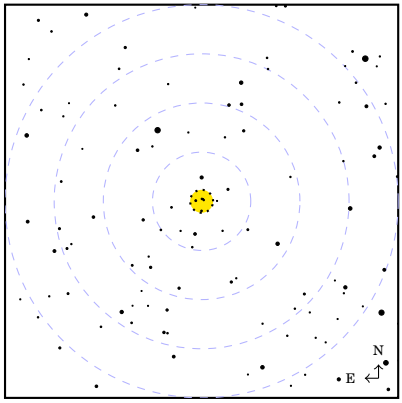


U32

NGC 2281

RA : 06.8^h	PSA: 23	Type: OC
Dec: +41°	OSA: N6	Mag :
Con:	CSA:	Size : 14.0′

See:



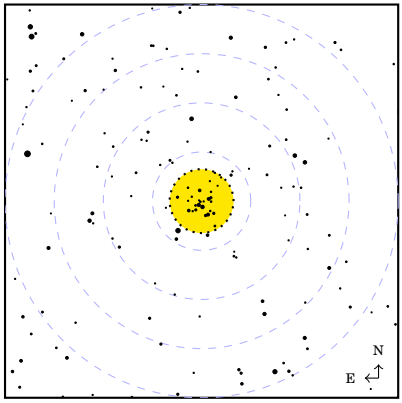
U33

NGC 2287 = M41 = BSC 21

RA : 06.8^h	PSA: 27	Type: OC
Dec: −21°	OSA: E6	Mag : 4.5
Con: CMa	CSA: 9/15	Size : 39.0′

See:

α CMa Sirius (mag −1.5)
→ 4° S to M41/U33

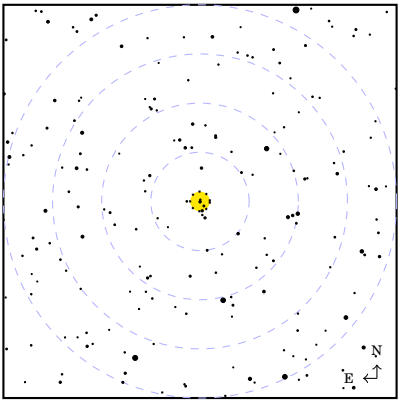


U34

NGC 2301

RA : 06.9^h	PSA: 25/27	Type: OC
Dec: +00°	OSA: E6	Mag :
Con:	CSA:	Size : 12.0′

See:



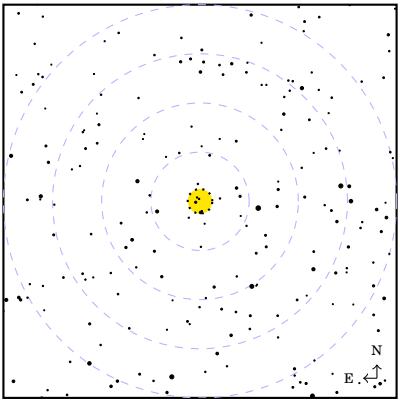
U35

NGC 2323 = M50 = BSC 44

RA : 07.0^h	PSA: 27	Type: OC
Dec: −08°	OSA: E8	Mag : 5.9
Con: Mon	CSA: 9	Size : 15.0′

See:

α CMa Sirius (mag −1.5)
→ 3° E to ι CMa
→ 5° N passing μ CMa to θ CMa
→ 4° N to asterism of three stars
→ 2° E to M50/U35



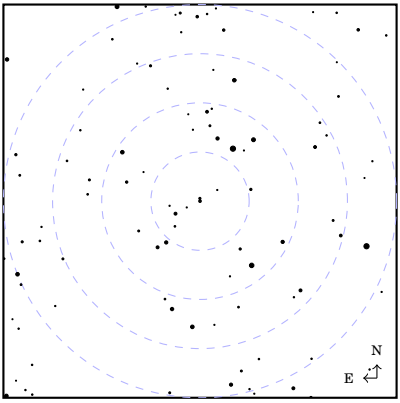
U36

NGC 2392 = C39 = Eskimo Nebula

RA : 07.5^h	PSA: 25	Type: PN
Dec: +21°	OSA: E7	Mag : 9.2
Con:	CSA:	Size : 0.8′×0.7′

See:

Notes: The open cluster NGC 2420 is 2° N (see DSC:TCO).

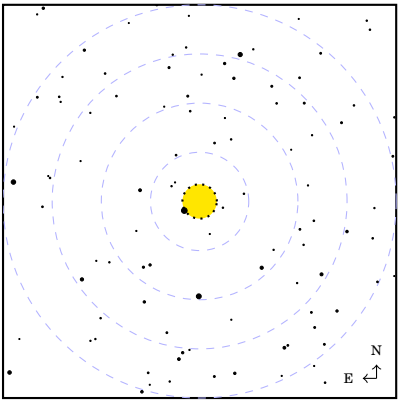


U37

NGC 2539 = Mel 83

RA : 08.2^h	PSA: 26	Type: OC
Dec: −13°	OSA: E8	Mag :
Con:	CSA:	Size : 21.0′

See:

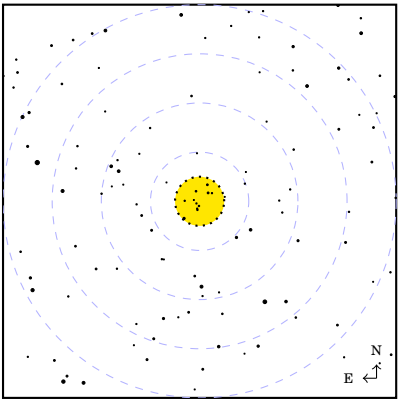


U38

NGC 2548 = M48 = BSC 37

RA : 08.2^h	PSA: 26	Type: OC
Dec: −06°	OSA: E10	Mag : 5.8
Con: Hya	CSA: 9/10	Size : 30.0′

See:

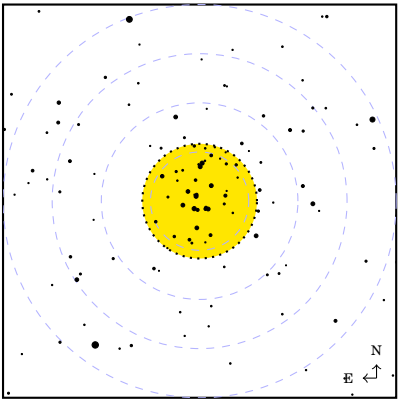


U39

NGC 2632 = M44 = BSC 12 = Mel 88 = Beehive Cluster

RA : 08.7^h	PSA: 24	Type: OC
Dec: +20°	OSA: E9	Mag : 3.1
Con: Cnc	CSA: 3/4/9/10	Size : 70.0′

See:

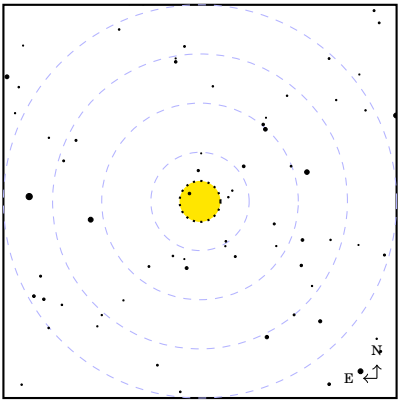


U40

NGC 2682 = M67 = BSC 51

RA : 08.9^h PSA: 24 Type: OC
Dec: +12° OSA: E9 Mag : 6.0
Con: Cnc CSA: 10 Size : 25.0′

See:



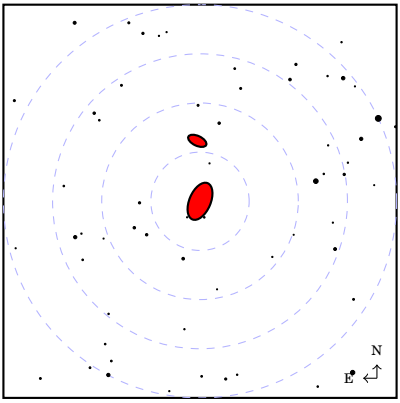
U41

NGC 3031 = M81 = BSC 72

RA : 09.9^h PSA: 31 Type: Gal
Dec: +69° OSA: N8 Mag : 6.9
Con: UMa CSA: 1/4 Size : 24.0′ ×
13.0′

See:

Notes: M82/U42 is 0.6° N.



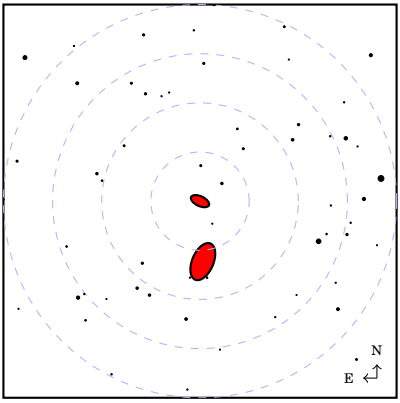
U42

NGC 3034 = M82 = BSC 122

RA : 09.9^h PSA: 31 Type: Gal
Dec: +70° OSA: N8 Mag : 8.4
Con: UMa CSA: 1/4 Size : 12.0′ ×
5.6′

See:

Notes: M81/U41 is 0.6° S.

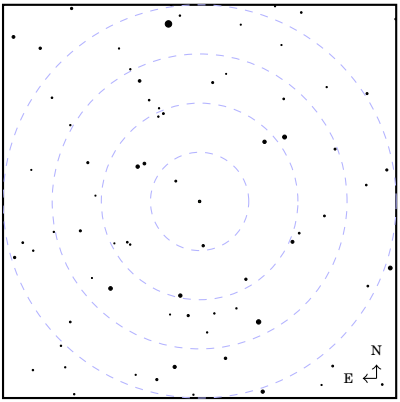


U43

NGC 3242 = C59 = BSC 125 = Ghost of Jupiter

RA : 10.4^h	PSA: 36/37	Type: PN
Dec: $-19°$	OSA: E10	Mag : 7.3
Con:	CSA:	Size : $0.8' \times 0.6'$

See:



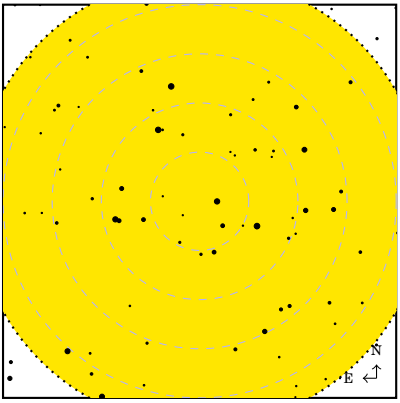
U44

Mel 111 = Cr 256 = Coma Star Cluster

RA : 12.4^h	PSA: 45	Type: OC
Dec: $+26°$	OSA: E13	Mag :
Con:	CSA:	Size : 4.6°

See:

Notes: Not shown in the PSA.



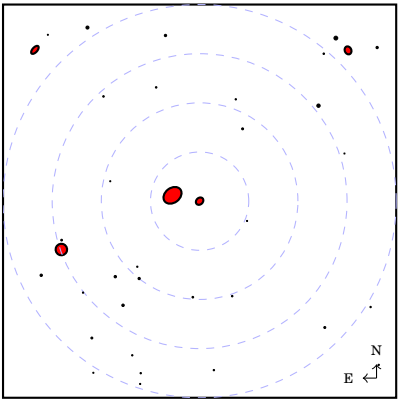
U45

NGC 4374 = M84

RA : 12.4^h	PSA: 45/C	Type: Gal
Dec: $+13°$	OSA: E14	Mag : 9.1
Con: Vir	CSA: 4/10/11	Size : $5.1' \times 4.1'$

See:

Notes: In the Virgo Cluster.



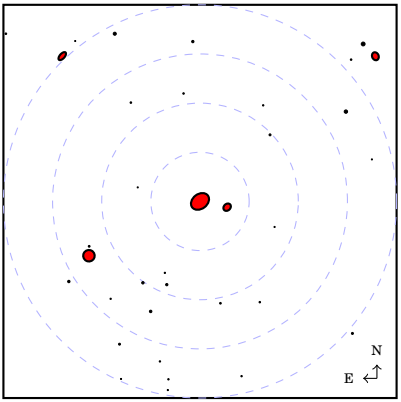
U46

NGC 4406 = M86

RA : 12.4^h	PSA: 45/C	Type: Gal
Dec: +13°	OSA: E14	Mag : 8.9
Con: Vir	CSA: 4/10/11	Size : 12.0′ × 9.3′

See:

Notes: In the Virgo Cluster.



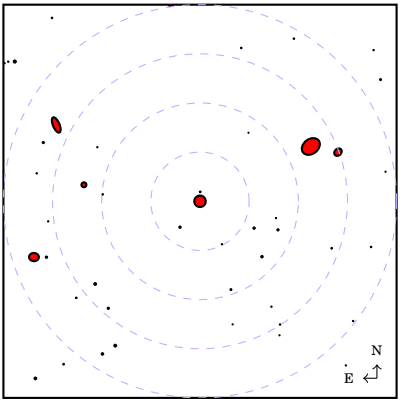
U47

NGC 4486 = M87

RA : 12.5^h	PSA: 45/C	Type: Gal
Dec: +12°	OSA: E14	Mag : 8.6
Con: Vir	CSA: 10/11	Size : 7.1′

See:

Notes: In the Virgo Cluster.

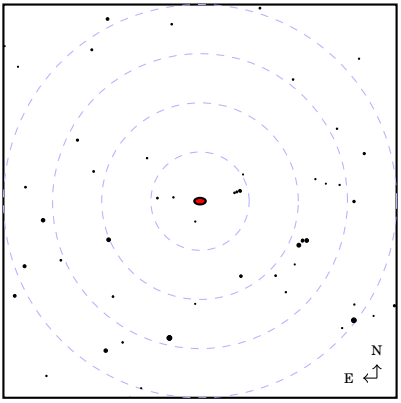


U48

NGC 4594 = M104 = BSC 133 = Sombrero Galaxy

RA : 12.7^h	PSA: 47	Type: Gal
Dec: −12°	OSA: E12	Mag : 8.0
Con: Vir	CSA: 10/11	Size : 7.1′×4.4′

See:

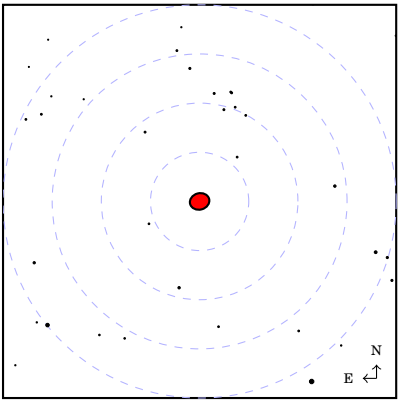


U49

NGC 4736 = M94 = BSC 134

RA : 12.8^h PSA: 43 Type: Gal
Dec: +41° OSA: N12 Mag : 8.2
Con: CVn CSA: 4/5 Size : 11.5′ × 10.0′

See:

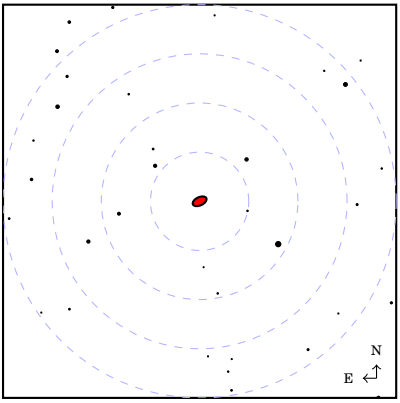


U50

NGC 4826 = M64 = BSC 149 = Black-Eye Galaxy

RA : 12.9^h PSA: 45 Type: Gal
Dec: +22° OSA: E13 Mag : 8.5
Con: Com CSA: 5/11 Size : 9.2′ × 4.6′

See:

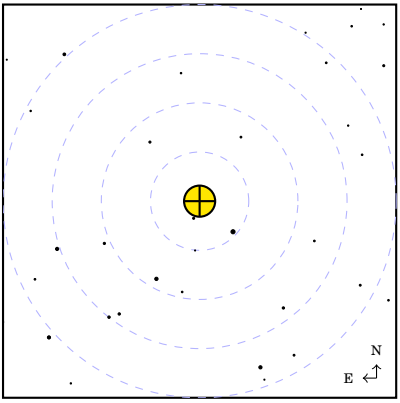


U51

NGC 5272 = M3 = BSC 53

RA : 13.7^h PSA: 43/44 Type: GC
Dec: +28° OSA: E15 Mag : 5.9
Con: CVn CSA: 5 Size : 19.0′

See:

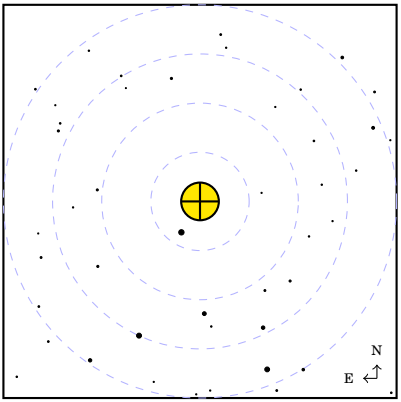


U52

NGC 5904 = M5 = BSC 66

RA : 15.3^h	PSA: 55/57	Type: GC
Dec: $+02°$	OSA: E15	Mag : 5.7
Con: Ser	CSA: 11/12	Size : $23.0'$

See:

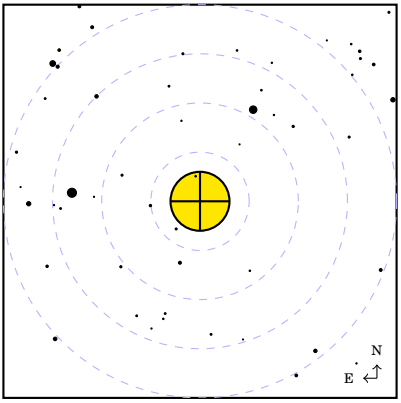


U53

NGC 6121 = M4 = BSC 43

RA : 16.4^h	PSA: 56/58	Type: GC
Dec: $-27°$	OSA: E18	Mag : 5.4
Con: Sco	CSA: 11/12/17/18	Size : $36.0'$

See:



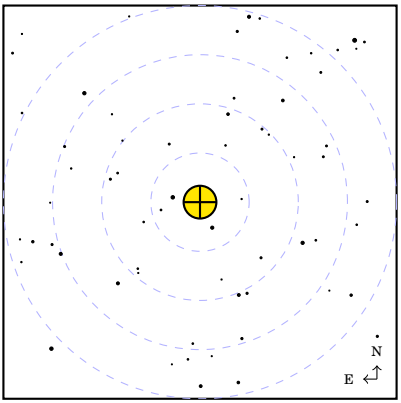
U54

NGC 6205 = M13 = BSC 40 = Hercules Cluster

RA : 16.7^h	PSA: 52	Type: GC
Dec: $+36°$	OSA: N14	Mag : 5.3
Con: Her	CSA: 5/6	Size : $20.0'$

See:

ζ Her (mag 2.8)
$\rightarrow 5°$ N (2/3 distance to η Her) to M13/U54

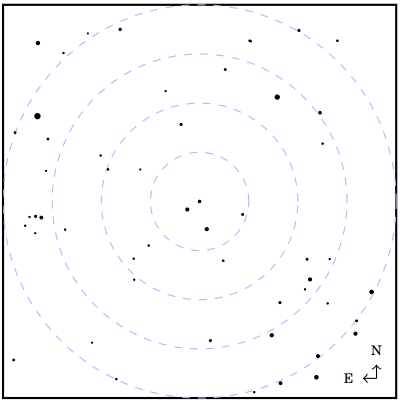


U55

NGC 6210

RA : 16.7^h	PSA: 54	Type: PN
Dec: +24°	OSA: E19	Mag :
Con:	CSA:	Size : 0.8′

See:

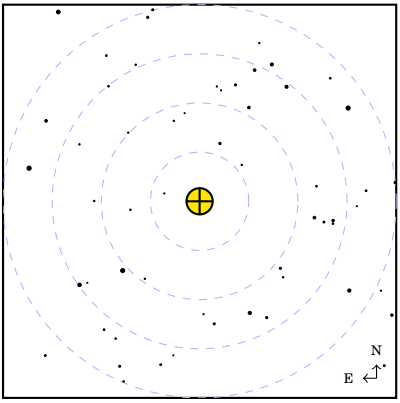


U56

NGC 6218 = M12 = BSC 94

RA : 16.8^h	PSA: 54/56	Type: GC
Dec: −02°	OSA: E17	Mag : 6.1
Con: Oph	CSA: 12	Size : 16.0′

See:

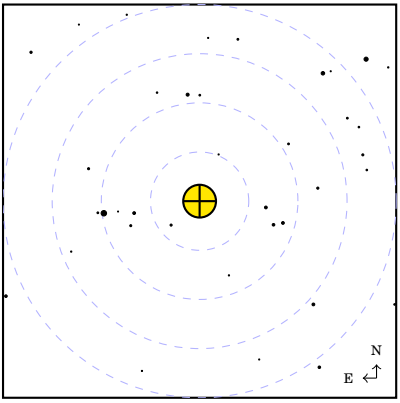


U57

NGC 6254 = M10 = BSC 61

RA : 17.0^h	PSA: 54/56	Type: GC
Dec: −04°	OSA: E17	Mag : 6.6
Con: Oph	CSA: 12	Size : 20.0′

See:



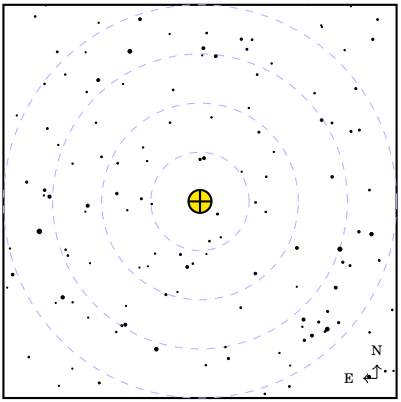
U58

NGC 6266 = M62 = BSC 81

RA : 17.0^h	PSA: 58	Type: GC
Dec: −30°	OSA: E18	Mag : 6.7
Con: Oph	CSA: 17/18	Size : 14.0′

See:

ε Sco (mag 2.3)
→ 2° E
→ 4° N to M63



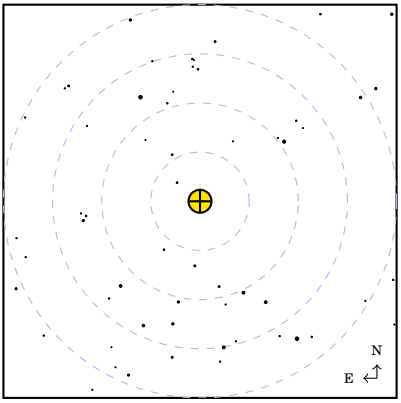
U59

NGC 6341 = M92 = BSC 59

RA : 17.3^h	PSA: 52	Type: GC
Dec: +43°	OSA: N14	Mag : 6.5
Con: Her	CSA: 5/6	Size : 14.0′

See:

π Her (mag 3.2)
→ 6.5° N to M92/U59



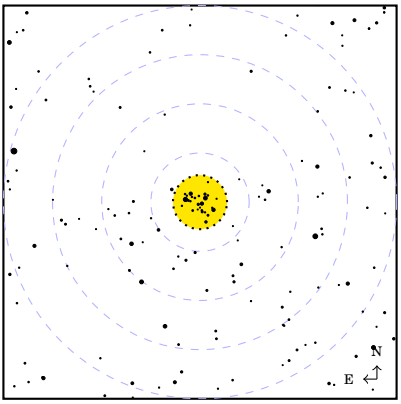
U60

NGC 6405 = M6 = BSC 18 = Butterfly Nebula

RA : 17.7^h	PSA: 58/69	Type: OC
Dec: −32°	OSA: E18	Mag : 4.2
Con: Sco	CSA: 18	Size : 33.0′

See:

λ Sco Shaula (mag 1.6)
→ 5° N
→ 1° E to M6/U60

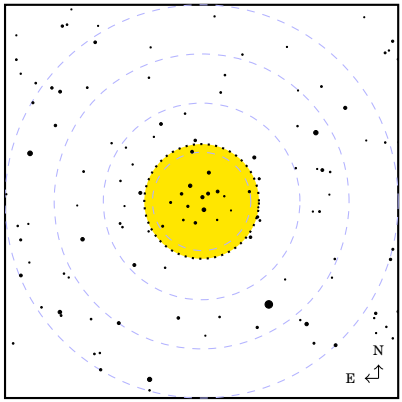


U61

IC 4665 = Mel 179 = Cr 349

RA : 17.8^h	PSA: 54	Type: OC
Dec: $+06°$	OSA: E17	Mag :
Con:	CSA:	Size : 70.0′

See:



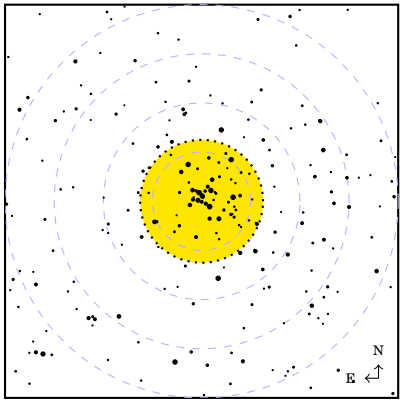
U62

NGC 6475 = M7 = BSC 10

RA : 17.9^h	PSA: 58/69	Type: OC
Dec: $-35°$	OSA: E18	Mag : 2.8
Con: Sco	CSA: 18	Size : 75.0′

See:

ε Sgr Kaus Australis (mag 1.8)
→ 6° W to M7/U62

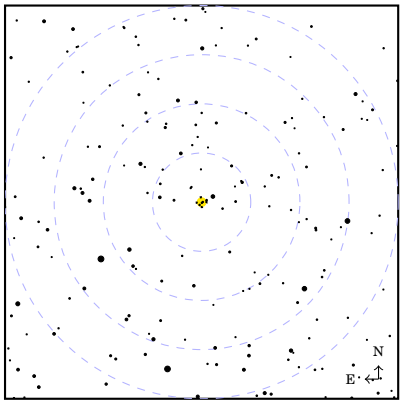


U63

NGC 6520

RA : 18.1^h	PSA: 67	Type: OC
Dec: $-28°$	OSA:	Mag :
Con:	CSA:	Size : 6.0′

See:



U64

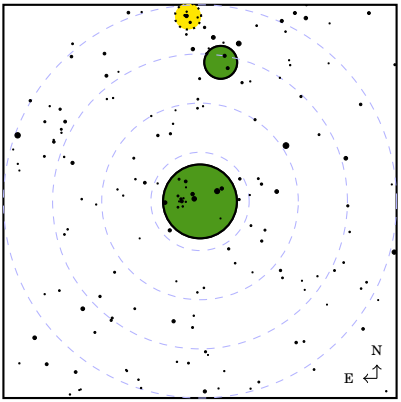
NGC 6523/6530 = M8 = BSC 45 = Lagoon Nebula

RA : 18.1^h	PSA: 67	Type: BN	
Dec: $-24°$	OSA: E20	Mag : 3.0	
Con: Sgr	CSA: 12/18	Size : 45.0′	×
		30.0′	

See:

Notes: M20 is 1.5° N. M21 is 2.0° N.

γ Sgr Alnasl (mag 3.0)
→ 6° N to M8/U64



U65

NGC 6618 = M17 = BSC 47 = Omega Nebula

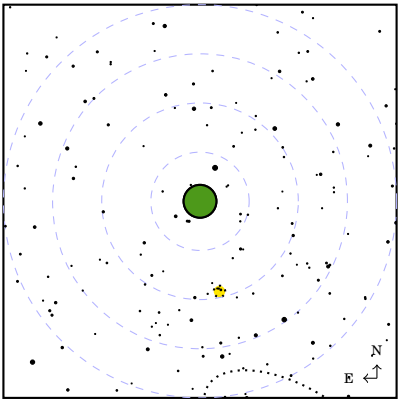
RA : 18.3^h	PSA: 67	Type: BN	
Dec: $-16°$	OSA: E20	Mag : 6.0	
Con: Sgr	CSA: 12	Size : 20.0′	×
		15.0′	

See:

Notes: M18 is 1° S. M24 is 3° SW.

ν Oph (mag 3.3)
→ 5° E
→ 6° S to M17/U65

μ Sgr (mag 3.8)
→ 5° N passing M24 and M18
→ 2° E to M17

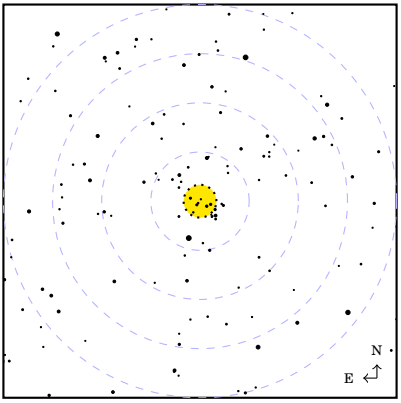


U66

NGC 6633

RA : 18.5^h	PSA: 65	Type: OC
Dec: $+07°$	OSA: E19	Mag :
Con:	CSA:	Size : 20.0′

See:

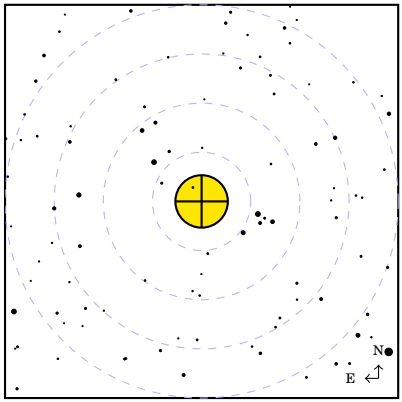


U67

NGC 6656 = M22 = BSC 28

RA : 18.6^h	PSA: 67	Type: GC
Dec: −24°	OSA: E20	Mag : 5.2
Con: Sgr	CSA: 12/18	Size : 32.0′

See:

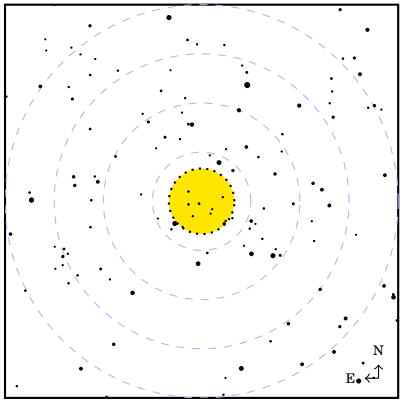


U68

IC 4756 = Mel 210

RA : 18.6^h	PSA: 65	Type: OC
Dec: +05°	OSA: E19	Mag :
Con:	CSA:	Size : 40.0′

See:

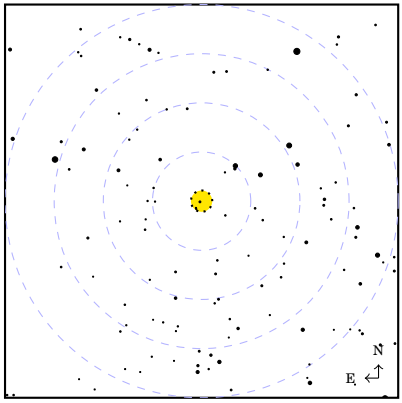


U69

NGC 6705 = M11 = Mel 213 = Cr 391 = Wild Duck Cluster

RA : 18.9^h	PSA: 65/67	Type: OC
Dec: −06°	OSA: E19	Mag : 5.3
Con: Sct	CSA: 12	Size : 13.0′

See:

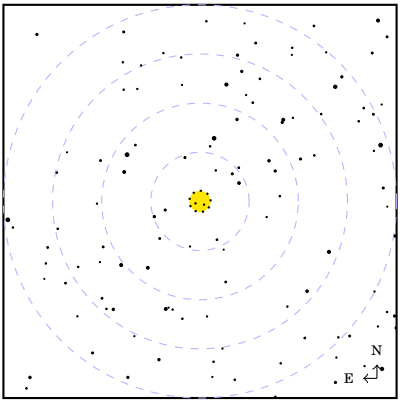


U70

NGC 6709

RA : 18.9^h	PSA: 65	Type: OC
Dec: +10°	OSA:	Mag :
Con:	CSA:	Size : 13.0′

See:



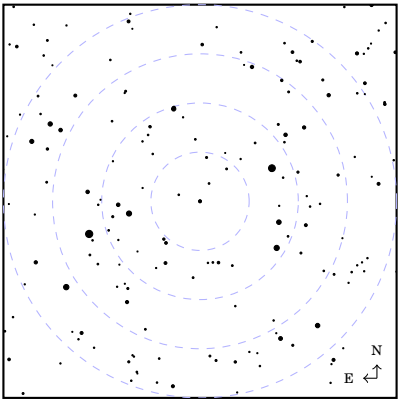
U71

NGC 6720 = M57 = BSC 127 = Ring Nebula

RA : 18.9^h	PSA: 63	Type: PN
Dec: +33°	OSA: N18	Mag : 8.8
Con: Lyr	CSA: 6/7	Size : 1.1′

See:

γ Lyr Sulaphat (mag 3.2)
→ 1.2° W
→ 0.4° N to M57/U71

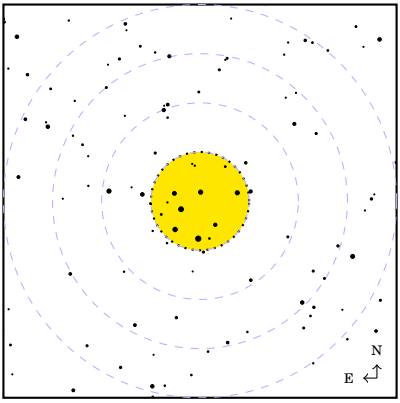


U72

Cr 399 = Brocchi's Cluster

RA : 19.4^h	PSA: 64/65	Type: OC
Dec: +20°	OSA:	Mag :
Con:	CSA:	Size : 60.0′

See:



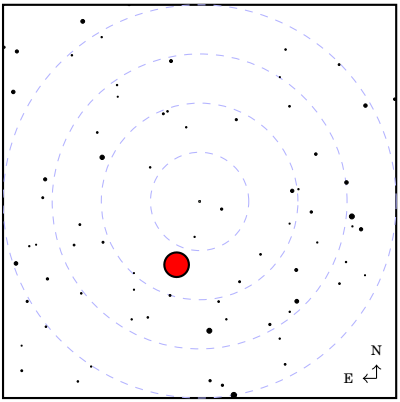
U73

NGC 6818

RA : 19.7^h	PSA: 66	Type: PN
Dec: $-14°$	OSA: E22	Mag :
Con:	CSA:	Size : $0.4' \times 0.3'$

See:

Notes: C57 is 0.7° SSE.

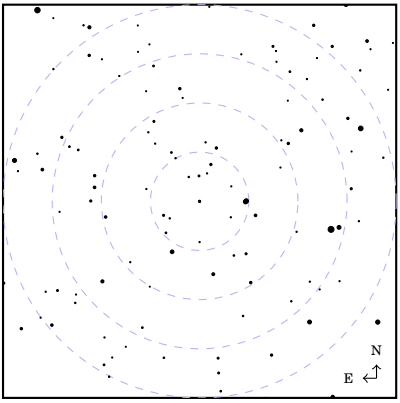


U74

NGC 6826

RA : 19.7^h	PSA: 62	Type: PN
Dec: $+51°$	OSA: N18	Mag :
Con:	CSA:	Size : $0.5' \times 0.4'$

See:

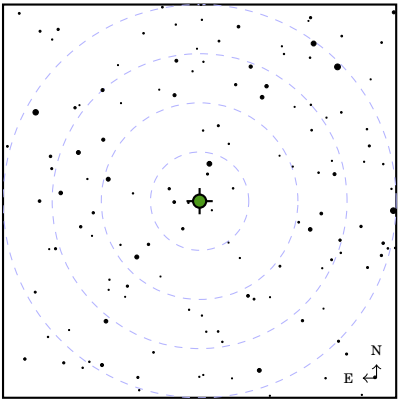


U75

NGC 6853 = M27 = BSC 88 = Dumbell Nebula

RA : 20.0^h	PSA: 64	Type: PN
Dec: $+23°$	OSA: E21	Mag : 7.3
Con: Vul	CSA: 6/7/12/13	Size : $8.0' \times 6.0'$

See:

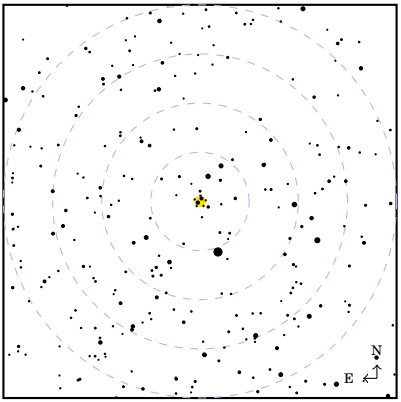


U76

NGC 6910

RA : 20.4^h	PSA: 62	Type: OC
Dec: +41°	OSA:	Mag :
Con:	CSA:	Size : 7.0′

See:

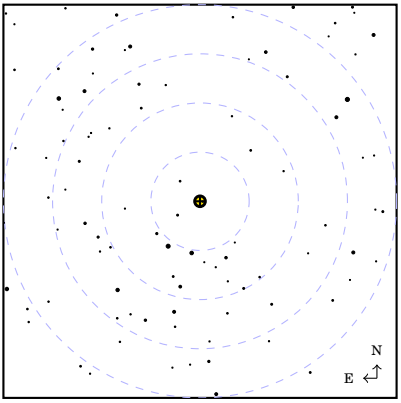


U77

NGC 6934

RA : 20.6^h	PSA: 64	Type: GC
Dec: +07°	OSA:	Mag :
Con:	CSA:	Size : 5.9′

See:

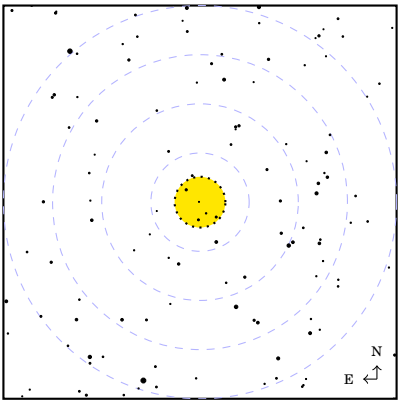


U78

NGC 6940 = Mel 232

RA : 20.6^h	PSA: 62	Type: OC
Dec: +28°	OSA: N20	Mag :
Con:	CSA:	Size : 31.0′

See:

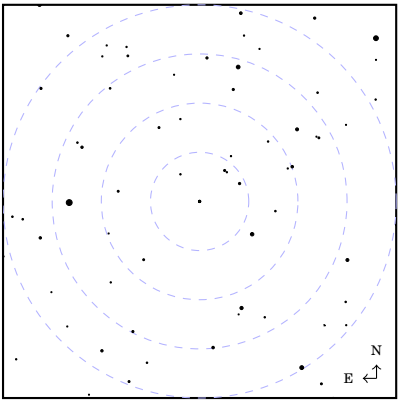


U79

NGC 7009

RA : 21.1^h	PSA: 77	Type: PN
Dec: $-11°$	OSA: E24	Mag :
Con:	CSA:	Size : $0.7' \times 0.4'$

See:

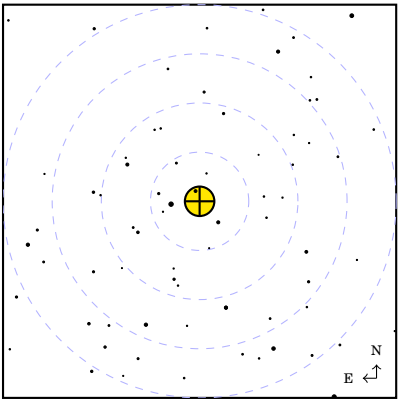


U80

NGC 7078 = M15 = BSC 54

RA : 21.5^h	PSA: 75	Type: GC
Dec: $+12°$	OSA: E23	Mag : 6.0
Con: Peg	CSA: 13	Size : 18.0′

See:

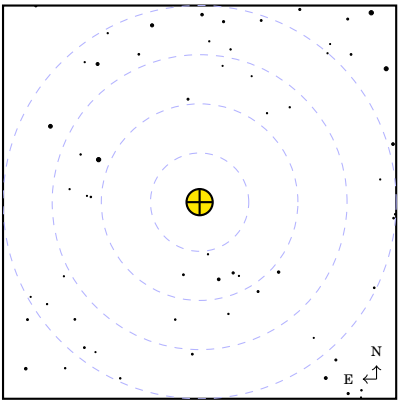


U81

NGC 7089 = M2 = BSC 56

RA : 21.6^h	PSA: 75/77	Type: GC
Dec: $-01°$	OSA: E24	Mag : 6.3
Con: Aqr	CSA: 13	Size : 16.0′

See:

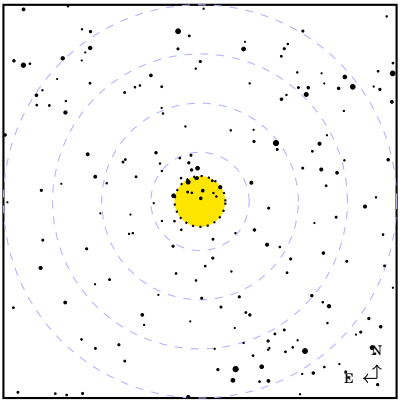


U82

NGC 7092 = M39 = BSC 36

RA : 21.5^h	PSA: 73	Type: OC
Dec: +48°	OSA: N24	Mag : 4.6
Con: Cyg	CSA: 1/6/7	Size : 31.0′

See:

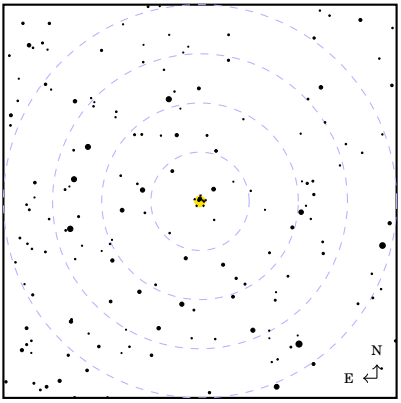


U83

NGC 7160

RA : 21.9^h	PSA: 71	Type: OC
Dec: +63°	OSA:	Mag :
Con:	CSA:	Size : 7.0′

See:



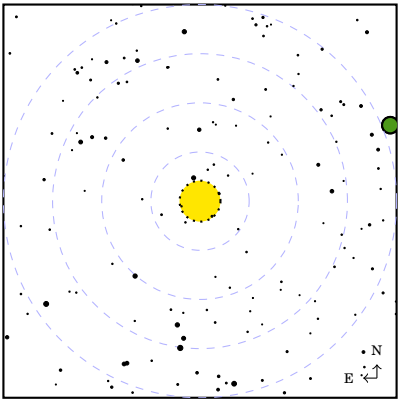
U84

NGC 7209

RA : 22.1^h	PSA: 73	Type: OC
Dec: +47°	OSA: N24	Mag :
Con:	CSA:	Size : 25.0′

See:

Notes: C19 is 2° WNW.

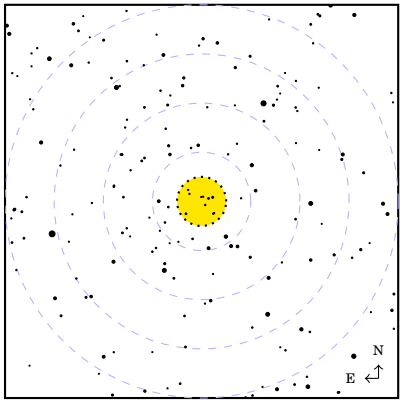


U85

NGC 7243 = C16 = BSC 63

RA : 22.3^h	PSA: 73	Type: OC
Dec: +50°	OSA: N24	Mag : 6.4
Con:	CSA:	Size : 30.0′

See:

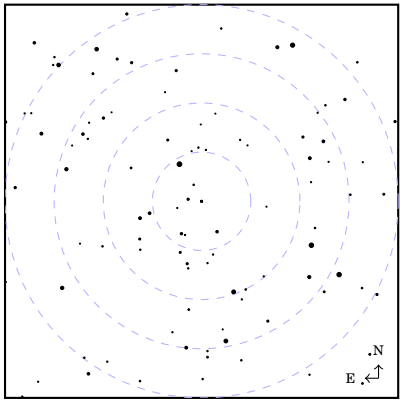


U86

NGC 7662

RA : 23.4^h	PSA: 72	Type: PN
Dec: +43°	OSA: N24	Mag :
Con:	CSA:	Size : 0.5′

See:

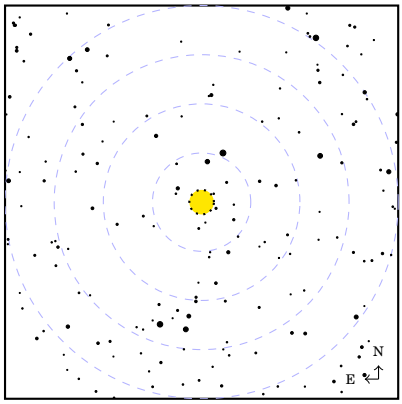


U87

NGC 7789 = Mel 245

RA : 23.9^h	PSA: 72	Type: OC
Dec: +57°	OSA: N22	Mag :
Con:	CSA:	Size : 15.0′

See:



Bibliography

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